51 Mechanisms of Progression in Chronic Kidney Disease

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CHAPTER OUTLINE

INTRODUCTION, 1742
STRUCTURAL AND FUNCTIONAL ADAPTATION OF THE KIDNEY TO NEPHRON LOSS, 1743
ADAPTATION OF SPECIFIC TUBULE FUNCTIONS IN RESPONSE TO NEPHRON LOSS, 1752
LONG-TERM ADVERSE CONSEQUENCES OF ADAPTATIONS TO NEPHRON LOSS, 1757
INSIGHTS FROM MODIFIERS OF CHRONIC KIDNEY DISEASE PROGRESSION, 1772
FUTURE DIRECTIONS, 1788

KEY POINTS

- Although the usefulness of the term "chronic kidney disease" has been questioned, the concept of CKD is strongly supported by evidence that in response to nephron loss, a common pathway of mechanisms provokes a vicious cycle of progressive kidney damage that eventuates in further nephron loss and explains the progressive nature of CKD of diverse causes.
- Glomerular hemodynamic adaptations to nephron loss resulting in glomerular capillary hypertension and glomerular hyperfiltration are key factors that drive CKD progression. This hypothesis is strongly supported by trials showing that treatment with both RAAS inhibitors and SGLT2 inhibitors, drugs that act by different mechanisms to reduce glomerular capillary hydraulic pressure, affords renoprotection.
- Nonhemodynamic factors, including proteinuria, tubulointerstitial fibrosis, oxidative stress, acidosis,and renal hypertrophy, act in concert with hemodynamic factors to provoke progressive kidney damage.
- In human CKD, superimposed episodes of acute kidney injury likely play a significant role in accelerating nephron loss in many patients.
- A thorough understanding of the multiple factors that contribute to common pathway mechanisms of CKD progression is essential to inform strategies for achieving renoprotection. Because of the complexity of the interacting mechanisms, it is clear that to achieve optimal renoprotection, the common pathway must be blocked at multiple points.

INTRODUCTION

The introduction of a definition for chronic kidney disease (CKD) by the National Kidney Foundation Kidney Disease Outcomes Quality Initiative (K/DOQI) and its adoption worldwide have made a valuable contribution to raising awareness of the global burden of disease due to CKD.[1] Importantly the adoption of a classification system for CKD that divides the spectrum of CKD into stages has emphasized the progressive nature of CKD and facilitated the development of stage-specific strategies for slowing progression. as well as treating the complications of chronic renal failure. These developments highlight the importance of understanding the mechanisms that contribute to CKD progression to inform strategies for slowing such progression. Central to these mechanisms are the adaptations observed in the kidney when nephrons are lost.

The kidney's primary function of maintaining constancy of the extracellular fluid (ECF) volume and composition is remarkably well preserved until late in the course of CKD. When nephrons are lost through disease or surgical ablation, those remaining or are least affected undergo remarkable physiologic responses, resulting in hypertrophy and hyperfunction that combine to compensate for the acquired loss of renal function. Effective kidney function requires close integration of glomerular and tubular functions. Indeed, the preservation of glomerulotubular balance seen until the terminal stages of CKD is fundamental to the intact nephron hypothesis of Bricker, which essentially states that as CKD advances, kidney function is supported by a diminishing pool of functioning (or hyperfunctioning) nephrons, rather than relatively constant numbers of nephrons, each with diminishing function.[2] This concept has important implications for the mechanisms of disease progression in CKD.

Several decades ago, clinical studies of patients with CKD established that once the glomerular filtration rate (GFR) fell below a critical level, a relentless progression to end-stage renal failure usually ensued, even when the initial disease activity had abated. The rate of decline of the GFR in a given individual often followed a near-constant linear relationship with time, enabling remarkably accurate predictions of the date at which end-stage renal failure would be reached and renal replacement therapy required. Among patients with diverse renal diseases, the slope of the GFR-time relationship was found to be a characteristic of individual patients rather than typical of their specific renal diseases. This observation suggested that the progressive nature of renal disease could be attributed to a final common pathway of mechanisms, independent of the primary cause of nephropathy.[3] Within this framework, Brenner and colleagues formulated a unifying hypothesis for renal disease progression based on the physiologic adaptations observed in experimental models of CKD.[4] The central tenets of the common pathway theory state that CKD progression occurs, in general, through focal nephron loss. and that the adaptive responses of surviving nephrons, although initially serving to increase single-nephron GFR and offset the overall loss in clearance, ultimately prove detrimental to the kidney. Over time, glomerulosclerosis and tubular atrophy further reduce nephron number, fueling a self-perpetuating cycle of nephron destruction, culminating in uremia. Although recent clinical studies have reported that the progression of CKD is variable and nonlinear in some persons,[5] these common pathway mechanisms remain relevant.

In this chapter, the functional and structural adaptations observed in remaining nephrons following substantial reductions in functioning renal mass and the mechanisms thought to be responsible for them are described in detail. How these changes may, in time, prove maladaptive and contribute to the progressive renal injury described earlier are then considered. Given the growing worldwide burden of CKD that causes substantial morbidity and mortality in individuals and threatens to overburden health care systems, it could be argued that the further elucidation of the mechanisms of CKD progression resulting in more effective interventions to slow its advance should remain among the highest priorities for nephrologists and health care systems today.

STRUCTURAL AND FUNCTIONAL ADAPTATION OF THE KIDNEY TO NEPHRON LOSS

ALTERATIONS IN GLOMERULAR PHYSIOLOGY

Glomerular hemodynamic responses to nephron loss have been studied largely in animals subjected to surgical ablation of renal mass. It was recognized several decades ago that unilateral nephrectomy in rats resulted in a rapid increase in function of the remaining kidney, detectable 3 days after nephrectomy, such that the GFR achieved a maximum of 70% to 85% of the previous two-kidney value after 2 to 3 weeks. More recent observations in conscious rats have reported a maximal increase of approximately 50% in the GFR of a single kidney at 8 days after uninephrectomy and a 300% increase in the GFR of the remnant kidney at 16 days after 5/6 nephrectomy (Fig. 51.1).[6] Because no new nephrons are formed in mature rodents, the observed rise in the GFR represents an increase in the filtration rate of remaining nephrons.

A detailed study of glomerular hemodynamics was facilitated by the identification of a rat strain, Munich-Wistar, which is unique in regularly bearing glomeruli on the kidney surface. This allowed micropuncture of the glomerulus and direct measurement of intraglomerular pressures, as well as sampling of blood from afferent and efferent arterioles. These techniques made possible the study of mechanisms underlying the compensatory rise in the GFR after renal mass ablation. Increases in whole-kidney GFR at 2 to 4 weeks after unilateral nephrectomy were attributable to an increase in the single-nephron GFR (SNGFR) averaging 83%, achieved in large part by a rise in the glomerular plasma flow rate (Q_A), which, in turn, resulted from dilation of afferent and, to a lesser extent, efferent arterioles. Although the systemic blood

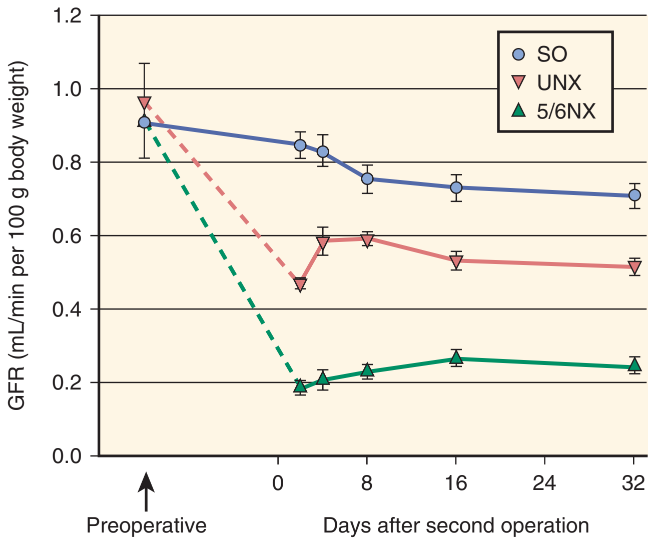

Fig. 51.1 Glomerular filtration rate *(GFR)* in conscious rats before and after sham operations *(SO)*, uninephrectomy *(UNX)*, or 5/6 nephrectomy *(5/6NX)*. Values are mean ± standard error of the mean. (From Chamberlain RM, Shirley DG: Time course of the renal functional response to partial nephrectomy: measurements in conscious rats. *Exp Physiol.* 2007;92:251–262.)

pressure (BP) was not elevated, glomerular capillary hydraulic pressure (P_{GC}) and the glomerular transcapillary pressure difference (ΔP) were increased significantly postuninephrectomy, accounting for an estimated 25% of the rise in SNGFR.[7] The glomerular ultrafiltration coefficient, K_f (the product of glomerular hydraulic permeability and surface area available for filtration), was unaltered at this stage but may become elevated later.[8]

With more extensive nephron loss, even greater compensatory increases in SNGFR were observed. In Munich-Wistar rats studied 7 days after unilateral nephrectomy and infarction of 5/6 of the contralateral kidney, SNGFR in the remnant was more than double that of two-kidney controls. This increment was again attributable to large increases in Q_A and a substantial rise in P_{GC}. Efferent and afferent arteriolar resistances were reduced but the decrease in afferent arteriolar resistance was again proportionately greater, accounting for the observed rise in P_{GC}.[9] Comparison of renal infarction versus surgical excision models of 5/6 nephrectomy subsequently found that changes in arteriolar resistance were similar, but that P_{GC} was significantly more elevated in the infarction model, indicating that glomerular transmission of elevated systemic BP (absent in the surgical excision model) also contributes to the increase in P_{GC}.[10] Changes in K_f after extensive renal mass ablation appeared to be time-dependent, with a decrease reported at 2 weeks after surgery[11] and an increase at 4 weeks.[12] Further studies have indicated that glomerular hemodynamic responses to nephron loss seem to be similar between the superficial cortical and juxtamedullary nephrons.[13] The rise in SNGFR associated with renal mass ablation is often referred to as *glomerular hyperfiltration* and the elevated P_{GC} is termed *glomerular hypertension*. Together these terms encompass the central concepts underlying the hemodynamic adaptations in the remnant kidney.

Glomerular hemodynamic adaptations to nephron loss may show interspecies variation. In dogs, increases in SNGFR observed 4 weeks after 3/4 or 7/8 nephrectomy were attributable largely to increases in Q_A and K_f. In contrast to the findings in rodents, ΔP in dogs was only modestly elevated, although after ablation of 7/8 of their renal mass, P_{GC} did increase significantly, independent of arterial pressure, again as a result of relatively greater relaxation of afferent versus efferent arterioles.[14]

In humans, the effects of nephron loss on the physiology of the remnant kidney have been studied mainly in healthy individuals undergoing donor nephrectomy for kidney transplantation. Inulin clearance studies of the earliest kidney donors have revealed that the total GFR in the donor's remaining kidney had increased to 65% to 70% of the previous two-kidney value by 1 week postnephrectomy. A metaanalysis of data from 48 studies that included 2988 living kidney donors has estimated that the GFR decreased, on average, by only 17 mL/min/1.73 m^2 after uninephrectomy.[15] These observations imply that the single-kidney GFR (and therefore also the average SNGFR) increases by 30% to 40% after uninephrectomy in humans. There is currently no method for measuring SNGFR or P_{GC} in humans, but detailed studies in 21 healthy kidney donors have reported that the observed 40% increase in single-kidney GFR could be accounted for by the observed increase in renal plasma flow and a rise in K_f resulting from glomerular hypertrophy without the need for an increase in P_{GC}.[16]

MEDIATORS OF THE GLOMERULAR HEMODYNAMIC RESPONSES TO NEPHRON LOSS

The specific factors that are sensed after renal mass ablation and serve as signals to initiate the adjustments in glomerular hemodynamics responsible for the increase in remnant kidney GFR remain to be identified. However, the effector mechanisms have been studied extensively; the hemodynamic changes can be attributed to the net effects of complex interactions of several factors, each having specific and sometimes opposing actions on the various determinants of glomerular ultrafiltration. Several vasoactive substances, including angiotensin II (Ang II), aldosterone, natriuretic peptides (NPs), endothelins (ETs) eicosanoids, and bradykinin, have been implicated. Moreover, sustained increases in the SNGFR also require resetting of the autoregulatory mechanisms that normally govern the GFR and renal plasma flow (RPF). For a detailed discussion of vasoactive peptides in the kidney, see Chapter 11.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Ang II appears to play a critical role in the development of glomerular capillary hypertension following renal mass ablation and may also contribute to changes in K_f. The acute infusion of Ang II in normal rats results in a rise in P_{GC} due to a greater increase in efferent than afferent resistance and reductions in Q_A and K_f.[17,18] Chronic administration of Ang II for 8 weeks resulted in systemic hypertension, lowered single-kidney GFR and, with the exception of K_f, elicited similar glomerular hemodynamic changes to those observed after acute infusion in both normal and uninephrectomized rats.[8] The importance of the influence of endogenous Ang II on glomerular hemodynamics in remnant kidneys has been revealed by studies with pharmacologic inhibitors of the renin-angiotensin aldosterone system (RAAS). Chronic treatment of 5/6 nephrectomized rats with either an angiotensin-converting enzyme (ACE) inhibitor [19,20] or Ang II (subtype 1) receptor blocker (ARB)[21,22] results in normalization of P_{GC} through a reduction in systemic BP and dilation of both afferent and efferent arterioles. SNGFR, however, remains elevated due to an increase in K_f. Furthermore, acute infusion of an ACE inhibitor or saralasin, a peptide analogue receptor antagonist of Ang II, was found to normalize P_{GC} in 5/6 nephrectomized rats through efferent arteriolar dilation, without affecting the mean arterial pressure (MAP).[11,23] It is unclear why these findings could not be confirmed with the ARB losartan.[24]

These effects of RAAS inhibition imply that there is increased local activity of endogenous Ang II, yet plasma renin levels show only a transient increase following 5/6 nephrectomy.[10,25] This suggests differential regulation of the systemic versus intrarenal RAAS and that Ang II is formed locally. Detailed studies have identified that all components of the RAAS are expressed in the kidney.[26] Renin mRNA and protein levels are both increased in glomeruli adjacent to the infarction scar in 5/6 nephrectomized rats.[27–29] Furthermore renal renin mRNA levels are increased at day 3 and 7 after renal mass ablation by infarction but not when a renal mass is excised surgically, suggesting that renal infarction activates the RAAS by creating a margin of ischemic tissue around the organizing infarct and explaining the greater severity of hypertension and glomerulosclerosis associated with the infarction model.[10]

Detailed studies of intrarenal Ang II levels following 5/6 nephrectomy achieved by infarction have confirmed these findings by showing higher Ang II levels in the periinfarct portion of the kidney than the intact portion at all time points.[25] On the other hand, the studies also showed that the rise in intrarenal Ang II levels following 5/6 nephrectomy was transient. Whereas Ang II levels in the periinfarct portion were elevated compared with sham-operated controls at 2 weeks after surgery, they were not statistically different at 5 or 7 weeks. In the intact portion of the remnant kidney, Ang II levels were similar to controls at 2 and 5 weeks and were lower at 7 weeks.[25] Sustained increases in intrarenal Ang II levels are therefore not required to maintain the hypertension and progressive renal injury characteristic of this model. Nevertheless, subsequent studies have shown that the renoprotective effects of ACE inhibitor and ARB treatment are associated with a reduction in intrarenal Ang II levels in both the peri-infarct and intact portions of the remnant kidney.[30] In contrast, treatment with the dihydropyridine calcium antagonist, nifedipine, did not reduce proteinuria, despite lowering BP to the same levels as the RAAS antagonists and was associated with an increase in intrarenal Ang II.[30] Thus intrarenal Ang II appears to play a central role in the pathogenesis of hypertension and renal injury in this model, even in the absence of sustained increases in Ang II levels. Further research is required to explain these findings fully. It could be argued that apparently normal intrarenal Ang II levels are inappropriately high in the context of the hypertension and ECF volume expansion seen in these animals or that the total intrarenal Ang II levels measured may have failed to detect important local elevations of Ang II. Ongoing research has elucidated several novel aspects of the intrarenal RAAS, including regulation by several other factors, including prorenin receptor, prostaglandin E2, and Wnt/β-catenin (positive regulators), as well as Klotho, vitamin D receptor, and liver X receptor (negative regulators).[31]

Attention has focused on the potential role of aldosterone in progressive renal injury. In addition to evidence that aldosterone may exert profibrotic effects in the kidney (see later), these observations have suggested that it may also have important glomerular hemodynamic effects. Previous observations that the deoxycorticosterone (DOCA)–salt model of hypertension is associated with glomerular capillary hypertension prompted detailed studies of microperfused rabbit afferent and efferent arterioles; these found dose-dependent constriction of both arterioles in response to nanomolar concentrations of aldosterone, with greater sensitivity observed in efferent arterioles.[32] These effects were not inhibited by spironolactone and were still present with albumin-bound aldosterone, indicating that they may be mediated by specific membrane receptors rather than the intracellular receptors responsible for most of the actions of aldosterone. Interestingly aldosterone may also counteract rabbit afferent arteriolar vasoconstriction via a nitric oxide (NO)–dependent pathway, an action that would also be expected to increase P_{GC}.[33,34]

ENDOTHELINS

ETs are potent vasoconstrictor peptides that act via at least two receptor subtypes, ET_A and ET_B. ET receptors have been identified throughout the body and are most abundant in the lungs and kidneys. ET_A receptors are primarily located on vascular smooth muscle cells and mediate vasoconstriction, as well as cellular proliferation. ET_B receptors are expressed on vascular endothelial and renal epithelial cells and appear to pay a role as clearance receptors, as well as mediating endothelium-dependent vasodilation via NO.[35–37] Renal production of ETs is increased after 5/6 nephrectomy, raising the possibility that they may also contribute to the observed glomerular hemodynamic adaptations.[38,39] Acute and chronic infusion of ET elicits dose-dependent reductions in RPF and GFR in normal rats.[40–43]

Observations regarding the relative effects of ETs on afferent and efferent arterioles are to some extent contradictory, possibly reflecting different experimental conditions. Despite some differences, most studies in intact animals have reported greater increases in efferent than afferent arteriolar resistance, resulting in an increase in P_{GC}. The ultrafiltration coefficient (K_f) was significantly reduced, and thus the SNGFR was unchanged or decreased.[44–47] On the other hand, observations in microperfused arterioles have found that ETs cause greater constriction of the afferent than efferent arteriole. Studies with selective ET_A and ET_B receptor antagonists and ET_B receptor knockout mice were somewhat contradictory, suggesting a complex interaction between ET_A and ET_B receptors in determining the response.[48,49] Comparative studies in rat juxtamedullary nephron preparations have shown that ET is a more potent vasoconstrictor of both afferent and efferent arterioles than Ang II, arginine vasopressin, norepinephrine, sphingosine-1-phosphate, and adenosine triphosphate (ATP).[50] In short-term studies of human subjects with CKD, ET receptor antagonists increased renal blood flow but had little effect on the GFR due to a decrease in filtration fraction, observations consistent with a greater vasodilatory effect on the efferent arteriole.[51,52] Interestingly these observations were made in subjects already receiving treatment with an ACE inhibitor or ARB. The potential interaction between ETs and other vasoactive molecules is further illustrated by observations that chronic infusion of Ang II results in increased production of ET[53] and that ET-1 transgenic mice are not hypertensive but show induction of inducible nitric oxide synthase (iNOS), resulting in increased NO production as a probable counterregulatory mechanism to maintain normal BP.[54] Furthermore some of the glomerular hemodynamic effects of ETs appear to be modulated by prostaglandins.[47] Detailed micropuncture studies to elucidate the role of endothelins in remnant kidney hemodynamics have not yet been published. These studies should be facilitated by the ongoing development of specific ET_A and ET_B receptor antagonists.

NATRIURETIC PEPTIDES

Atrial natriuretic peptide (ANP) and other structurally related NPs mediate, in large part, the functional adaptations in tubular sodium reabsorption that maintain sodium excretion in 5/6 nephrectomized rats[55] but also exert important hemodynamic effects. Circulating ANP levels are elevated in 5/6 nephrectomized rats, and acute administration of an NP antagonist elicited profound decreases in the GFR and RPF in 5/6 nephrectomized rats on a high-salt (but not low-salt) diet, indicating that NP play an important role in the observed hemodynamic responses to 5/6 nephrectomy.[56]

In another study, brain natriuretic peptide (BNP) levels were found to be elevated after 3/4 nephrectomy in the absence of cardiac dysfunction or upregulation of myocardial BNP gene expression.[57] Further insights into the renal hemodynamic effects of NP have been gained from observations in normal rats infused with a synthetic ANP. Whole-kidney and single-nephron GFR increased by approximately 20% due entirely to a rise in P_{GC}, resulting from significant afferent arteriolar dilation and efferent arteriolar constriction.[58] In the previous experiments, some residual elevation in remnant kidney GFR appeared to persist, even after the NP system was suppressed by sodium restriction or a NP receptor antagonist, suggesting that factors other than NP contribute to glomerular hyperfiltration following renal mass ablation. The potential interaction between NP and other vasoactive molecules is illustrated by the observation that ANP infusion in normal rats induces an increase in renal nitric oxide synthase activity.[59]

EICOSANOIDS

Eicosanoids, another family of potent vasoactive molecules present in abundance in the kidney, may also play a role in mediating glomerular hyperfiltration. Urinary excretion per nephron of both vasodilator and vasoconstrictor prostaglandins (PGs) is increased in rats and rabbits after renal mass ablation.[60–62] Infusion of PGE2, PGI2, or 6-keto-PGE1 into the renal artery elicits significant renal vasodilation.[63] Whereas acute inhibition of PG synthesis by infusion of the cyclooxygenase (COX) inhibitor indomethacin has no effect on GFR or glomerular hemodynamics in normal rats, indomethacin lowers both the SNGFR and Q_A after 3/4 or 5/6 nephrectomy.[60,61] On the other hand, chronic treatment with a selective COX-2 inhibitor attenuates the systemic and glomerular hypertension observed in 5/6 nephrectomized rats but has no effect on the GFR.[64]

The relative effects of PG synthesis inhibitors on afferent and efferent arterioles may vary with time postnephrectomy. Afferent arteriolar constriction was the predominant finding reported at 24 hours postsurgery, whereas constriction of both afferent and efferent arterioles was observed at 3 to 4 weeks.[60,61] Some contribution of thromboxanes to glomerular hemodynamic adjustments after 5/6 nephrectomy in rats is suggested by the increase in the GFR seen after acute infusion of a selective thromboxane synthesis inhibitor.[62] Thus different eicosanoids appear to exert opposite effects, but the general impression is that the combined effects of vasodilator PGs outweigh those of the vasoconstrictors. This interaction is illustrated by the observation that perfusion of isolated glomeruli with bradykinin resulted in vasodilation of the efferent arteriole that was completely blocked by an indomethacin but that this blockade was reversed by a specific antagonist of 20-hydroxyeicosatetraenoic acid (20-HETE), a vasoconstrictor eicosanoid, indicating that the glomerulus produces both vasodilator and vasoconstrictor eicosanoids.[65]

NITRIC OXIDE

The extremely short half-life of NO precludes direct measurement of NO levels or the administration of exogenous NO in experimental models. The actions of NO have thus been inferred from experiments with inhibitors of nitric oxide synthase (NOS). Intravenous infusion of NOS inhibitors results in systemic and renal vasoconstriction, as well as a reduction in the GFR, in normal rats.[66,67] Thus NO appears to exert a tonic effect on the physiologic maintenance of systemic BP and renal perfusion under resting conditions. It is unclear, however, whether NO plays a specific role in the adaptive hemodynamic changes that follow renal mass ablation. Indeed, renal expression of NOS and renal NO generation are reduced in 5/6 nephrectomized rats, whereas systemic production of NO is increased.[68,69] Mean arterial pressure and renal vascular resistance increase, whereas renal blood flow (RBF) and the GFR decrease to a similar extent after acute infusion of an endothelial NOS (eNOS) inhibitor, NG-monomethyl-L-arginine (L-NMMA), irrespective of whether given to normal rats or 3 to 4 weeks after unilateral or 5/6 nephrectomy.[67] Chronic NOS inhibition with nitro-L-arginine methyl ester (L-NAME) produces elevations in systemic BP and P_{GC} in 5/6 nephrectomized rats without affecting the GFR,[70] whereas chronic treatment with aminoguanidine, an inhibitor of inducible NOS, has no effect on the GFR, RPF, or P_{GC}.[69]

Similarly, greater increases in BP and proteinuria were observed after 5/6 nephrectomy in eNOS knockout versus wild-type mice.[71] On the other hand, renal NOS expression and activity are increased early after unilateral nephrectomy, and pretreatment of rats with a subpressor dose of L-NAME prevents the early increase in RBF and decrease in renal vascular resistance usually observed after unilateral nephrectomy.[72,73] It therefore appears that NO plays a role in early hemodynamic adaptations to nephron loss, resulting in an increase in RBF, but in the longer term, NO retains a tonic influence on systemic and renal hemodynamics without being a specific determinant of the adaptive changes in glomerular hemodynamics.

BRADYKININ

Bradykinin is a potent vasodilatory peptide that is elevated in the remnant kidney[25] and may therefore contribute to hemodynamic adaptations after nephron loss. Acute and chronic infusion of bradykinin result in increased RPF but have no effect on the GFR.[74,75] Micropuncture studies in intact animals are lacking, but studies of isolated perfused afferent arterioles have shown that bradykinin induces a biphasic response with vasodilation at low concentrations and vasoconstriction at higher concentrations. Both effects appear to be mediated by products of COX.[76] Similar experiments with efferent arterioles have found dose-dependent vasodilation (no biphasic response) that was dependent on cytochrome P450 metabolites but independent of COX products or NO.[77] When glomeruli were perfused with bradykinin, vasodilation of efferent arterioles was again observed but was inhibited by a COX inhibitor, indicating that bradykinin induces glomerular production of COX metabolites (PGs) that also contribute to efferent arteriolar dilation.[65] Further studies are required to elucidate the role of bradykinin after nephron loss.

UROTENSIN II

Urotensin II (UII) is the most potent vasoconstrictor identified to date, but its actions appear to vary in different vascular territories and, in some vessels, it may even produce vasodilation. Infusion of exogenous UII has been reported to increase or decrease the GFR in normal rodents in different experiments (see Chapter 11). The vasoconstrictor effects of UII may be mediated, at least in part, by upregulation of renal renin and aldosterone synthase gene expression.[78] UII

Table 51.1 Hemodynamic Effects of Vasoactive Molecules Mediating Glomerular Hemodynamic Adaptations After Partial Renal Mass Ablation

Parameter	R_A	R_E	P_{GC}	Q_A	K_f	SNGFR	RPF	GFR
Angiotensin II	↑	↑↑	↑	↓	↓↔	↓↔	↓	↔
Aldosterone	↑	↑↑	↑	?	?	?	?	?
Endothelins	↑↔	↑	↑↔	↓	↓↔	↓↔	↓	↓↔
Natriuretic peptides	↓	↑ (?)	↑	↔	↔	↑	↑↔	↑
Prostaglandins	↓	↓	↔	↑	↑	↑	↑	↑
Bradykinin	↓↑	↓	?	?	?	?	↑	↔
Observed changes after partial renal ablation	↓↓	↓	↑	↑	↑↓	↑	–	↓

GFR, Glomerular filtration rate; *K_f,* glomerular ultrafiltration coefficient; *P_{GC},* glomerular capillary hydraulic pressure; *Q_A,* glomerular plasma flow rate; *R_A,* afferent arteriolar resistance; *R_E,* efferent arteriolar resistance; *RPF,* renal plasma flow; *SNGFR,* single-nephron GFR.

is produced in the kidney, and urotensin receptors have been localized to glomerular arterioles.[79] Increased renal expression of mRNA for urotensin-related protein (which also binds to the urotensin receptor) and urotensin receptor has been reported in rats after 5/6 nephrectomy,[80] but the potential role of UII in the hemodynamic adaptations that follow nephron loss has yet to be investigated.

ADJUSTMENTS IN RENAL AUTOREGULATORY MECHANISMS

After extensive renal mass ablation, there is a marked readjustment of the autoregulatory mechanisms that control RPF and the GFR.[81–83] The role of myogenic mechanisms is uncertain, but detailed studies of afferent arteriolar myogenic responses have suggested that their primary role is to protect the glomerulus from elevations in SBP.[84] This notion has been supported by animal experiments that have reported that rat strains with impaired myogenic response show greater transmission of perfusion pressure to glomerular capillaries and are more susceptible to glomerular injury than strains with an intact myogenic response.[85] The tubuloglomerular feedback system is reset after renal mass ablation to permit and sustain the elevations in SNGFR and P_{GC} described previously.[86,87] Studies have indicated that connecting tubule glomerular feedback mediated by the epithelial sodium channel (ENaC) plays a key role in this process after uninephrectomy.[88] Resetting appears to occur as early as 20 minutes after unilateral nephrectomy,[89] in proportion to the extent of renal ablation. The adjustments observed after uninephrectomy are of lesser magnitudes than those seen after 5/6 nephrectomy.[86]

INTERACTION OF MULTIPLE FACTORS

As is readily appreciated from the previous discussion, the adjustments in glomerular hemodynamics seen after renal mass ablation represent the net effect of several endogenous vasoactive factors. Natriuretic peptides and vasodilator PGs dilate the preglomerular vessels, whereas bradykinin dilates both afferent and efferent arterioles. On the other hand, Ang II, vasoconstrictor PGs, and possibly ETs constrict both afferent and efferent arterioles, with a greater effect on the latter. A net fall in preglomerular vascular resistance is observed, whereas efferent arteriolar resistance decreases to a lesser extent. Together with greater transmission of the raised systemic BP to the glomerular capillary network, these alterations in microvascular resistances result in the observed elevations in Q_A, P_{GC}, ΔP, and SNGFR (Table 51.1). The importance of multiple vasoactive factors is illustrated by the observation that treatment of 5/6 nephrectomized rats with omapatrilat, an inhibitor of both ACE and neutral endopeptidase that results in reduced Ang II production, as well as increased NP and bradykinin levels, lowers P_{GC} more than ACE inhibition alone.[90]

The complexity of factors involved is further illustrated by observations that other molecules involved in the modulation of progressive renal injury may exert hemodynamic effects by influencing the mediators discussed previously. Acute infusion of hepatocyte growth factor (HGF) has been shown to induce a decline in BP and GFR, an effect that is mediated by a short-term increase in ET-1 production.[91] In isolated perfused preparations, platelet-activating factor (PAF) at picomolar concentrations has been shown to induce glomerular production of NO, resulting in dilation of preconstricted efferent arterioles, whereas at nanomolar concentrations, PAF constricts efferent arterioles through local release of COX metabolites.[92] The potential role of other recently identified vasoactive molecules such as UII remains to be elucidated.

RENAL HYPERTROPHIC RESPONSES TO NEPHRON LOSS

The notion that a single kidney enlarges to compensate for the loss of its partner has been entertained since antiquity. Aristotle (384–322 BC) noted that a single kidney was able to sustain life in animals, and that such kidneys were enlarged. In preparation for the first human nephrectomy in 1869, a German surgeon, Gustav Simon, uninephrectomized dogs and noted a 1.5-fold increase in the size of the remaining kidney at 20 days.[93] Compensatory renal hypertrophy has been studied in a variety of species, including toads, mice, rats, guinea pigs, rabbits, cats, dogs, pigs, and baboons. Most experimental work has been conducted in rodents subjected to uninephrectomy, but hypertrophic responses have also been studied in response to unilateral ureteric obstruction or after nephrotoxin administration.[94]

WHOLE-KIDNEY HYPERTROPHIC RESPONSES

Among the earliest responses to unilateral nephrectomy are biochemical changes that precede cell growth. Increased incorporation of choline, a precursor of cell membrane phospholipid, has been detected as early as 5 minutes,[95,96] and increased choline kinase activity has been observed at 2 hours after nephrectomy.[97] Activity of ornithine decarboxylase, the enzyme catalyzing the first step of polyamine synthesis, is elevated at 45 to 120 minutes, and polyamine levels peak at 1 to 2 days postnephrectomy.[98–100] Early alterations in mRNA metabolism have also been observed. Although there are no changes in the half-life or cytoplasmic distribution of mRNA, a near-25% increase in the fraction of newly synthesized poly(A)-deficient mRNA occurs within 1 hour of uninephrectomy, and total RNA synthesis in the kidney increases by 25% to 100% relative to that in the liver.[101–103] Ribosomal RNA synthesis is increased by 40% to 50% at 6 hours.[104] The rate of protein synthesis is increased at 2 hours and is nearly doubled at 3 hours.[105] Data on cyclic nucleotide levels, which are thought to affect cell growth and proliferation, are conflicting. Some studies have reported elevated levels of cyclic guanosine monophosphate (cGMP) in the remaining kidney as early as 10 minutes after surgery,[106–108] whereas others have found no consistent changes in cyclic adenosine monophosphate (cAMP) or cGMP levels.[94,109] Genome-wide analysis of gene expression using cDNA microarrays in remaining rat kidneys up to 72 hours after uninephrectomy has revealed the dominant response to be suppression of the genes responsible for inhibition of growth and apoptosis.[110]

Early biochemical changes are followed by a period of rapid growth. DNA synthesis is increased at 24 hours, and increased numbers of mitotic figures are evident at 28 to 36 hours. Both reach a maximum five- to tenfold increase at 40 to 72 hours.[111–114] In rats, kidney weight is increased at 48 to 72 hours after uninephrectomy and increases by 30% to 40% at 2 to 3 weeks (Fig. 51.2).[73,94] The nephron number is fixed shortly before birth in most species, so this gain in kidney weight is attributable to increased nephron size. Growth is thought to occur largely through cell hypertrophy, accounting for 80% of the increase in renal mass seen in adult rats and, to a lesser extent, through hyperplasia.[105] Renal mass continues to rise for 1 to 2 months until a 40% to 50% increase is achieved. The degree of compensatory growth is a function of the extent of renal ablation. Uninephrectomy has been shown to provoke an 81% increase of residual renal mass at 4 weeks compared with an increase of 168% after 70% renal ablation. Normal controls gained 31% in kidney weight over the same period.[115]

Age diminishes renal hypertrophic responses. After uninephrectomy, greater increases in kidney weight and more extensive hyperplasia were observed in 5- versus 55-day-old rats,[116] and aging rats exhibited gains in kidney weight of only one-third to three-quarters of those seen in younger controls.[94,117–119]

In humans, assessment of renal hypertrophy after nephrectomy is dependent on radiologic studies. Ultrasound studies have reported increases of 19% to 100% in kidney volume,[120] and in computed tomography (CT) studies have found an increase of 30% to 53% in renal cross-sectional area after nephrectomy.[121,122] Contrast-enhanced CT has been used to measure renal parenchymal volume post–unilateral nephrectomy. One study reported increases of 12.1% and 8.9% at 1 week and 6 months, respectively.[123] The degree of hypertrophy correlated positively with the function of the kidney removed

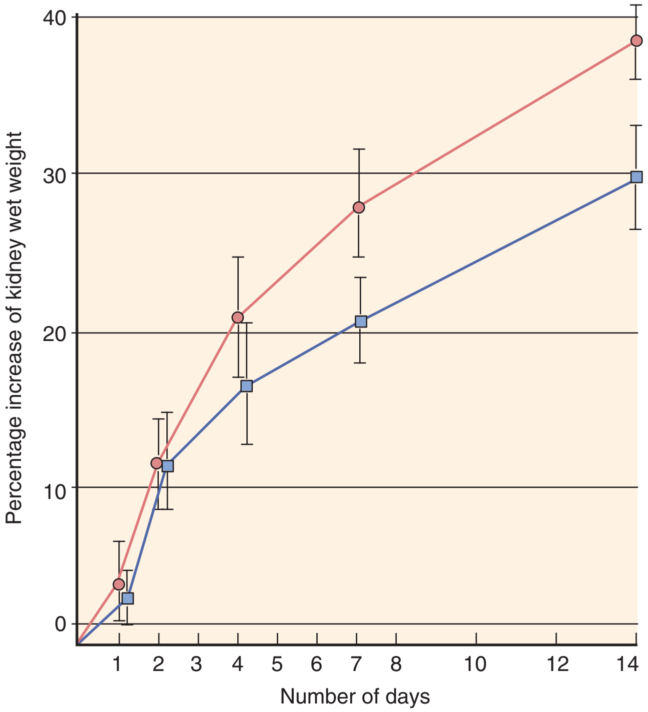

Fig. 51.2 Rate of compensatory renal growth after unilateral nephrectomy *(circles)* and ureter ligation *(squares)*. (From Dicker SE, Shirley DG. Compensatory hypertrophy of the contralateral kidney after unilateral ureteral ligation. *J Physiol (Lond).* 1972;220:199–210.)

and negatively with patient age. One relatively large study of living kidney donors observed a 27.6% ± 9.7% increase in the remaining kidney volume at 6 months post–donor nephrectomy,[124] and a recent detailed study has reported an increase in renal cortical volume of 27% at a median of 0.8 years postnephrectomy that increased to 35% after a median of 6.1 years.[16] Other studies using CT scans have reported increases in renal volume of 16.5 to 18.5% at 1 year after nephrectomy for renal cell carcinoma[125] and 22.4% ± 23.2% at 6 to 12 months after donor nephrectomy.[126] In a detailed study that used magnetic resonance imaging (MRI) to measure renal volume, an increase was observed from 198 ± 87 mL preoperatively to 329 ± 175 mL at 3 months postnephrectomy for renal cell carcinoma. The change in renal volume correlated positively with changes in renal blood flow at 1 week postnephrectomy ($r = 0.6$).[127] Differences among these studies are likely attributable to the relatively small number of subjects enrolled, wide variation in the time intervals between nephrectomy and assessment of renal size, and differing indications for nephrectomy.

GLOMERULAR ENLARGEMENT

The principal morphometric change observed in glomeruli after uninephrectomy is an increase in volume. Glomerular enlargement appears to parallel whole-kidney growth and has been detected as early as 4 days after surgery.[128] The degree of enlargement of superficial and juxtamedullary glomeruli is similar. Proportionally similar increases in number and size of all cell types occur, with preservation of the relative volumes of different glomerular cells.[94] There is consensus that glomerular capillaries increase in length and number (i.e., more branching), but most studies have shown that diameter or cross-sectional surface area of the glomerular capillaries remains constant or increases only minimally.[129,130] Transplantation of hypertrophied kidneys into uninephrectomized rats has demonstrated regression of glomerular hypertrophy within 3 weeks, yet the increase in capillary length was maintained.[130]

Glomerular hypertrophy, as evidenced by elevated RNA/DNA and protein/DNA ratios, as well as by increased glomerular volume (V_G) on electron microscopy, has been detected at 2 days after 5/6 nephrectomy in rats.[131] The initial increase in V_G was due almost entirely to increases in visceral epithelial cell volume, whereas at 14 days the increase in V_G was largely accounted for by mesangial matrix expansion. Although several studies have reported glomerular capillary lengthening after 5/6 nephrectomy, few have detected any increase in cross-sectional area or diameter of the glomerular capillaries.[132–135] These observations should, however, be considered in the light of important technical considerations. In vitro perfusion of isolated glomeruli has demonstrated that V_G increases as perfusion pressure is raised through physiologic and pathophysiologic ranges. Moreover, glomerular capillary "compliance" in these studies was a function of the baseline V_G, and glomeruli obtained from remnant kidneys post–5/6 nephrectomy had a higher compliance than those from control animals.[136] These findings have two important implications. First, although glomerular pressures are only minimally elevated after uninephrectomy, the glomerular capillary hypertension associated with more extensive renal ablation is likely to contribute significantly to the increase in V_G. Second, estimates of V_G in tissues that have not been perfusion-fixed at the appropriate BP should be interpreted with caution. Direct comparison of V_G in perfusion-fixed versus immersion-fixed kidney from the same rats yielded estimates of V_G in immersion-fixed samples that were 61% lower than those from perfusion-fixed kidneys.[137]

MECHANISMS OF RENAL HYPERTROPHY

Despite more than a century of research that identified a large number of mediators or modulators of renal hypertrophy, the identities of the specific factors that regulate hypertrophy and the stimuli to which these factors respond remained elusive until relatively recently. Renal innervation does not appear to play a role because kidneys transplanted into bilaterally nephrectomized rats exhibit the same degree of hypertrophy after 3 weeks as kidneys remaining after uninephrectomy.[138] The absence of any reduction in renal hypertrophy when rats are treated with an ACE inhibitor after uninephrectomy indicates that the renin-angiotensin system also does not play a major role.[139] Several hypotheses were advanced to account for the observed changes associated with renal hypertrophy and have been discussed in detail in other publications[93,94] but are summarized later.

SOLUTE LOAD

The notion that hypertrophy after uninephrectomy is stimulated by the need for the remaining kidney to excrete larger amounts of metabolic waste products, necessitating more excretory "work," was proposed by Sacerdotti in 1896. Subsequently, it became apparent that urea excretion is largely a function of glomerular filtration, whereas the main energy-requiring function of the renal tubules is reabsorption of filtered electrolytes (principally sodium), and water. The hypothesis was therefore modified to view hypertrophy as a response to the increased demand for water and solute reclamation imposed by an increased SNGFR (solute load hypothesis). Several lines of evidence have supported the concepts underlying the solute load hypothesis. After uninephrectomy, RBF increased by 8% in the remaining kidney and preceded hypertrophy, and treatment with a subpressor dose of the NOS inhibitor L-NAME prevented the rise in RBF and substantially attenuated increases in renal weight, as well as glomerular and proximal tubule areas, at 7 days postnephrectomy.[73] In the remnant kidney, proximal tubule sodium absorption increased in parallel with GFR (glomerulotubular balance),[140] and tubules continued to display enhanced fluid reabsorption in vitro, implying that the adaptive changes were intrinsic to the tubular epithelial cells (TECs).[141,142] In chronic glomerulonephritis, a lesion characterized by marked heterogeneity in the SNGFR, there was preservation of the SNGFR to proximal fluid reabsorption ratio and a close correlation between glomerular and proximal tubule hypertrophy. Moreover, sustained increases in the GFR in the absence of renal mass ablation result in renal hypertrophy in some conditions, including pregnancy (in some but not all studies) and diabetes mellitus.[93]

On the other hand, experimental maneuvers dissociating renal solute load from hypertrophy appear to contradict the solute load hypothesis. Total diversion of urine from one kidney into the peritoneum by ureteroperitoneostomy was associated with an increase in the GFR in the contralateral kidney of similar magnitude to that seen after uninephrectomy, but no increase in renal mass or mitotic activity.[143] In another example, potassium depletion resulted in renal hypertrophy without any increase in GFR.[144] Moreover, the findings that some of the early biochemical changes associated with hypertrophy precede increases in glomerular filtration or sodium reabsorption argue against a causal association of hypertrophy and increased solute load. Despite these conflicting data, there is nevertheless considerable evidence of an association between the GFR and proximal tubule hypertrophy that may play a role in stimulating renal growth in the remnant kidney.[93]

RENOTROPIC FACTORS

Failure of the solute load hypothesis to explain all the experimental data has led others to propose instead that the primary stimulus for renal hypertrophy is a change in renal mass, and that renal growth is under the control of specific growth and/or inhibitory factors. Evidence in support of this theory was derived from three types of experiments.[145]

In the first, a stable connection was established between the extracellular space and microcirculation of two animals (parabiosis), and the effects of renal mass ablation in one animal were assessed in the intact kidneys of its partner.[93] Despite some inconsistencies due to variations in methodology, these experiments generally found that uninephrectomy in one animal resulted in hypertrophy of the contralateral kidney and, to a lesser extent, of both kidneys of the parabiotic partner. Bilateral nephrectomy in one partner or a triple nephrectomy produced incremental degrees of hypertrophy in the remaining kidney(s). Furthermore, the hypertrophy was rapidly reversed following cessation of cross circulation.

A second strategy was to inject serum or plasma from uninephrectomized animals into intact subjects and then assess renal hypertrophy by radiolabeled thymidine uptake or mitotic count.[93] Although studies using single small intraperitoneal or subcutaneous doses were negative, the administration of repeated large doses by an intraperitoneal or intravenous route elicited renal hypertrophy in most subjects studied.

The data that most consistently supported the existence of a renotropic factor were derived from in vitro experiments in which renal tissues were incubated in the presence or absence of plasma or serum from rats subjected to renal mass ablation. Evidence for hypertrophy was generally assessed by the incorporation of radiolabeled thymidine or uridine into DNA or RNA, respectively. In general, these experiments showed increased uptake of radiolabeled nucleotides after incubation with serum from uninephrectomized animals. This effect appeared to be organ- but not species-specific. That a tissue factor produced by the kidneys and upregulated after nephrectomy may be required for the activity of a circulating renotropin was suggested by experiments in which kidney extract from rats taken 20 hours after uninephrectomy, in the presence of normal rat serum, was found to stimulate ^{3}H-thymidine incorporation in normal renal cortex. However, addition of the same extract in the absence of the serum tended to depress ^{3}H-thymidine uptake.[146] Serum taken from bilaterally nephrectomized animals lacked renotropic effects, but these were restored after dialysis of the serum, suggesting the presence of renotropin inhibitory factors that accumulated in the absence of renal function.[146] Although the specific identity of renotropin has remained elusive, several lines of evidence suggested that it was a small protein. Retention of activity after ultrafiltration, dialysis, and removal of albumin from serum implied that renotropin was a molecule of 12 to 25 kDa in size, with no significant binding to albumin.[93,94]

Several hypotheses were advanced to reconcile the previously observed effects and operation of a putative renotropic system.[93,94] The following were variously proposed: (1) renotropin is a circulating substance normally catabolized or excreted by the kidneys; (2) renal growth is regulated by a specific renotropin-producing tissue inhibited by a factor produced by normal kidneys; and (3) renal growth is tonically inhibited by a substance produced by normal kidneys, a decrease in the levels of which induce an enzyme in the renal cortex that cleaves a circulating precursor of renotropin to produce the active molecule.

ENDOCRINE EFFECTS

Several of the major endocrine systems influence renal growth but each lacks selective effects on the kidney. There is little evidence that any of these systems represents the specific mediators of compensatory renal hypertrophy. Whereas early experiments suggested that hypophysectomy inhibits compensatory hypertrophy after uninephrectomy, later studies that controlled for the reduction in renal mass that usually accompanies hypopituitarism found a degree of hypertrophy comparable to that seen in normal rats.[147]

Nevertheless, specific renotropic activity has been identified in a subfraction of ovine pituitary extract associated with a lutropin-like substance.[93,94] Uninephrectomy is accompanied by a transient increase in the pulsatile release of growth hormone (GH) in male but not female rats, suggesting a role for this hormone in the early phase of hypertrophy in males.[148] When the increase in GH was prevented by the administration of an antagonist to GH-releasing factor or the effects of GH were blocked by a GH receptor blocker, renal hypertrophy was significantly attenuated.[149,150] Adrenal hormones appear to play only a small role in renal hypertrophy. Adrenalectomy does not inhibit compensatory growth after uninephrectomy.[151] Whereas renal weight relative to body weight is reduced in hypothyroidism and increased by excess thyroid hormone, compensatory hypertrophy still occurs in thyroidectomized rats.[152] Progesterone and estradiol in excess or ovariectomy has little effect on renal weight, but testosterone appears to play a role, as evidenced by a fall in kidney and body weight after orchidectomy and an increase in kidney weight with excess testosterone.[153,154] Whereas orchidectomy does not inhibit hypertrophy after uninephrectomy, exogenous testosterone does increase the degree of hypertrophy observed in some but not all studies.[93,94]

GROWTH FACTORS

Of the numerous growth factors and their receptors that have been localized in the kidney, at least four are associated with renal hypertrophy.[155,156] Several lines of evidence suggest a role for insulin-like growth factor-1 (IGF-1). Renal tissue IGF-1 levels were elevated at 1 to 5 days after uninephrectomy and started to decline within days in some[157,158] but not all studies.[159] In one study, the level of renal IGF-1 expression correlated significantly with the extent of renal mass ablation.[160] On the other hand, Shohat and associates have found an increase in serum IGF-1 levels only at 10 days postnephrectomy, which was still present on day 60.[157] That IGF-1 may be induced independently of GH in the setting of renal hypertrophy is illustrated by preservation of the increase in renal IGF-1 in hypophysectomized[161] and GH-deficient rats.[162]

Other molecules related to IGF function are also upregulated. Renal IGF-1 receptor gene expression was increased two- to fourfold in female rats after uninephrectomy[148]; IGF-1 binding protein mRNA was upregulated in the remnant kidney at 2 weeks after 5/6 nephrectomy[163]; and analysis of the genome-wide transcriptional response to unilateral nephrectomy identified IGF-2 binding protein as one of the activated genes.[110] Further evidence has suggested that IGF-1 may in turn promote production of vascular endothelial growth factor (VEGF), implying that VEGF may be a downstream mediator of IGF-1 effects, at least in the pathogenesis of diabetic retinopathy.[164] That VEGF is important for compensatory renal hypertrophy has been confirmed by the observation that treatment of mice with VEGF antibodies after uninephrectomy completely prevents glomerular hypertrophy and inhibits renal growth at 7 days.[158] Epidermal growth factor (EGF) in the remaining kidney is increased on day 1 in mice[165] and by day 5 in rats.[166] In addition, EGF has been shown to induce IGF-I mRNA production in collecting duct cells in vitro, suggesting the existence of a local paracrine system.[167] Increased mRNA levels for both HGF and its receptor, c-met, have been demonstrated in the remaining kidney as early as 6 hours after uninephrectomy.[168,169] In another study, the rise in HGF message was found to be nonspecific, occurring in both liver and kidney, and also in sham-operated rats, whereas the increase in mRNA for c-met was specific for the outer renal medulla.[170]

Despite these associations, the timing of the changes in growth factor levels remains unclear. Whereas some investigators have reported early increases,[165,171] several others reported changes only at time points when significant hypertrophy is already present, thus failing to provide convincing evidence that they represent the proximal effectors in a renotropic system.[157,166]

MESANGIAL CELL RESPONSES, A UNIFYING HYPOTHESIS

Mesangial cells play a central role in glomerular function, modulating glomerular capillary blood flow and ultrafiltration surface area. In addition, mesangial cells are both a source of and target for vasoactive molecules, growth factors, cytokines, and extracellular matrix proteins. Studies have suggested that they also play a major role in compensatory renal hypertrophy. In vitro experiments have found that when mesangial cells from a remaining kidney after uninephrectomy are cultured with serum obtained from rats after uninephrectomy, their conditioned medium induces hypertrophy in tubule cells.[172] Uninephrectomy induces significant transient proliferation of mesangial cells, reaching a peak at 24 hours and ceasing within 72 hours. This proliferation occurs in an environment of increased circulating as well as renal growth factors and cytokines, such as GH, IGF-1, and interleukin-10 (IL-10), as well as reduced levels of antiproliferative factors such as transforming growth factor-β (TGF)-β and ANP. A reduction in mesangial cell proliferation occurs in parallel with the onset of tubule cell hypertrophy. IL-10 and TFG-β have been identified as the major mediators of the mesangial regulation of tubule cell hypertrophy. Mesangial cells are the main source of IL-10 in the kidney, where it acts as an autocrine growth factor and induces the expression of TFG-β.[173] Mesangial cells are the only resident renal cells known to produce and activate TFG-β, a process that is regulated by multiple factors, including Ang II, IGF-1, HGF, basic fibroblast growth factor (bFGF), tumor necrosis factor-α (TNF-α), EGF, and platelet-derived growth factor (PDGF), all of which are produced by mesangial cells. IL-10 expression starts to increase in the remaining kidney within hours of uninephrectomy, peaks at 24 hours, and returns to normal within several days. In contrast, circulating and renal TFG-β levels fall in the first 24 hours after nephrectomy and start to rise from 72 hours, reaching a peak at 1 week.[173]

The importance of IL-10 was confirmed by experiments in which inhibition of IL-10 production resulted in lower TFG-β levels and reduced tubular hypertrophy, resulting in a 20% to 25% reduction in the weight of the remaining kidney.[173] IL-10 has no direct effect on tubule cells, whereas TFG-β has been identified as an important mediator of tubule cell hypertrophy.[174] The data presented are therefore consistent with the hypothesis that hyperfiltration after unilateral nephrectomy induces mesangial cell proliferation as well as the production of IL-10 and other growth factors that induce expression and activation of TFG-β in mesangial cells. TFG-β in turn stimulates tubule cell hypertrophy[175] (Fig. 51.3). This goes a long way to unifying the components of the solute load and renotropin hypotheses into a single paradigm to explain the mechanisms of compensatory renal hypertrophy.

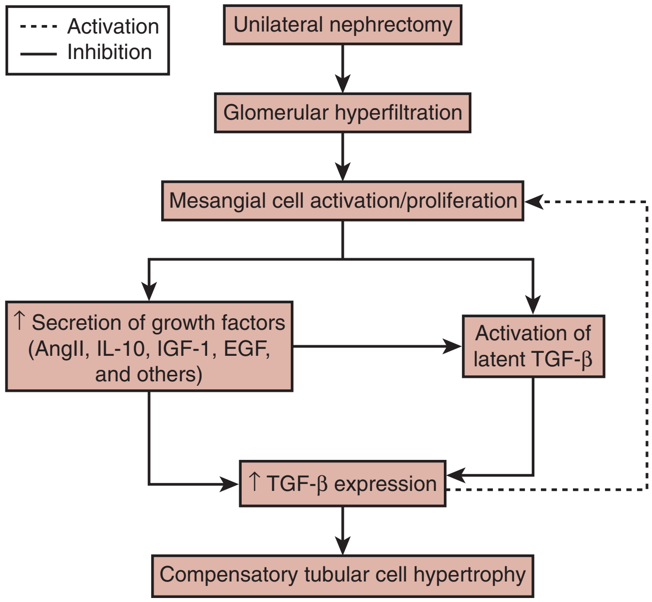

Fig. 51.3 Possible mechanisms involved in compensatory renal growth after unilateral nephrectomy. *Ang II,* Angiotensin II; *EGF,* epidermal growth factor; *IGF-1,* insulin-like growth factor-1; *IL-1,* interleukin-1; *TGF- β,* transforming growth factor-β. (From Sinuani I, Beberashvili I, Averbukh Z, et al. Mesangial cells initiate compensatory tubular cell hypertrophy. *Am J Nephrol.* 2010;31:326–331.)

TUBULE CELL RESPONSES

Detailed investigation of cellular responses to renal mass reduction has begun to elucidate some of the mechanisms involved in compensatory hypertrophy at the cellular level. Renal hypertrophy is achieved by modulation of the cell cycle, which becomes arrested in the late G1 phase and therefore does not progress to the S phase, resulting in cell hypertrophy instead of hyperplasia. This is achieved through activation of cyclin-dependent kinase (CDK) 4–cyclin D without subsequent engagement of CDK 2–cyclin E, a process thought to be regulated by TGF-β and CDK inhibitor proteins $p21^{Waf1}$, $p27^{kip1}$, and $p57^{kip2}$. Activity of CDK 4–cyclin D complexes increases at 4, 7, and 10 days postnephrectomy, and CDK 2–cyclin E increases at days 2, 4, and 7 to 14, implying that $p21^{Waf1}$, $p27^{kip1}$, and $p57^{kip2}$ may play an important role in regulating tubule cell hypertrophy.[176–178] The required increase in RNA and protein synthesis appears to be mediated by mammalian target of rapamycin (mTOR), a protein kinase that controls protein synthesis as well as cell growth and metabolism.

In cells, mTOR exists in two distinct multiprotein complexes, mTORC1 and mTORC2. mTORC1 acts through multiple mediators to regulate protein synthesis and cell size. Two important downstream effectors of mTORC1 are 4E-binding protein 1 and protein kinase p70S6 kinase 1 (S6K1). The importance of mTORC1 has been confirmed by the observation that pretreatment of rats with rapamycin, an inhibitor of mTORC1, inhibits renal hypertrophy after uninephrectomy.[179] Furthermore, experiments in S6K1 knockout mice have found inhibition of 60% to 70% of the hypertrophy observed after uninephrectomy, indicating that S6K1 plays a major role in compensatory hypertrophy.[180]

ADAPTATION OF SPECIFIC TUBULE FUNCTIONS IN RESPONSE TO NEPHRON LOSS

As noted previously, the bulk of the increase in renal mass following uninephrectomy is due to hypertrophy of the proximal nephron. The more distal nephron segments also enlarge, but to a lesser extent. In uninephrectomized rats, the proximal convoluted tubule is increased on average by 17% in luminal diameter and 35% in length, yielding a 96% increase in total volume; the distal convoluted tubule is enlarged by 12% in luminal diameter and 17% in length, yielding a 25% increase in total volume.[181] Maintenance of homeostasis for various solutes in the presence of a declining GFR requires highly integrated responses from each tubule segment. Whereas some solutes, including creatinine and urea, are chiefly cleared by glomerular filtration and therefore rise gradually in plasma with a declining GFR, for others, the tubule solute handling adapts so that plasma levels remain constant, virtually until end–stage renal failure is reached (Fig. 51.4).

ADAPTATION IN PROXIMAL TUBULE SOLUTE HANDLING

In renal ablation models, as with the increase in remnant kidney SNGFR, the extent to which the proximal tubule enlarges is inversely proportional to the remnant kidney mass. Proximal tubule enlargement is associated with an increase in proximal fluid reabsorption. In studies of animals and humans with reduced renal mass, the increase in proximal fluid reabsorption observed was found to be proportional both to the increase in the remnant kidney GFR and the increase in tubular volume.[182] Similarly, in proximal tubules isolated from remnant kidneys, the observed increase in transtubular fluid flux was proportional to the increases in size and protein content of the tubule epithelial cells.[141,183] Folding of the basolateral membrane of the proximal tubule epithelium was also found to increase, resulting in augmentation of the basolateral surface area, in proportion to the increase in cell volume.[184] This increase in surface area was accompanied by an increase in activity of Na^+-K^+-ATPase, the membrane pump that generates the main driving force for proximal tubule solute and water transport.[184]

Increases in proximal tubule size and surface area are not, however, the only determinants of increased transport activity in this nephron segment. Fluid reabsorption in isolated proximal tubule segments increases within 24 hours of nephrectomy—that is, when the GFR is already increasing, but well before significant hypertrophy occurs—implying an intrinsic TEC adaptation to nephron loss.[185] This observation also raises the possibility that the increase in proximal fluid reabsorption occurring in response to nephron loss is driven by the increase in the SNGFR.[93,94] Because solute reclamation is an energy-requiring process, it is not surprising that in uninephrectomized rabbits, the increase in proximal tubule volume is accompanied by a proportional increase in mitochondrial volume.[186] The observation that the increase in renal mass is outstripped by the rise in the GFR in models of progressive nephron loss implies that renal energy consumption per unit of remnant renal mass increases as renal function declines.[181]

The rise in the SNGFR that occurs in the remnant kidney presents increased loads of glucose, amino acids, and other solutes that should be reabsorbed entirely in the proximal tubule, provided that the maximal transport capacity was not exceeded. Maximal proximal tubular reabsorptive capacities for glucose and amino acids have been shown to increase in proportion to tubule mass after partial renal ablation.[187] Some metabolic functions of proximal tubules are also augmented in the remnant kidney to maintain adequate plasma levels of important metabolites, including citrulline, arginine, and serine.[188] Other proximal tubule functions, however, are not adjusted in proportion to proximal tubule mass; fractional phosphate reabsorption is decreased, whereas ammoniagenesis increases.[187,189,190] These adaptations are appropriate homeostatic responses that permit continued excretion of daily phosphate and acid loads as the number of functioning nephrons declines.

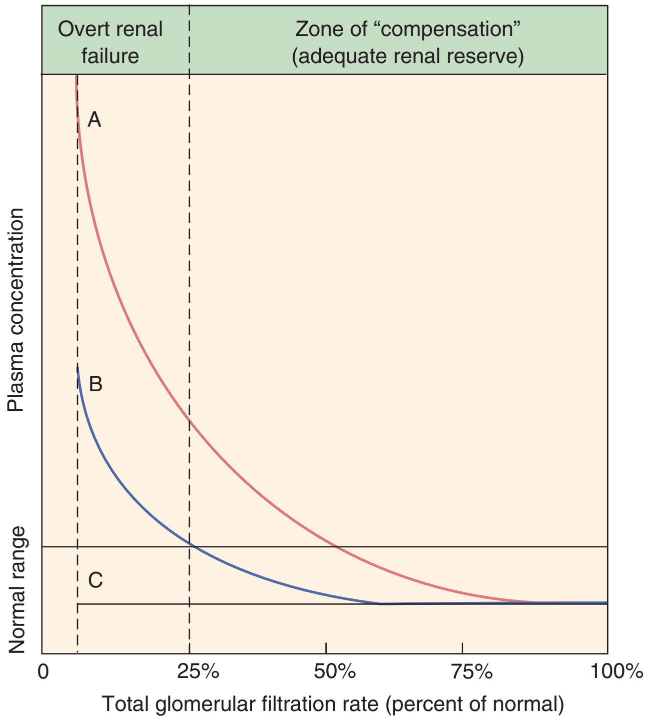

Fig. 51.4 Representative patterns of adaptation for different types of solutes in body fluids in chronic renal failure. (A) Rise in serum concentration with each permanent reduction in the glomerular filtration rate (GFR) (e.g., creatinine). (B) Rise in serum concentration only after the GFR falls below a critical value due to adaptive increases in tubular secretion (e.g., phosphate). (C) Serum concentration remains normal throughout almost the entire period of progression of renal failure (e.g., sodium). (Modified from Bricker NS, et al. in Brenner BM, Rector FC, eds. *The Kidney,* 2nd ed. Philadelphia: WB Saunders; 1981.)

LOOP OF HENLE AND DISTAL NEPHRON

Although there is little change in cross-sectional area in the thick ascending limb of the loop of Henle, fluid reabsorption in this segment also increases in proportion to the SNGFR.[181] In contrast, both the distal tubule and cortical collecting duct enlarge in response to nephron loss.[181] Unlike the proximal tubule, however, where the increased reabsorptive capacity is chiefly due to increased tubule dimensions, the increased reabsorptive capacity observed in the distal segments is far greater than would be expected for the corresponding increase in tubule volume, implying a major adaptive increase in active solute transport.[181] Levels of mRNA for the sodium-dependent *myo*-inositol cotransporter (SMIT) and $Na^+/Cl^-/$betaine gamma-aminobutyric acid transporter (BGT-1) are increased in the cortex and outer medulla of remnant kidneys from 5/6 nephrectomized rats.[191] Likewise, potassium secretion by the distal nephron increases in compensation for nephron loss, facilitated by an increase in the basolateral surface area of cortical collecting duct principal cells and an increase in Na^+-K^+-ATPase activity.[192,193]

GLOMERULOTUBULAR BALANCE

Micropuncture studies have confirmed that proximal fluid reabsorption remains proportional to glomerular filtration over a wide range of SNGFR values in both glomerular and tubulointerstitial diseases.[194,195] This glomerulotubular balance is critical to the physiologic integrity of remnant nephron function and hence ECF homeostasis. Compensatory increases in the SNGFR in surviving nephrons must be accompanied by similar increases in proximal tubular solute and water reabsorption to avoid overwhelming the distal nephron transport capacity and disrupting its regulation of the volume and composition of the final urine. Conversely, reductions in SNGFR in damaged nephrons must be matched by similar reductions in proximal fluid reabsorption to maintain adequate solute and water delivery to the distal tubule, again permitting excretion of urine of appropriate volume and composition.

Glomerulotubular balance is maintained as follows. The degree of single-nephron hyperfiltration occurring as a consequence of nephron loss determines the passive Starling forces operating in the postglomerular microcirculation, which, in turn, govern net transtubular solute reabsorption.[196] Increases in the SNGFR associated with an increased filtration fraction result in elevated postglomerular capillary protein concentrations, which determine nonlinear increases in oncotic pressure, Π_E, the major determinant of peritubular capillary reabsorptive force (P_r). Reductions in the SNGFR, in contrast, result in a lowered peritubular oncotic pressure, thereby reducing P_r. Thus the SNGFR and proximal fluid reabsorption remain in direct proportion to one another.

Prevention of hyperfiltration by dietary protein restriction has been shown to abrogate the increase in proximal fluid reabsorption in the remnant kidney, underscoring the dependence of proximal tubular function on the level of glomerular filtration.[196] In the remnant kidney of rats subjected to extensive renal mass ablation, absolute fluid reabsorption was found to be markedly increased in proximal portions of superficial and juxtamedullary nephrons, yet fluid delivery to the more distal segments of the nephron was also somewhat increased.[197] In the setting of nephron loss, sodium reabsorption by the loop of Henle has been shown to remain proportional to sodium delivery to that segment, indicating preservation of tubulotubular balance, a mechanism that maintains appropriate distal solute and water delivery in the presence of progressive nephron loss. Until the adaptive capacities of these mechanisms are finally exhausted, the operation of glomerulotubular balance and tubulotubular balance ensures that the distal tubule mechanisms that determine final urine volume and composition are not overwhelmed by unregulated distal delivery of water and solute.[198] In keeping with these physiologic observations, morphologic studies have shown that within the same kidney, nephrons associated with damaged glomeruli are usually atrophic and presumably hypo functioning or nonfunctioning, whereas those associated with healthier glomeruli are usually hypertrophic and hyperfunctioning.[199]

To maintain homeostasis in the presence of continued food and water intake, specific mechanisms that enhance single-nephron water and solute excretion must come into play, in addition to adjustments in the SNGFR and tubular reabsorption that occur in response to nephron loss. These mechanisms are not unique to the setting of renal insufficiency, however, and are also engaged when the normal kidney is challenged to excrete extraordinary loads of solute and water. In general, the adaptive physiology of the chronically injured kidney is adequate to preserve homeostasis for many solutes under baseline conditions, but the adaptive capacity may easily become overwhelmed by fluctuations in fluid intake, especially by increases in electrolyte and acid loads. Patients with chronic renal failure are therefore susceptible to develop volume overload, volume loss, hyperkalemia, and acidosis when the excretory capacity of the kidney is challenged by relatively modest changes in excretory demands.

SODIUM EXCRETION AND EXTRACELLULAR FLUID VOLUME REGULATION

In chronic renal failure, ECF volume is often maintained very close to normal until end-stage is reached.[200] This remarkable feat is accomplished by an increase in fractional sodium excretion (FE_{Na}) in inverse proportion to decline in the GFR.[201] Many studies have attempted to identify which nephron segments are responsible for the decrease in sodium reabsorption:

- Micropuncture studies in uninephrectomized rats have shown that tubule fluid transit times, as well as the half-time for reabsorption of a stationary saline droplet in the proximal tubule lumen, were not different from controls.[181]
- In remnant kidneys of rats receiving high normal or low-sodium diets, absolute sodium reabsorption was found to increase, but fractional sodium and fluid reabsorption were found to decrease in all three uremic groups.[202]
- Micropuncture studies in dogs and rats have failed to detect significant reductions in fractional proximal tubule fluid and sodium reabsorption.[203]
- Distal sodium delivery was found to be markedly increased in the rat remnant kidney.[202]
- Increased solute transport activity has been demonstrated in the distal tubule of uninephrectomized rats.[181]
- Under conditions of hydropenia and salt loading, sodium reabsorption by the medullary collecting duct of the rat remnant kidney was markedly reduced.[204]

Investigators have also used a computational model of epithelial transport to investigate changes after nephron loss. Similar to experimental data, the simulations found that after uninephrectomy or 5/6 nephrectomy, fractional reabsorption of sodium and water remains normal from the proximal tubule to distal convoluted tubule but increases in the collecting tubule and collecting duct to maintain sodium balance.[205]

Taken together, these data suggest that proximal fractional reabsorption remains largely unchanged, and that in the setting of renal insufficiency, adjustments in sodium excretion occur predominantly in the loop and distal nephron segments.[206] These physiologic observations were supported by studies investigating changes in sodium transporters after 5/6 nephrectomy. At 4 weeks after surgery, a substantial increase in abundance of the Na^+-K^+-$2Cl^-$ and Na^+-Cl^- cotransporters (expressed chiefly in the loop of Henle and distal tubule, respectively) was observed, whereas marked decreases were observed in both at 12 weeks.[207] The expression of ENaC alpha increased throughout the observation period.[207]

In addition to load-dependent tubular adaptations in sodium handling, sodium excretion is also modulated by hormonal influences. Levels of NPs are elevated in chronic renal failure as a result of reduced clearance and in response to alterations in sodium and volume status.[56,208] In rats with extensive renal mass ablation, plasma ANP levels may be restored toward normal levels by dietary sodium restriction but, in response to increases in sodium intake, they rise progressively along with sodium excretion.[209] The notion that ANP plays an important role in mediating adaptive changes in sodium excretion in the setting of renal ablation has been confirmed by observations that the administration of a NP receptor antagonist reduces both FE_{Na} and GFR in 5/6 nephrectomized rats receiving normal or high- salt diets, but does not alter these variables in rats fed low-salt diets.[210] Significantly, NPs not only modulate sodium excretion but may also contribute to the attendant glomerular hyperfiltration and thereby further exacerbate renal injury (see previously).

Systemic hypertension has also been proposed by Guyton and associates to contribute to the increase in FE_{Na} observed with renal insufficiency.[211] They hypothesized that a constant sodium intake in the presence of a reduced number of functioning nephrons leads to positive sodium balance as a result of reduced excretory capacity. Positive sodium balance leads to an increase in ECF volume and a rise in systemic BP that in turn leads to an increase in FE_{Na} and reestablishes the steady state. In support of this hypothesis, salt intake has been shown to be critical to the development of hypertension in subtotally nephrectomized dogs,[212] and uremic patients have been found to exhibit marked sodium retention when treated with vasodilating antihypertensive agents.[213] On the other hand, a lowered salt intake in 5/6 nephrectomized rats does not prevent the development of systemic hypertension,[134] suggesting that sodium excretion and hypertension are not always interdependent in the setting of extensive renal mass ablation. Sodium conservation, on the other hand, is also impaired with renal insufficiency and, in response to an acute reduction in sodium intake, most patients were unable to reduce sodium excretion below 20 to 30 mEq/day.[214] The salt-losing tendency associated with CKD appears to be dependent on the salt load per nephron and may therefore be reversible with adequate dietary sodium restriction. Other factors modulating FE_{Na} in the setting of renal insufficiency include changes in sympathetic nervous system activity, aldosterone, PGs, and parathyroid hormone (PTH) levels.[206,215] Sodium homeostasis and volume regulation are discussed in further detail in Chapters 6 and 10.

URINARY CONCENTRATION AND DILUTION

ECF homeostasis is usually well maintained until renal insufficiency is far advanced, when the ability of the kidney to excrete a volume load becomes significantly reduced.[206] Normal generation of solute-free water is about 12 mL/100 mL of the GFR and is dependent on dilution of tubule fluid in the thick ascending limb, maintenance of low water permeability in the distal nephron segments in the absence of antidiuretic hormone (ADH), and decreased hypertonicity of the medullary interstitium during water diuresis. Although the single- nephron capacity to excrete free water per milliliter of the GFR is not reduced in patients with advanced renal disease,[216] the absolute reduction in the GFR reduces the overall capacity of the kidney to excrete a water load. Patients with chronic renal failure therefore cannot adequately dilute their urine and are prone to water intoxication and hyponatremia. Hypothetically, in addition to the excretion of the equivalent of 2 L of so-called isotonic urine per day (obligatory excretion of 600 mOsm/day), normal kidneys, with a GFR of 150 L/day, can excrete up to 18 L of free water/day, whereas failing kidneys, with a GFR of 15 L/day, can only excrete about 1.8 L of free water/day. The minimum urinary osmolality achievable by normal kidneys would therefore approach 30 mOsm/L (600 mOsm/20 L), whereas that of diseased kidneys would be 160 mOsm/L (600 mOsm/3.8 L).

Urinary concentration is also impaired in renal insufficiency. Normal urinary concentration requires preservation of the countercurrent mechanism to maintain hypertonicity of the medullary interstitium and normal water transport across the distal nephron segments in response to ADH. The maximal urinary osmolality in normal subjects is about 1200 mOsm/L. As the GFR decreases, however, maximal urinary osmolality falls and, with a GFR of 15 mL/min/1.73 m^2, s reduced to about 400 mOsm/L.[217] A normal individual can therefore excrete the obligatory daily 600 mOsm in as little as 0.5 L of urine, whereas the patient with a GFR of 15 mL/min/1.73 m^2 can excrete the same load in a minimum of 1.5 L. Part of the defect in urinary concentration observed with renal damage may be attributed to the high solute load imposed per surviving nephron. In patients with chronic renal failure, however, the osmotic effect of urea was shown to be inadequate to account fully for the reduction in maximal urine concentration, indicating that factors other than osmotic diuresis contribute to reduction in urine-concentrating ability in these patients.[217]

Furthermore, in patients with chronic glomerulonephritis, reduction in urine concentrating capacity was found to correlate significantly with the degree of medullary fibrosis on renal biopsy,[218] suggesting that disruption of the medullary architecture, with the consequent loss of medullary hypertonicity, may result in disproportionate impairment of urinary concentrating ability at any given level of the GFR. Consistent with this observation, patients with primary tubulointerstitial injury (e.g., analgesic nephropathy, sickle cell disease), have markedly impaired urine-concentrating abilities, even early in the course of their illness.[217,219,220] Similarly, in animal experiments, surgical exposure of the renal papilla in intact hydropenic rats was found to lead to a reduction in urinary osmolality because of the accompanying alterations in vasa recta flow and ensuing washout of medullary solutes.[221] Interestingly, similar exposure of papillae in rats with remnant kidneys did not affect urinary osmolality, presumably because medullary solute washout had already occurred due to the adaptive responses to nephron loss.

Urinary concentration also depends on water reabsorption in the distal nephron segments in the remnant kidney. Reduction in water reabsorption may be the result of several mechanisms in the failing kidney. A defective cAMP-mediated response to ADH may render the cortical collecting duct resistant to the effects of ADH, resulting in increased water delivery to the papillary collecting duct.[222] Urinary osmolality is inversely proportional to fractional water delivery to the papillary collecting duct in 5/6 nephrectomized rats, despite an increase in absolute water reabsorption per functioning collecting tubule when compared with controls.[221] Patients with renal insufficiency are therefore prone to volume depletion in the presence of water deprivation or impaired thirst mechanisms. More commonly, the inability to concentrate urine becomes manifest as nocturia, which develops as renal function deteriorates. Urinary concentrating and diluting mechanisms are discussed in further detail in Chapter 10.

POTASSIUM EXCRETION

To maintain potassium homeostasis in the presence of continued dietary intake and a reduced number of functioning nephrons, potassium excretion per nephron must increase. In both normal and diseased kidneys, almost all the filtered potassium is reabsorbed in the proximal tubule and loop of Henle. Potassium excretion is therefore determined predominantly by distal secretion,[206] although a reduction in potassium reabsorption by the loop of Henle has been shown to contribute to increased potassium excretion in rats with reduced renal mass.[223] In both normal and partially nephrectomized dogs, urinary potassium excretion was found to correlate directly with serum potassium concentration.[224] Similarly, in intact and uninephrectomized rats, net potassium secretion in the distal convoluted tubule occurred only during potassium infusion, whereas potassium secretion by cortical collecting tubules (CCTs) occurred under all conditions and was greater after uninephrectomy.[225] Other studies have confirmed that the CCT is an important site of potassium secretion in the remnant kidney.[192,204] Secretion of potassium by CCTs isolated from remnant kidneys of rabbits fed normal or high-potassium diets was shown to persist in vitro and to be directly related to the dietary potassium content,[192] indicating an intrinsic tubular adaptation to potassium load. This adaptation was absent in CCTs from rabbits in which dietary potassium had been reduced in proportion to the amount of renal mass lost. In addition to variations with the dietary potassium load, the increase in potassium secretion by remnant CCTs was also found to correlate with plasma aldosterone levels, but not with intracellular potassium concentration or Na^+-K^+-ATPase.[192] In contrast, however, others have reported an increase in cortical and outer medullary Na^+-K^+-ATPase activity in homogenates from rat remnant kidneys that was abrogated when potassium intake was reduced in proportion to the reduction in the GFR.[226] These findings have been largely confirmed using a computational model of epithelial transport. In simulations of an acute potassium load and chronic high-potassium diet, potassium secretion increased in the connecting tubule, but the increase was less after uninephrectomy (18% less with acute and 13% less with chronic loads) and 5/6 nephrectomy (42% less with acute and 31% less with chronic loads) than in sham controls. The increase in potassium secretion per tubule failed to compensate fully for the decrease in nephron number.[227] Finally, the frequent occurrence of hyperkalemia in patients with chronic renal failure after treatment with an aldosterone antagonist or an ACE inhibitor, suggests that "normal" aldosterone levels are required to maintain adequate potassium excretion in this population.[228] In general therefore the increase in potassium secretion by surviving nephrons appears to be predominantly determined by the rise in plasma potassium levels after potassium ingestion and by intrinsic tubular adaptation to the increased filtered potassium load.[224,225] In both dogs and patients with chronic renal failure, however, the kaliuretic response to an oral potassium load is attenuated compared to normals, despite higher serum potassium levels.[224,229] The eventual complete excretion of a potassium load therefore occurs at the expense of a sustained increase in serum potassium levels. Control of potassium excretion is discussed further in Chapters 6 and 17.

ACID-BASE REGULATION

Reduction of the GFR in patients with CKD is associated with the development of systemic metabolic acidosis due to a reduction in the serum bicarbonate concentration. Normal acid-base balance requires reabsorption of filtered bicarbonate, excretion of titratable acid, ammonia generation, and acidification of tubular luminal fluid by the distal nephron.[206] In chronic renal failure, acidosis develops as a result of varying degrees of impairment in each of these processes.[230]

A reduction in renal ammonia synthesis is the greatest limitation to acid excretion in CKD. Low serum bicarbonate levels result in maintenance of acidic urine, which stimulates proximal tubule ammoniagenesis and also promotes ammonia conversion, resulting in its entrapment as ammonium in the tubule lumen. Net ammonia production per hypertrophied proximal tubule has been shown to increase in response to nephron loss.[222] With a decreasing GFR, however, this increase becomes inadequate to compensate for further nephron loss, and absolute ammonia excretion falls.[190] In addition, disruption of the tubulomedullary ammonium concentration gradient because of structural injury may impair ammonia trapping and therefore reduce ammonium excretion.[190] Bicarbonate reabsorption by the nephron occurs predominantly in association with sodium reclamation in the proximal tubule and is dependent on the generation of a proton gradient in the distal nephron.

Conflicting data with respect to bicarbonate reabsorption in remnant kidneys may reflect species differences. In dogs with remnant kidneys, bicarbonate reabsorption was increased at both proximal and distal micropuncture sampling sites compared with intact controls.[231] In contrast, bicarbonate reabsorption per unit GFR is reduced in humans and rats with chronic renal failure,[206] and some patients with renal failure demonstrate bicarbonate wasting until the serum bicarbonate level drops below 20 mEq/L.[232] Bicarbonate reabsorption is also reduced in the setting of hyperkalemia, increased ECF volume, and hyperparathyroidism, all of which may be present in patients with chronic renal failure.[233–235] Distal urinary acidification tends to be relatively well preserved in patients with chronic renal failure, and urinary pH, although higher than in normal individuals with experimental acidosis, is usually around 5.[236] Urinary excretion of titratable acid is also generally well preserved in the setting of nephron loss as a consequence of increased fractional phosphate excretion.[187,190] As renal failure progresses, acid excretion becomes more dependent on the excretion of titratable acid. Renal acidification mechanisms are discussed more comprehensively in Chapters 9 and 16.

CALCIUM AND PHOSPHATE

Derangements of calcium and phosphate metabolism occurring with renal insufficiency are the result of not only impaired urinary excretion of these solutes but also associated abnormalities in vitamin D metabolism and PTH secretion. With progressive renal dysfunction, 1α-hydroxylation of vitamin D by the kidney decreases, calcium absorption from the gut decreases, serum calcium level tends to decrease, serum phosphate level tends to increase, and PTH secretion increases. In response to increased PTH, calcium is mobilized from bone, renal phosphate excretion is enhanced, and the steady state becomes reestablished, with secondary hyperparathyroidism as the trade-off.[237] In chronic renal failure (CRF), the serum phosphate level does not increase until the GFR falls below 20 mL/min/1.73 m^2, and phosphate balance is maintained predominantly by an increase in fractional phosphate excretion.[238]

With moderate renal failure, therefore, filtered phosphate is not greatly increased, and the increase in phosphate excretion must be achieved by a reduction in phosphate reabsorption per nephron.[239] With more severe reductions in the GFR, however, phosphate excretion is maintained by an increase in serum phosphate levels as well as by reduced reabsorption per nephron. Sodium-dependent phosphate transport measured in proximal tubular brush border membrane vesicles prepared from the remnant kidneys of dogs was shown to be decreased when compared with that in vesicles derived from normal dogs.[187] Interestingly, however, this decrease was abolished if the partially nephrectomized dog had also undergone parathyroidectomy, indicating that PTH plays an important role in proximal tubular adaptation to phosphate excretion. Studies of isolated proximal tubules from euparathyroid uremic rabbits have shown a reduction in net phosphate flux per unit of reabsorptive surface area, and an increase in sensitivity to PTH.[189] The authors postulated that the number of PTH receptors per tubule must increase in the remnant kidney, concomitant with tubular hypertrophy. The levels of mRNA encoding the sodium-coupled phosphate transporter, NaPi-2, are reduced by approximately 50% in remnant kidneys from 5/6 nephrectomized rats.[240] In contrast, tubules from hyperparathyroid uremic rabbits demonstrated reduced PTH sensitivity, consistent with the downregulation or persistent occupancy of the PTH receptors.

On the other hand, studies in animals with reduced renal mass subjected to parathyroidectomy have shown that fractional excretion of phosphate remains inversely proportional to the reduction in the GFR,[241] indicating that phosphate excretion is not entirely dependent on the presence of PTH. Fibroblast growth factor 23 (FGF23) has been identified as a major mediator of increased phosphaturia after nephron loss.[242] First identified as the primary mediator of autosomal dominant hypophosphatemia, increased FGF23 expression has been shown in transgenic models to increase urinary phosphate excretion through the downregulation of NaPi-2a expression in proximal tubules.[243] Further experiments using gene deletion models have found that decreased expression of NaPi-2a and NaPi-2c by exogenous FGF23 is mediated predominantly by FGF receptor 1.[244] Activation of FGF receptors by FGF23 is critically dependent on binding with its coreceptor, Klotho.[242] After 5/6 nephrectomy in rats, investigators observed hyperphosphatemia after a high-phosphate diet, despite high levels of FGF23. This was explained by a marked reduction in levels of Klotho that was prevented by treatment with calcitriol. Further experiments in cultured cells have indicated that Klotho is suppressed by phosphaturia through Wnt/β-catenin signaling.[245] Circulating levels of FGF 23 become elevated early in the course of CKD, with levels rising once the GFR falls below 70 mL/min/1.73 m^2 in humans,[242] although this may depend to some extent on vitamin D status. In one study, FGF23 was elevated early in persons with CKD who were vitamin D–replete, but PTH was more frequently elevated than FGF23 in those with vitamin D insufficiency.[246] That FGF23 is important in mediating increased phosphaturia after nephron loss has been demonstrated by experiments in which the administration of neutralizing anti-FGF23 antibodies after renal mass reduction resulted in decreased fractional excretion of phosphate and increase in serum phosphate levels.[247] Whereas most of the reduction in phosphate reabsorption is achieved in the proximal tubule, there is also some evidence of increased fractional phosphate excretion by the distal tubule in uremic dogs and rats.[248]

As renal failure advances, renal 1α-hydroxylation of vitamin D decreases, and as a result, calcium absorption from the gut is reduced.[249] In addition to its effects on renal phosphate excretion, FGF23 inhibits renal 1α-hydroxylase activity, thereby reducing levels of 1,25-OH vitamin D.[243,247] In renal failure, fractional intestinal calcium absorption is inversely proportional to blood urea nitrogen levels.[249] Calcium excretion, on the other hand, varies widely in patients with renal disease, probably due to differences in diet, heterogeneity of vitamin D production, and predominance of glomerular versus tubulointerstitial injury.[250] In normal individuals, calcium excretion is mediated by the suppression of PTH-induced reabsorption in the distal nephron and by the suppression of PTH-independent mechanisms in the thick ascending limb. In patients with CRF, fractional calcium excretion remains unchanged until the GFR falls below 25 mL/min/1.73 m^2, when fractional excretion increases due to the obligatory solute diuresis.[206] Absolute calcium excretion, however, remains low. Hypocalciuria in patients with CRF has been shown to be due in part to the attendant hyperparathyroidism.[251] Similar findings were obtained in rats with reduced renal mass, in which parathyroidectomy resulted in increased calcium excretion compared with nonparathyroidectomized controls.[252] Renal calcium clearance is increased in patients with tubulointerstitial disease and in rats with surgical papillectomy, suggesting that regulation of calcium reabsorption depends on intact medullary structures and that regulation of calcium excretion may be largely modulated by the distal nephron segments.[206] The potential contributions of calcium and phosphate to renal disease progression are discussed below. Calcium and phosphate metabolism are also discussed in greater detail in Chapters 7 and 18.

LONG-TERM ADVERSE CONSEQUENCES OF ADAPTATIONS TO NEPHRON LOSS

The functional and structural adaptations to nephron loss described previously may be regarded as a beneficial response that minimizes the resultant loss of the total GFR. It has been appreciated for several decades, however, that rats subjected to partial nephrectomy subsequently develop hypertension, albuminuria, and progressive renal failure. Detailed histopathologic studies in rat remnant kidneys after 5/6 nephrectomy have revealed mesangial accumulation of hyaline material that progressively encroaches on capillary lumina, obliterating Bowman's space and finally resulting in global sclerosis of the glomerulus. These findings, together with the observation that sclerosed glomeruli are a common finding in human CKD of diverse causes, have led to the hypothesis that glomerular hyperfiltration ultimately results in damage to the remaining glomeruli and contributes to a vicious cycle of progressive nephron loss. The 5/6 nephrectomy model has been extensively studied, and considerable progress has been made in elucidating how the physiologic adaptations of remaining nephrons, which initially permit greatly augmented function per nephron, ultimately produce a complex series of adverse effects that eventuate in progressive renal injury and an inexorable decline in function.[9]

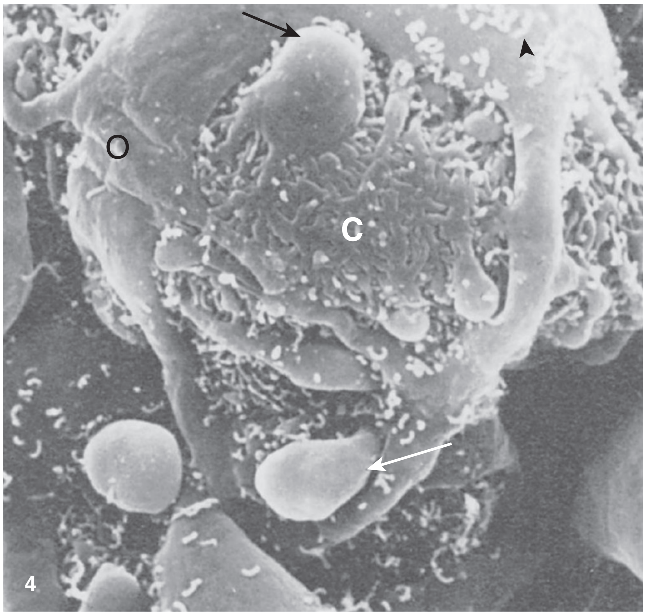

Fig. 51.5 Scanning electron micrograph of a glomerulus from a rat following 5/6 nephrectomy; view from urinary space. Cytoplasmic blebs *(arrows),* numerous microvilli *(arrowhead),* focal obliteration (O), and coarsening (C) of foot processes are seen. (×3600.) (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am. J. Physiol.* 1981;241: F85–F93.)

HEMODYNAMIC FACTORS

As early as 1 week after extensive renal mass ablation, glomerular hyperfiltration and glomerular capillary hypertension were found to be associated with morphologic changes in glomerular cells, including visceral epithelial cell cytoplasmic attenuation, protein reabsorption droplets and foot process fusion, mesangial expansion, and focal lifting of endothelial cells from the basement membrane (Figs. 51.5 and 51.6).[9] Evidence that these morphologic changes were a consequence of the glomerular hemodynamic alterations was provided by studies in rats fed a low-protein diet after 5/6 nephrectomy. This intervention prevented the hemodynamic changes, effectively normalizing Q_A, P_{GC}, and the SNGFR, and abrogated the structural lesions observed in rats on standard diet.[9] Similar findings were subsequently described in a variety of animal models of CKD, including diabetic nephropathy[253,254] and DOCA-salt hypertension.[255]

Together, these observations led Brenner and colleagues to propose that the hemodynamic adaptations following renal mass ablation ultimately prove injurious to glomeruli and initiate processes that result in glomerulosclerosis. The resulting obliteration of further glomeruli would induce hyperfiltration in remaining, less affected glomeruli, thereby establishing a vicious cycle of progressive nephron loss. These mechanisms constitute a common pathway for renal damage that could account for the inexorable progression of CKD, independent of the cause of the initial renal injury.[4] This hypothesis also explains the finding of both atrophic and hypertrophic nephrons typically encountered in chronically diseased kidneys. Further evidence supportive of the so-called hyperfiltration hypothesis was gleaned from the study of experimental diabetic nephropathy, in which glomerular hyperfiltration was also found to be a forerunner of glomerular pathology.[9,254] Maneuvers such as unilateral nephrectomy, which exacerbates hyperfiltration in the remaining kidney, were also found to exacerbate diabetic renal injury.[256] Furthermore, when the kidney was shielded from elevated perfusion pressure and from glomerular capillary hypertension by creating unilateral renal artery stenosis, the ipsilateral kidney was protected against the development of diabetic injury, which progressed unabated in the contralateral kidney.[257] In addition, when glomerular hyperfiltration was reversed in 5/6 nephrectomized rats by transplantation of an isogeneic kidney, hypertension and proteinuria were ameliorated, and glomerular injury was limited.[258]

Similarly, augmenting renal mass in the Fisher-Lewis rat transplant model normalized P_{GC} and greatly reduced the development of chronic renal allograft injury.[259,260] Similarly, it is noteworthy that the phase of transition from an acute, nonhypertensive experimental injury, induced by puromycin aminonucleoside (PAN) administration, to a chronic nephropathy characterized by proteinuria and glomerulosclerosis, is also associated with the development of glomerular capillary hypertension.[261]

Direct evidence that similar mechanisms may operate in human kidneys has been derived from a study of 14 patients with solitary kidneys who had undergone varying degrees of partial nephrectomy of the remaining kidney for malignancy.[262] Before renal-sparing surgery, proteinuria was absent in all patients. Although serum creatinine levels remained stable after an initial rise of 50% in 12 patients, the 2 patients subjected to the most extensive nephrectomy (75% and 67%, respectively) developed progressive renal failure and required long-term dialysis. Moreover, among the remaining patients, 7 developed proteinuria, the levels of which were inversely related to the amount of renal tissue preserved. Renal biopsy

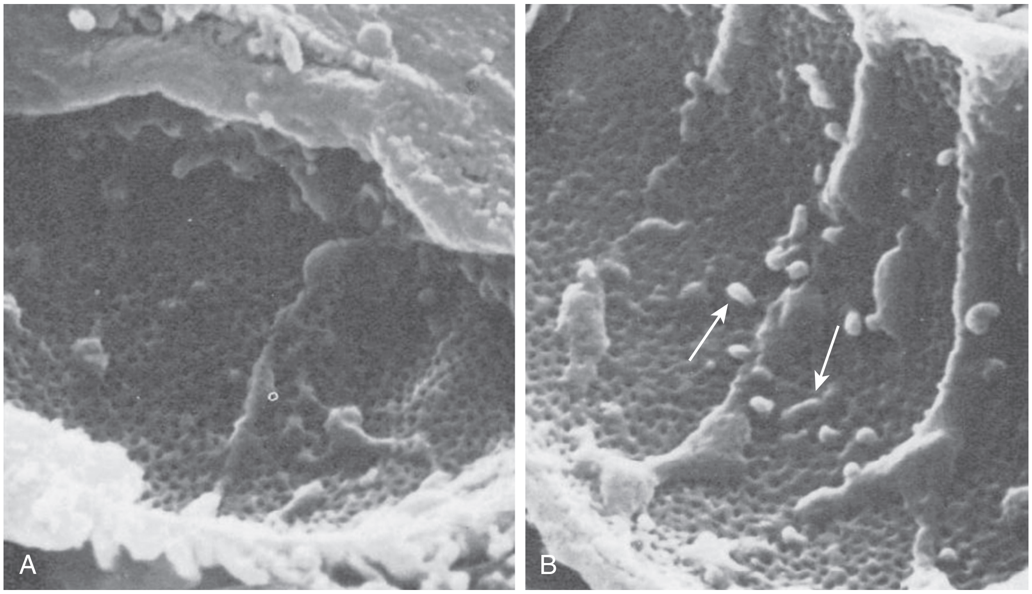

Fig. 51.6 Scanning electron micrographs of glomerular capillaries. (A) Normal endothelial appearance. (B) Rats post–5/6 nephrectomy. Scattered endothelial blebs *(arrows)* are often present in this group. (×18,000.) (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am. J. Physiol.* 1981;241:F85–F93.)

specimens in 4 patients with moderate to severe proteinuria showed focal segmental glomerulosclerosis (FSGS),[262] which later morphometric analysis revealed involvement of virtually all glomeruli examined.[263] The importance of renal mass in humans was further illustrated by an observational study of 749 patients who underwent radical nephrectomy or nephron-sparing surgery for removal of a renal mass. Those who had nephron-sparing surgery had a significantly lower incidence of reduced the GFR (16.0% vs. 44.7%) and proteinuria (13.2% vs. 22.2%) compared with those who underwent radical nephrectomy.[264] Similarly, a large metaanalysis that included data from 31,729 people who had a radical nephrectomy and 9,281 people who had a partial nephrectomy for cancer found a 61% reduction in risk of developing CKD stages 3 to 5 after partial nephrectomy.[265]

Finally, whereas unilateral donor nephrectomy is associated with an excellent prognosis in most patients, two case-controlled studies have reported a small absolute increase in the long-term risk of end-stage kidney disease (ESKD).[266,267] One detailed study in 51 living donors has reported that hypertension in kidney donors 50 years of age and older was associated with a lower estimated number of functioning nephrons per kidney; it is therefore possible that donors with a low nephron number are at increased risk of subsequent kidney damage.[268]

The importance of glomerular hemodynamic factors in the development of progressive renal injury was further illustrated by studies that reported dramatic protective effects against the development of glomerulosclerosis after chronic inhibition of the RAAS with ACE inhibitor or ARB treatment in 5/6 nephrectomized rats.[19–22] Micropuncture studies have shown that like the low-protein diet, the renoprotective effects of RAAS inhibition are associated with near-normalization of the P_{GC}, yet, in contrast with the effects of dietary protein restriction, the SNGFR remained elevated.[269] This suggested that glomerular capillary hypertension, rather than hyperfiltration per se, was the key factor in the initiation and progression of glomerular injury.

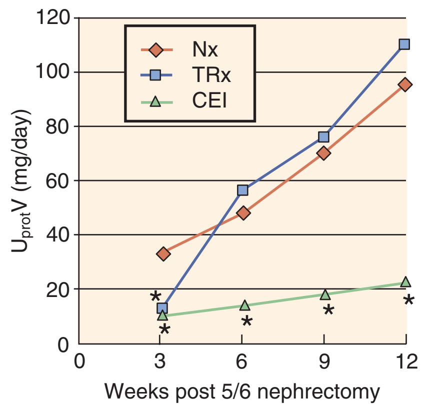

Fig. 51.7 Proteinuria levels following 5/6 nephrectomy in untreated rats *(Nx)* versus treatment with triple therapy (reserpine, hydralazine, and hydrochlorothiazide *[TRx]*), Nx + TRx, or enalapril (NX + CEI). Despite equivalent levels of blood pressure control, enalapril therapy almost completely prevented proteinuria and glomerulosclerosis, whereas triple therapy afforded no renoprotection. $U_{prot}V$, 24-hour urine protein excretion; *P < .05 vs. Nx at the same time point. (From Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest.* 1986;77:1993–2000.)

Confirmation of this view has come from an experiment in which rats were treated with a combination of reserpine, hydralazine, and hydrochlorothiazide (triple therapy) to lower arterial pressure to levels similar to those obtained with an ACE inhibitor. In contrast to the glomerular hemodynamic effects of the ACE inhibitor, triple therapy did not alleviate glomerular hypertension or proteinuria, and glomerular injury progressed[20,21] (Fig. 51.7). Interestingly, within the context of pharmacologic inhibition of the RAAS, the level to which systemic BP is reduced remains a critical determinant of the extent of the renal protection conferred.[270] The effectiveness of both ACE inhibitor and ARB in lowering glomerular pressure and ameliorating glomerular injury has since been observed in several other animal models of chronic kidney disease, including diabetic nephropathy,[253,271,272] hypertensive kidney disease,[273,274] experimental chronic renal allograft failure (a model that lacks systemic hypertension but exhibits glomerular capillary hypertension),[275–277] age-related glomerulosclerosis,[278,279] and obesity-related glomerulosclerosis.[280]

That similar mechanisms are relevant in human CKD progression has been strongly suggested by the results of clinical trials showing substantial renoprotective effects with ACE inhibitor and ARB treatment.[281–285] The importance of glomerular capillary hypertension has been further illustrated by studies of the effects of omapatrilat, a vasopeptidase inhibitor. Micropuncture studies after 5/6 nephrectomy have shown an even greater lowering of P_{GC} with omapatrilat than with ACE inhibitor treatment, despite equivalent effects on systemic BP. In subsequent chronic studies, omapatrilat produced more effective renoprotection than the ACE inhibitor.[90] Thus among the determinants of glomerular hyperfiltration, glomerular capillary hypertension has been identified as a critical factor in the initiation and progression of glomerular injury.

The development a new class of drug for the treatment of diabetes, the sodium glucose cotransporter 2 (SGLT2) inhibitors, has provided further evidence to support the importance of glomerular hemodynamic factors in CKD progression in the context of diabetes. These drugs act by inhibiting the reabsorption of filtered glucose and sodium in the proximal tubule, resulting in glycosuria that reduces hyperglycemia, and modest natriuresis. Importantly, it has been proposed that increased delivery of sodium chloride to the macula densa results in increased tubuloglomerular feedback, which reduces glomerular hyperfiltration by provoking the constriction of afferent arterioles.[286–288] In rats with streptozotocin-induced diabetes, treatment with an SGLT2 inhibitor resulted in a marked reduction in SNGFR that was inversely proportional to the distal tubular chloride concentration, indicating activation of tubuloglomerular feedback.[289] Similarly, in persons with type 1 diabetes, treatment with an SGLT2 inhibitor reduced glomerular hyperfiltration to normal levels in conjunction with a decrease in effective renal plasma flow and an increase in renal vascular resistance, observations consistent with preglomerular (afferent arteriole) vasoconstriction.[290]

Thus treatment with SGLT2 inhibitors ameliorates glomerular hyperfiltration and (by implication) glomerular hypertension in persons with diabetes. In large clinical trials in those with type 2 diabetes and the GFR greater than 30 mL/min/1.73 m^2, treatment with an SGLT2 inhibitor was associated with multiple renoprotective benefits, including reduced progression to macroalbuminuria, reduced doubling of serum creatinine levels, and a lower incidence of ESKD and death from renal causes,[291,292] confirming the importance of glomerular hyperfiltration (and glomerular hypertension) in CKD progression. Nevertheless, numerous other potential renoprotective effects of SGLT2 inhibitors have been identified, including lowering of systemic BP, reduction of albuminuria, weight loss, amelioration of arterial stiffness, reduction of renal oxygen demand, and induction of hypoxia-inducible factor 1 (HIF1).[286–288]

To investigate the potential benefit of SGLT2 inhibitors in persons with diabetes and CKD further, the CREDENCE trial enrolled 4,401 adults with type 2 diabetes, estimated GFR (eGFR) of 30 to 89 mL/min/1.73 m^2, and urine albumin-to-creatinine ratio (ACR) of 300 to 5000 mg/g, despite treatment with the maximum dose of an ACE inhibitor or ARB. In July 2018, the sponsors announced that the trial had been stopped early following an interim analysis that found benefit in the canagliflozin treatment arm with respect to the composite primary outcome of ESKD, doubling of serum creatinine levels, and cardiovascular or renal death,[293] although the results had not yet been published at the time of this writing. Further trials are now underway to evaluate the renoprotective potential of SGLT2 inhibitors across a broader spectrum of CKD, including nondiabetic CKD. Interestingly, however, treatment with an SGLT2 inhibitor (without the addition of an ACE inhibitor or ARB) did not afford renoprotection in rats after 5/6 nephrectomy.[294] The SGLT2 inhibitors are reviewed in more detail in relation to the treatment of diabetic kidney disease in Chapter 39.

MECHANISMS OF HEMODYNAMICALLY INDUCED INJURY

MECHANICAL STRESS

Several mechanisms have been proposed whereby glomerular hypertension and hyperperfusion may result in glomerular cell injury. Experiments in isolated perfused rat glomeruli have reported significant increases in glomerular volume, with increases in perfusion pressure over the normal and relevant abnormal range.[136] These increases in wall tension and glomerular volume likely result in stretching of glomerular cells. Experimental evidence has suggested that such stretching may have adverse consequences for all three major cell types in the glomerulus. Furthermore, advances in the study of cellular responses to mechanical stress raised the possibility that glomerular hyperperfusion may also promote the development of glomerulosclerosis through more subtle and complex pathways that induce profibrotic phenotypic alterations in glomerular cells.[295]

ENDOTHELIAL CELLS

The vascular endothelium serves a number of complex functions, including acting as a dynamic barrier to leukocytes and plasma proteins, secretion of vasoactive factors (prostacyclin, NO, and endothelin), conversion of Ang I to Ang II, and expression of cell adhesion molecules. It is also the first cellular structure in the kidney that encounters the mechanical forces imparted by glomerular hyperperfusion. After 5/6 nephrectomy, endothelial cells are activated or injured, resulting in detachment and exposure of the basement membrane (Fig. 51.6). This in turn may induce platelet aggregation, deposition of fibrin, and intracapillary microthrombus formation.[9,296] It has been recognized for some time that segmental glomerulosclerosis is associated with focal obliteration of capillary loops[297] and that interstitial fibrosis is associated with a loss of peritubular capillaries.[298] Furthermore, it has been shown that this loss of capillaries in the remnant kidney is associated with a decrease in endothelial cell proliferation and reduced constitutive expression of VEGF by podocytes and renal tubule cells, as well as increased expression of the antiangiogenic factor, thrombospondin-1, by the renal interstitium.[299]

Because VEGF is an important endothelial cell angiogenic, survival, and trophic factor, these findings suggest that capillary loss may be caused in part by failure of recovery from hemodynamically mediated endothelial cell injury. Indeed inhibition of endothelial and mesangial cell proliferation by treatment with everolimus after 5/6 nephrectomy resulted in increased proteinuria, glomerulosclerosis, and interstitial fibrosis associated with reduced glomerular expression of VEGF.[300] Furthermore, short-term treatment of rats with VEGF ameliorated both glomerular and peritubular capillary loss after 5/6 nephrectomy.[301] This preservation of capillaries was associated with a trend toward less glomerulosclerosis and significantly less interstitial deposition of type III collagen, as well as better preservation of renal function. Further evidence of the importance of endothelial cells in the preservation of function after nephron loss has been provided by experiments in which bone marrow–derived endothelial progenitor cells (EPCs) were administered to mice after 5/6 nephrectomy. EPC-treated mice showed better preservation of renal function, less proteinuria, and relatively preserved renal structure, as well as decreased expression of proinflammatory molecules and restored levels of the angiogenic molecules VEGF, VEGF receptor-2, and thrombospondin 1.[302] Long-term studies are required to evaluate further the potential benefits of improving renal angiogenesis in the setting of progressive renal injury.

Endothelial cells bear numerous receptors that allow them to detect and respond to changes in mechanical forces. Thus exposure of endothelial cells to changes in shear stress, cyclic stretch, or pulsatile barostress that result from glomerular hyperperfusion may induce changes in the expression of genes involved in inflammation, cell cycle control, apoptosis, thrombosis, and oxidative stress.[303] The in vitro responses of endothelial cells to mechanical forces have largely been studied in the context of vascular remodeling and atherosclerosis, but it can readily be appreciated that similar responses may affect the development of inflammation and fibrosis in the remnant kidney. Of particular interest are observations that shear stress can stimulate endothelial expression of adhesion molecules[304] and proinflammatory cytokines.[305] It is clear therefore why biomechanical activation has emerged as an important paradigm in endothelial cell biology,[306] but further studies focusing on glomerular endothelial responses to mechanical stress are required to elucidate the role of these mechanisms in progressive renal injury.

MESANGIAL CELLS

Mesangial cells are closely associated with the capillaries in the glomerulus and are therefore also exposed to mechanical forces. Evidence from in vitro studies has indicated that mesangial cells respond to changes in these mechanical forces in ways that may promote inflammation and fibrosis. Subjecting mesangial cells to cyclical stretch or strain has been shown to induce the proliferation[307] and synthesis of extracellular matrix constituents.[308,309] Cyclic stretch also activates the transcription factor nuclear factor-kappa B (NF-κB),[310] stimulates protein (monocyte chemoattractant protein 1 [MCP-1]),[311] and TGF-β[312] and its receptor,[313] as well as connective tissue growth factor (CTGF).[314] Cyclic stretch also activates the RAAS in cultured mesangial cells,[315] and Ang II in turn may induce TGF-β synthesis.[316] In vitro studies have identified several signaling pathways responsible for stretch-induced signal transduction in mesangial cells.

Cyclic stretch activates the serine-threonine kinase Akt through a mechanism requiring phosphatidylinositol-3-kinase and transactivation of EGF receptor and is necessary for the increased synthesis of collagen type 1A1 observed in the mesangial cells. Furthermore Akt activation was observed in remnant kidney glomeruli, indicating that these mechanisms are also present in vivo.[317] The actin cytoskeleton is an important transmitter of mechanical signals, and the GTPase, RhoA, a central regulator of the cytoskeleton, is activated by cyclic stretch. Stretch-induced activation of the mitogen-activated protein kinase Erk, linked to increased matrix production, is dependent on the activation of RhoA.[318] Mesangial cells cultured at ambient pressures of 50 to 60 mm Hg (i.e., levels corresponding to glomerular capillary hypertension) also show enhanced synthesis and secretion of extracellular matrix when compared with cells grown at “normal” pressures of 40 to 50 mm Hg.[319] Exposure of mesangial cells to barostress, achieved by culture under increased barometric pressure, stimulates the expression of cytokines, including platelet-derived growth factor (PDGF)-B[320] and MCP-1.[321] Transduction of mechanical forces by mesangial cells has been associated with tyrosine phosphorylation[322] and protein kinase C–induced increases in S-6 kinase activity.[323]

PODOCYTES

A growing body of evidence attests to the importance of podocyte injury in a variety of renal diseases and in CKD progression.[324] Podocytes display morphologic evidence of injury as early as 1 week after 5/6 nephrectomy (see Fig. 51.5)[9] and 6 months after uninephrectomy.[325] Increased numbers of podocytes have been observed in the urine from rats after 5/6 nephrectomy and in human CKD.[324] In 5/6 nephrectomized rats, the number of podocytes correlated with the severity of proteinuria, as well as mean arterial BP, suggesting that podocyte loss may contribute to CKD progression.[326] The importance of podocyte injury in CKD progression is further supported by the observation that amelioration of glomerular damage in 5/6 nephrectomized rats treated with 1,25-dihydroxyvitamin D3 is associated with preservation of podocyte number, as well as prevention of podocyte hypertrophy and injury.[327] Detailed in vitro studies have shown that early podocyte injury is associated with dysregulation of calcium homeostasis and disruption of the actomyosin contractile apparatus. The Rho family of small guanosine triphosphatases plays a key role in this process. Calcium influx via transient receptor potential canonical type 5 (TRPC5) channels results in the activation of Rac1 and is associated with increased podocyte migration and proteinuria, whereas calcium influx via TRCP type 6 activates RhoA, resulting in the preservation of stress fibres, prevention of podocyte migration, and maintenance of the filtration barrier.[328] However, subsequent experiments in TRCP6-deficient podocytes have found that TRCP6 is not required for mechanotransduction.[329] Rather, mechanical stretch was found to induce ATP release from podocytes by a process dependent on cholesterol and podocin. ATP release in turn activated the purinergic channel $P2X_4$, resulting in an influx of calcium and increased disorganization of the actin cytoskeleton.[329]

Thus purinergic channels appear to be key mechanotransducers in podocytes. Because podocytes are attached to the outer aspect of the glomerular basement membrane, it is reasonable to expect that they would be exposed to increased mechanical forces resulting from glomerular hypertension. Confirmation that podocytes respond to such physical forces has been derived from several in vitro experiments that examined podocyte responses to stretching. Activation of a voltage-sensitive potassium channels was observed in response to stretching of the podocyte cell membrane,[330] and culture of podocytes under constant stretch induced activation of the protein kinases Erk1/2 and JNK via the EGF receptor, as well as other changes in cell signaling.[331] Mechanical stretch inhibited podocyte proliferation[332] and, in common with TGF-β, reduced $\alpha_3\beta_1$-integrin expression and podocyte adhesion in vitro.[333] Exposure to cyclic stretching that mimics pulsatile strain within the glomerulus has been shown to cause reorganization of the actin cytoskeleton,[334] upregulation of COX-2, and E prostanoid (EP)-4 receptor expression,[335] as well as podocyte hypertrophy.[336] Subsequent experiments using mice with podocyte-specific depletion or overexpression of EP4 receptors have found that prostaglandin E2 acting via EP4 receptors contributes to the development of proteinuria after 5/6 nephrectomy and therefore probably contributes to podocyte injury.[337]

In another experiment, cyclic stretching of podocytes was associated with increased production of Ang II and TGF-β, as well as upregulation of angiotensin subtype 1 (AT1) receptors, resulting in increased Ang II–dependent apoptosis.[338] Cyclic stretch of podocytes also resulted in a 50% reduction of nephrin (a key component of the slit diaphragm) mRNA and protein levels via an Ang II–dependent mechanism that was inhibited by the peroxisome proliferator-activated receptor-gamma (PPAR-γ) agonist, rosiglitazone, which prevented AT1 receptor upregulation.[339] Taken together, these data suggest that stretch-induced podocyte injury is a further mechanism whereby glomerular hypertension contributes to glomerular injury.

CELLULAR INFILTRATION IN REMNANT KIDNEYS

Despite the lack of an obvious immune stimulus, an inflammatory cell infiltrate composed predominantly of macrophages and smaller numbers of lymphocytes is observed in remnant kidneys after 5/6 nephrectomy.[340] Interestingly, similar observations have been reported in a rat model of spontaneous renal agenesis associated with a 60.2% reduction in nephron endowment.[341] As discussed earlier, it is possible that the glomerular hemodynamic adaptations to nephron loss may provoke an inflammatory cell response through the effects of mechanical forces on endothelial and mesangial cells. Thus upregulation of renal endothelial adhesion molecules may facilitate the egress of leukocytes from the circulation into the mesangium, where they may participate in further renal injury. The recruited cellular infiltrate may constitute an abundant source of potent pleiotropic cytokine products, which in turn influence other infiltrating leukocytes, dendritic cells, and kidney cells, stimulating cell proliferation, elaboration of extracellular matrix components, and increased endothelial adhesiveness.[342]

Evidence is emerging that these proposed mechanisms, based largely on in vitro observations, are indeed relevant in vivo. In the two-kidney, one-clip model of renovascular hypertension, upregulated expression of adhesion molecules and TGF-β, as well as cell infiltration, is observed only in the nonclipped kidney that is exposed to the hypertensive perfusion pressure.[343,344] In the 5/6 nephrectomy model, coordinated upregulation of a variety of cell adhesion molecules, cytokines, and growth factors in association with macrophage infiltration has been observed at time points that precede the development of severe glomerulosclerosis.[345,346] Furthermore, the renoprotection afforded by an ACE inhibitor or ARB treatment in this model was associated with inhibition of cytokine upregulation and prevention of renal infiltration by macrophages.[346,347]

Infiltrating macrophages, although present in the glomeruli of remnant kidney, are chiefly distributed in the tubulointerstitial regions,[340,346] suggesting that they play a role in the development of the TIF that accompanies glomerulosclerosis. Further analysis of the cellular infiltrate has also identified mast cells in close proximity to areas of TIF.[348] It is possible that interstitial infiltrates are recruited as the result of tubulointerstitial cell activation by the downstream effects of cytokines released in the glomeruli. Alternatively, it has been proposed that excessive uptake of filtered proteins by tubule epithelial cells stimulates expression of cell adhesion and chemoattractant molecules that recruit macrophages and other monocytic cells to tubulointerstitial areas[349] (see later for further discussion). The chemokine receptor CCR-1 has been shown to be important in interstitial but not glomerular recruitment of leukocytes. Treatment with a nonpeptide CCR-1 antagonist has been shown to reduce interstitial macrophage infiltration and ameliorate interstitial fibrosis in the unilateral ureteric obstruction (UUO) model, but data are still lacking in the 5/6 nephrectomy model.[350] Furthermore, antagonism of MCP-1 signaling through gene therapy—induced production of a mutant form of MCP-1 by skeletal muscle resulted in reduced interstitial macrophage infiltration and amelioration of interstitial fibrosis in mice after UUO.[351]

The identification of renal tubule cells expressing alpha smooth muscle actin after 5/6 nephrectomy has raised the possibility that tubule cells may undergo transdifferentiation to a myofibroblast phenotype that contributes to interstitial fibrosis.[352] Furthermore the renoprotection observed with mycophenolate treatment in 5/6 nephrectomized rats is associated with reductions in interstitial myofibroblast infiltration and collagen type III deposition.[353] There is, however, ongoing debate regarding the origin of interstitial myofibroblasts in CKD. Fate-mapping studies have indicated that mesenchymal cells called *pericytes* are the predominant source of myofibroblasts, and that epithelial to mesenchymal transdifferentiation is not a source of myofibroblasts or accounts for only a small minority.[354] Other cells that may give rise to interstitial myofibroblasts include resident fibroblasts, endothelial cells, and bone marrow–derived cells.[355,356]

Several lines of evidence have suggested that this cellular infiltrate contributes to renal injury and is not merely a consequence of it.[357] In one study, multiple linear regression analysis identified glomerular macrophage infiltration in the remnant kidney as a major determinant of mesangial matrix expansion and adhesion formation between Bowman's capsule and glomerular tufts.[340] Furthermore, depletion of leukocytes by irradiation in rats delayed the onset of glomerular injury after renal ablative surgery.[358] Several studies have reported amelioration of the cellular infiltrate and renal injury in the 5/6 nephrectomy model following treatment with the immunosuppressive agent mycofenolate mofetil.[359–362] One study has found that mycophenolate also lowers P_{GC}, which may account for some of its renoprotective effects.[363] Several other antiinflammatory interventions have been shown to ameliorate renal injury after 5/6 nephrectomy. Treatment with the antiinflammatory agent nitroflurbiprofen, also a NO donor, was associated with moderate renoprotection.[364] Rats treated with a PPAR-γ receptor agonist evidenced significant attenuation of the proteinuria and glomerulosclerosis observed in untreated rats, despite the failure of treatment to lower BP.[365] This renoprotection was observed in association with marked reductions in glomerular cell proliferation, glomerular macrophage infiltration, and renal expression of plasminogen activator inhibitor-1 (PAI-1), as well as TGF-β. The authors speculated that some of these effects may have resulted from the known actions of PPAR-γ receptor activation to antagonize the activities of the transcription factors activator protein-1 (AP-1) and NF-κB.

Administration of a sphingosine-1-phosphate (S1P) receptor agonist, a novel immunosuppressant that inhibits the egress of T cells and B cells from lymph nodes, attenuated the increase in chemokine receptor expression observed in untreated rats and tended to normalize regulated on activation, normal T cell–expressed and secreted (RANTES) and MCP-1 gene expression.[366] Glomerular and interstitial inflammation, as well as fibrosis, were also reduced. Overexpression of the antiinflammatory cytokine interleukin-10 (IL-10) in rats was associated with reduced interstitial inflammation and lower levels of MCP-1, RANTES, interferon-γ (IFN-γ), and IL-2 expression, as well as attenuation of proteinuria, glomerulosclerosis, and TIF.[367] Finally, overexpression of the gene for angiostatin, an antiangiogenic factor that also inhibits leukocyte recruitment as well as neutrophil and macrophage migration, was associated with the inhibition of macrophage and T cell infiltrates in glomeruli, as well as the interstitium, reduced MCP-1 expression, and attenuation of glomerulosclerosis and interstitial fibrosis.[368] Taken together, these findings strongly support the hypothesis that in addition to direct glomerular cell injury, glomerular hemodynamic adaptations to nephron loss provoke a complex series of proinflammatory and profibrotic responses that further contribute to renal damage. Treatments that antagonize the mediators of these responses may therefore be of benefit in slowing the rate of CKD progression.

NONHEMODYNAMIC FACTORS IN THE DEVELOPMENT OF NEPHRON INJURY FOLLOWING EXTENSIVE RENAL MASS ABLATION

The weight of evidence in support of the hypothesis that glomerular hemodynamic adaptations are central to progressive renal injury does not exclude the possibility that the kidney may also be affected by a variety of factors not directly attributable to hemodynamic changes. These nonhemodynamic factors have been extensively studied in recent years and may offer new therapeutic targets for future renoprotective interventions.

PROTEINURIA

> ***Clinical Relevance***
> **Proteinuria**
> Reduction of proteinuria should be regarded as a key therapeutic goal. Specifically, the dose of RAAS inhibitor treatment should be titrated up until proteinuria has been minimized or the maximum dose has been reached. If required, additional antihypertensive drugs should be added to RAAS inhibitor treatment to achieve blood pressure targets and minimize proteinuria.

Abnormal excretion of protein in the urine is the hallmark of experimental and clinical glomerular disease. Whereas immune complex deposition and resulting inflammation account for abnormal permeability of the glomerular filtration barrier to proteins in glomerulonephritis, studies in rats subjected to extensive renal ablation have shown loss of glomerular barrier function to proteins of similar molecular size, yet in the apparent absence of primary immune-mediated renal injury or inflammatory response. Sieving studies using dextrans and other macromolecules in rats 7 or 14 days after 5/6 nephrectomy have revealed the loss of both size and charge selectivity of the glomerular filtration barrier. Ultrastructural examination of the remnant kidneys has revealed detachment of glomerular endothelial cells and visceral epithelial cells from the glomerular basement membrane. In addition, protein reabsorption droplets and attenuation of cytoplasm resulting in bleb formation was observed in podocytes. The authors concluded that the altered permselectivity may be caused in part by the separation of endothelial cells from the glomerular basement membrane, allowing access of macromolecules and, in part, to loss of anionic sites in the lamina rara externa, resulting in loss of charge selectivity and detachment of podocytes.[369]

Studies have identified decreased nephrin expression in podocytes as a further mechanism contributing to proteinuria after 5/6 nephrectomy,[370] and in vitro studies have reported a 50% reduction in nephrin expression when podocytes were exposed to cyclic stretching.[339] A direct role for Ang II in modulating glomerular capillary permselectivity has been suggested by the observation of marked increases in urinary protein excretion during infusion of Ang II in normal rats. Although some investigators have attributed this to a direct effect of Ang II on the cellular components of the glomerular filtration barrier, resulting in the opening of interendothelial junctions and epithelial cell disruption, others have shown that the increase in proteinuria may be accounted for almost completely by the associated hemodynamic changes, principally a reduction in Q_A and an increase in filtration fraction.[371] On the other hand, the notion that Ang II may mediate changes in glomerular permselectivity, independent of its effects on glomerular hemodynamics, has been supported by studies in an isolated perfused rat kidney preparation in which infusion of Ang II augmented urinary protein excretion and enhanced the clearance of tracer macromolecules, independently of any change in filtration fraction.[372] Furthermore, Ang II and aldosterone have been shown to reduce nephrin expression in podocytes and may therefore directly affect glomerular permselectivity.[339,373,374]

Proteinuria, long considered simply a marker of glomerular injury, has also been implicated as an effector of injury processes involved in renal disease progression, especially those resulting in tubulointerstitial fibrosis.[375] In rats with aminonucleoside-induced nephrotic syndrome, the proteinuric phase of the disease was associated with an acute interstitial nephritis, the intensity of which correlated closely with the severity of the proteinuria.[349] Furthermore, in an overload proteinuria model induced by daily intraperitoneal administration of bovine serum albumin to uninephrectomized rats, proximal tubule cell injury and interstitial infiltration of macrophages and lymphocytes were evident after 1 week.[376] The severity of the proteinuria showed a positive correlation with the intensity of the infiltrate. At 4 weeks, focal areas of chronic interstitial inflammation were noted.[376] Other experiments have identified mast cells as a component of the inflammatory infiltrate observed after protein overload. The number of mast cells correlated with the severity of interstitial inflammation, as well as with levels of stem cell factor (SCF) and TGF-β.[377] A causative association between excessive proteinuria and interstitial inflammation has been suggested by in vitro studies of proximal tubule epithelial cells cultured in media supplemented with high concentrations of albumin, immunoglobulin G (IgG), or transferrin. Cellular uptake of these proteins by endocytosis was observed to increase secretion of ET-1,[378] MCP-1,[379] RANTES,[380] IL-8,[381] and fractaline.[382] Electrophoretic mobility shift assay of cell nucleus extracts has revealed intense activation of the transcription factor NF-κB that was dependent on the concentration of protein in the medium.[380] Furthermore, the liberation of these molecules was noted to be predominantly from the basolateral aspect of the cells. This would be in keeping with secretion into the renal interstitium in vivo, thereby contributing to the development of tubulointerstitial inflammation and fibrosis.

On the other hand, secretion of TGF-β by tubule cells in response to albumin was not inhibited by inhibitors of endocytosis, implying that a different mechanism must be responsible.[383] Exposure of tubule cells to albumin has also been shown to result in increased levels of intracellular reactive oxygen species and activation of the signal transducer and activator of transcription (STAT) signaling pathway.[384] The STAT pathway, in turn, mediates a variety of cellular responses, including proliferation and induction of cytokines, as well as growth factors. Preliminary evidence has suggested that exposure of tubule cells to albumin may also induce apoptosis.[385]

Other experiments have found apoptosis in tubule cells exposed to high-molecular-weight plasma proteins but not smaller proteins.[386] Albumin and transferrin exposure also induced complement activation in tubule cells and reduced binding of factor H, a natural inhibitor of the alternative complement pathway.[387] The involvement of immune cells in the processing of absorbed proteins has been shown in studies in which tubule cells were found to cleave albumin into an N-terminal, 24–amino acid peptide that was further processed by dendritic cells (DCs) into antigenic peptides, with binding sites for major histocompatibility complex (MCH) class I molecules that were capable of activating CD8+ T cells.[388] After 5/6 nephrectomy, DCs were found in the interstitium, peaking at 1 week and decreasing at 4 weeks, coinciding with their appearance in renal lymph nodes. DCs from the lymph nodes were able to activate CD8+ T cells in culture.

Despite the previous evidence, other investigators have raised concerns regarding the interpretation of these observations.[389] They pointed out that the concentrations of plasma proteins used in vitro were nonphysiologic and far exceeded those observed in proximal tubule fluid from experimental models of nephrotic syndrome. Furthermore, many of the experiments were performed in cells that were routinely cultured in the presence of high concentrations of protein (serum) that could significantly alter their phenotype. Not all investigators have been able to confirm these observations, however. In particular, some have found proliferative or profibrotic responses when proximal tubule cells are exposed to serum or serum fractions, but no response after exposure to purified forms of albumin or transferrin, suggesting that factors other than albumin or transferrin may be involved.[390,391] Furthermore experiments in mice with tubules deficient in megalin, the key molecule responsible for the endocytosis of filtered proteins by tubule cells, have found that tubulointerstitial inflammation is not inhibited in the absence of megalin, implying that it is not dependent on endocytosis.[392] It was proposed instead that tubulointerstitial inflammation is provoked by misdirection of protein-rich glomerular filtrate into the interstitium due to the formation of adhesions between the glomerular tuft and Bowman's capsule or, in the case of crescentic glomerulonephritis, encroachment of the crescent onto the proximal tubule, resulting in occlusion and tubule degeneration.[393] Moreover, it was proposed that the increase in cytokine and adhesion molecule production observed in tubule cells that have endocytosed excess protein may actually be a protective response.[392]

Several lines of evidence have suggested that filtered molecules other than albumin or immunoglobulin may play a role in the progression of chronic nephropathies. It has been proposed that free fatty acids (FFAs) bound to albumin may play an important role in provoking a proinflammatory response in tubule cells. In one experiment, albumin-bound fatty acids stimulated macrophage chemotactic activity, whereas delipidated albumin did not.[394] Albumin-bound FFAs have also been shown to activate PPAR-γ and induce apoptosis in proximal tubule cells.[395] High-density lipoprotein (HDL) and low-density lipoprotein (LDL) have been identified in the urine, renal interstitium, and tubule cells in renal biopsies of patients with nephrotic syndrome. In vitro, cultured human proximal TECs take up LDL and HDL.[396]

Oxidized LDL may cause tubular cell injury, and exposure of TECs to HDL is associated with increased synthesis of ET-1.[396,397] A role has also been proposed for other compounds bound to filtered proteins, such as IGF-1, which has been detected in increased amounts in the proximal tubular fluid of rats with adriamycin nephrosis.[398] Proximal TECs cultured in the presence of proximal tubular fluid from nephrotic rats exhibit enhanced cell proliferation and increased secretion of types I and IV collagen. Both effects were inhibited by neutralizing IGF-1 receptor antibodies. Other growth factors in plasma, including HGF and TFG-β, may also appear in glomerular ultrafiltrates with proteinuria and exert effects on tubule cells.[399] Furthermore, cytokines produced in injured glomeruli may have downstream proinflammatory effects. Whereas complement components are normally absent from tubular fluid, C3 and C5b-9 neoantigen were observed along the luminal border of TECs in the protein overload proteinuria model.

To examine the role of filtered complement in renal injury, rats with puromycin aminonucleoside nephrosis were subjected to complement depletion with cobra venom factor or inhibition of complement activation by the administration of soluble recombinant human complement receptor type 1, before the onset of proteinuria.[400] In control rats, proximal tubular degeneration, interstitial leukocyte infiltrate, and renal impairment (as assessed by inulin and *para*-aminohippurate [PAH] clearances) occurred at 7 days, together with positive staining for C3 and C5b-9 along the proximal tubule brush border. Both interventions were associated with significantly less tubulointerstitial pathology and greater clearance of PAH but not inulin; the severity of the proteinuria was unaffected, suggesting that filtered complement plays a significant role in the tubulointerstitial injury associated with proteinuria. A more selective approach, using recombinant complement inhibitory molecules targeted to proximal tubule cells with carrier antibodies to brush border antigen, resulted in a significant reduction of interstitial fibrosis in the same model.[401] Similarly, mice with C3 deficiency were protected against interstitial inflammation after protein overload, and wild-type kidneys transplanted into C3-deficient mice were protected, whereas kidneys from C3-deficient mice were not protected when transplanted into wild-type mice. This implies that filtered rather than locally synthesized C3 is important in the pathogenesis of interstitial inflammation associated with proteinuria.[402]

In experimental models of proteinuric renal disease, filtered proteins have also been found to accumulate in the glomerular mesangium[369] and may therefore contribute to glomerular a tubulointerstitial injury. Further support for this notion has been obtained from a metaanalysis of 57 studies of experimental CKD, which found a consistent positive correlation between the severity of proteinuria and the extent of glomerulosclerosis.[403] Lipoproteins, in particular, accumulate in the glomeruli of patients with glomerulonephritis.[404,405] Furthermore, LDL stimulates mesangial cells to proliferate in vitro[406,407] and enhances mesangial cell synthesis of the extracellular matrix protein fibronectin.[408] LDL exposure is also associated with increased mesangial cell mRNA levels for MCP-1[408] and PDGF.[407] Oxidation of LDL by mesangial cells or macrophages may enhance its toxicity.[406] Thus accumulation of proteins in the mesangium may stimulate a number of different mechanisms that contribute to glomerulosclerosis.

The relevance of these findings to the processes occurring in vivo has been borne out by studies in rats. In the protein overload model, the development of proteinuria at 1 week was associated with significant increases in TGF-β at both protein and mRNA levels in interstitial and proximal tubule cells.[376] Similarly, renal cortical mRNA levels encoding the macrophage chemoattractant, osteopontin, were increased on day 4, and immunofluorescence localized increased osteopontin staining to cortical tubules at day 7. MCP-1 and osteopontin mRNA and protein levels were elevated at 2 and 3 weeks.

Furthermore, a significant effect of proteinuria on molecules involved in extracellular matrix (ECM) protein turnover was observed. Although mRNA levels for various renal matrix proteins were variable, staining for the proteins in the cortical interstitium increased progressively. Levels of mRNA for the protease inhibitors PAI-1 and tissue inhibitor of metalloproteinases-1 (TIMP-1) were elevated at 2 weeks, at which time significant renal fibrosis was present.[376] Gene expression profiling has identified over 100 genes that are upregulated in the proximal tubule cells of mice exposed to overload proteinuria.[409] Consistent with the hypothesis that protein-bound FFAs are important, rats receiving FFA-replete bovine serum albumin (BSA) developed more severe tubulointerstitial injury and more extensive macrophage infiltration than those receiving FFA-depleted BSA.[410,411]

In other models of proteinuric renal disease, including 5/6 nephrectomy and passive Heymann nephritis, accumulation of albumin and IgG by proximal tubule cells occurred before infiltration of the interstitium by macrophages and MHC-II positive mononuclear cells.[412] The infiltrates localized to areas where proximal tubule cells stained positive for intracellular IgG or where luminal casts were present. Furthermore, proximal tubule cells that stained positive for IgG also showed evidence of increased osteopontin production. The IgG staining in proximal tubule cells was subsequently associated with the peritubular accumulation of macrophages and alpha smooth muscle actin-positive cells, as well as upregulation of TFG-β mRNA in the tubular and infiltrating cells.[413]

The importance of inflammatory factors in the development of interstitial fibrosis is illustrated by the observation that treatment of rats with experimental membranous nephropathy with rapamycin is associated with reduced expression of profibrotic and proinflammatory genes, as well as amelioration of interstitial inflammation and fibrosis.[414] Further studies in the 5/6 nephrectomy model have suggested that tubulointerstitial injury may play an important role in the decline of the GFR, especially in the late stages of progressive renal injury.[415] By examining serial sections of remnant kidneys, the investigators have shown that in association with a doubling in serum creatinine levels, there was a substantial increase in the proportion of glomeruli no longer connected to tubules (atubular glomeruli) or to atrophic tubules. Most of these glomeruli were not globally sclerosed, implying that the tubular injury was responsible for the final loss of function in these nephrons. The authors speculated that the absorption of excess filtered protein may play an important role in this tubular injury.[415] Finally, evidence has been accumulating for the role of proteinuria in the development of interstitial damage in human CKD. Among 215 patients with CKD, the urine ACR correlated with urinary MCP-1 levels and interstitial macrophage numbers.[416] Furthermore, urine ACR and interstitial macrophage numbers independently predicted renal survival.

Establishing a cause and effect relationship between proteinuria and renal damage in humans is difficult but several clinical studies have provided evidence to support this. A metaanalysis of 17 clinical studies of CKD has revealed a positive correlation between the severity of proteinuria and the extent of biopsy-proven glomerulosclerosis,[403] and data from a metaanalysis that included over 1 million participants identified albuminuria as a strong independent risk factor for progression to ESKD.[417] Observations from the Modification of Diet in Renal Disease (MDRD) trial have also suggested that proteinuria is an independent determinant of CKD progression. Greater levels of baseline proteinuria were strongly associated with more rapid declines in the GFR and reduction of proteinuria, independently of reduction in BP, was associated with lesser rates of decline in the GFR. Furthermore, the degree of benefit achieved by lowering BP below usual target levels was highly dependent on the level of baseline proteinuria.[418]

The severity of proteinuria at baseline has been shown to be the most important independent predictor of renal outcomes in randomized trials of ACE inhibitor or ARB treatment in those with diabetic nephropathy[419] and nondiabetic CKD.[282] Furthermore the percentage reduction in proteinuria over the first 3 to 6 months, as well as the absolute level of proteinuria at 3 or 6 months, are strong independent predictors of the subsequent rate of decline in the GFR among patients with diabetic nephropathy[419] and nondiabetic CKD.[420] A metaanalysis that included data from 1860 patients with nondiabetic CKD has confirmed these findings and has shown that during antihypertensive treatment, the current level of proteinuria is a powerful predictor of the combined endpoint of doubling of baseline serum creatinine levels or onset of ESKD (relative risk, 5.56 for each 1.0 g/day of proteinuria).[421] Similarly a metaanalysis of 21 randomized trials of drug treatment in CKD that included 78,342 participants has found that for each 30% initial reduction in albuminuria with treatment, the risk of ESKD decreased by 23.7% (95% confidence interval [CI], 11.4%– 34.2%) independently of the class of drug used for treatment.[422]

The Renoprotection of Optimal Antiproteinuric Doses (ROAD) study has provided the most direct evidence of the clinical benefit of proteinuria reduction to date. Subjects with proteinuric CKD were randomized to standard therapy with an ACE inhibitor or ARB (separate groups) or to ACE inhibitor or ARB therapy titrated to the maximum antiproteinuric dose (two further groups). Despite comparable BP control, subjects in the groups randomized to maximum antiproteinuric doses had 51% and 53% relative risk reductions in the combined primary endpoints of creatinine level doubling, ESKD, or death.[423]

Taken together, the evidence from experimental and clinical studies provides support for the hypothesis that impaired glomerular permselectivity results in excessive filtration of proteins and/or protein-bound molecules that contribute to kidney damage, but many questions regarding the tubulotoxic potential of filtered plasma proteins and the identity of the specific molecules involved remain unanswered. Despite these uncertainties, the close association between the severity of proteinuria and renal prognosis implies that reduction of proteinuria should be regarded as an important independent therapeutic goal in clinical strategies seeking to slow the rate of progression of CKD. The mechanisms and consequences of proteinuria are discussed further in Chapter 30.

TUBULOINTERSTITIAL FIBROSIS

Together with secondary FSGS, TIF constitutes a major component of the progressive renal injury observed in CKD. TIF is characterized by inflammatory cell and fibroblast infiltration, accumulation of ECM, tubule cell loss, and rarefaction of peritubular capillaries. Fibrogenesis starts at small sites of inflammation and then expands if a profibrotic milieu persists. The inflammatory infiltrate is composed of lymphocytes, macrophages, detritic cells, and mast cells.[424] Lymphocytes are recruited early in the process, and their importance is highlighted by the protection from fibrosis observed in Rag-2 null mice, which lack B and T lymphocytes.[425] Monocytes are recruited and transdifferentiate into macrophages and fibrocytes. The profibrotic role of macrophages is illustrated by observations that the extent of macrophage accumulation correlates closely with the severity of fibrosis,[426] and macrophage depletion attenuates fibrosis.[427] Nevertheless, it has been proposed that alternatively activated M2 macrophages exert antiinflammatory actions, and the infusion of cells enriched for M2 macrophages has been shown to reduce renal fibrosis in mice.[428] Myofibroblasts are the chief source of ECM production, and the accumulation of interstitial myofibroblasts is central to the pathogenesis of TIF.

There has been considerable controversy regarding the cellular origins of interstitial myofibroblasts in TIF. Different investigators have identified resident fibroblasts,[429] transdifferentiation from tubule epithelial cells and endothelial cells,[430,431] bone marrow–derived fibrocytes,[432] and pericytes[354] as possible sources.[356] Fate-mapping studies have subsequently indicated that pericytes are the predominant source of myofibroblasts and that epithelial to mesenchymal transdifferentiation (EMT) is not a source of myofibroblasts or accounts for only a small minority.[354] Further studies have helped resolve the controversy by showing that TECs undergo partial EMT, expressing both epithelial and mesenchymal markers, but remain attached to the tubular basal membrane. Moreover, prevention of EMT substantially ameliorated renal fibrosis in animal models, confirming that partial EMT of TECs contributes to the pathogenesis of renal fibrosis.[433,434]

The excess ECM is composed largely of collagen types I and II, as well as fibronectin. Investigators have sought to identify the relative contribution of different cell types to the production of collagen I using cell type–specific knockouts. Experiments have revealed that bone marrow–derived cells (fibrocytes) contribute 38% to 50% of collagen deposition in a UUO model. Furthermore, initial collagen I synthesis was attributable to resident mesenchymal fibroblasts and was beneficial in preserving renal function in this model, whereas bone marrow–derived cells were responsible for later collagen I production that did not affect renal function.[435] TECs were found not to contribute directly to collagen I production. Bone marrow–derived cells were similarly shown to contribute substantially to collagen I production in an adenine-induced nephropathy model and, in this case, greater deposition of collagen was associated with lower the GFR at all time points.[435] Targeting of bone marrow–derived cell collagen I production therefore represents an attractive therapeutic option to ameliorate renal fibrosis.

Matrix accumulation has been proposed to commence with the appearance of collagen nucleators in the interstitial fluid that act as a scaffold for the deposition of fibrillar collagens. Fibroblasts use collagen fibrils as a scaffold to move through damaged tissue along chemoattractant gradients. Fibrogenesis is promoted by the expression of key growth factors, principally TFG-β.[436] The role of tissue proteases in TIF is complex and has not been fully characterized. Whereas matrix metalloprotease (MMP) types 2 and 9 degrade collagen type IV (and possibly type I and III) in vitro, they do not consistently abrogate TIF in vivo.[437,438] Indeed, in some models, MMPs appear to exert profibrotic effects.[438]

Rarefaction of peritubular capillaries is a hallmark of TIF. In the early stages, capillaries may be damaged by transient ischemia that promotes apoptosis.[439] Further capillary loss is attributable to an imbalance between pro- and antiangiogenic factors[440,441] or loss of peritubular endothelial cells through endothelial-mesenchymal transdifferentiation.[430] Detailed studies have identified changes in peritubular capillary endothelium in animal models of CKD, including the development of subendothelial spaces, loss of fenestrations, and generation of caveolae and vesicles, as well as increased permeability and degeneration of the microvascular tree.[442] Tissue hypoxia results from the rarefaction of peritubular capillaries and accumulation of ECM, requiring oxygen to diffuse over greater distances to reach cells. Hypoxia contributes to TIF by promoting EMT and apoptosis of tubule cells, as well as fibroblast activation and ECM production.[443,444] The importance of hypoxia in provoking interstitial (and glomerular) injury has been demonstrated by studies in which induction of hypoxia-inducible factor, a key mediator of protective responses to hypoxia, was associated with amelioration of proteinuria, glomerulosclerosis, and TIF, as well as decreased macrophage infiltration and expression of type IV collagen and osteopontin.[445,446] Tubulointerstitial inflammation and fibrosis are discussed in more detail in Chapter 35.

MOLECULAR MEDIATORS OF RENAL FIBROSIS

TGF-β is associated with chronic fibrotic states throughout the body and is the central mediator of renal fibrosis.[447] Three isoforms of TGF-β have been described in mammals, but the focus of most research has been on TGF-β1. The active form of TGF-β is a dimer that binds to a transmembrane receptor, which is a heterodimer comprised of TGF-β receptor types I and II. Intracellular signaling occurs via Smad-dependent and Smad-independent pathways. Smad2 and Smad3 have been identified as the most important intracellular mediators of the actions of TGF-β, with Smad3 having profibrotic effects and Smad2 proposed to have counter-regulatory antifibrotic effects. In response to TGF-β receptor signaling, Smad2 and Smad3 are phosphorylated and form a complex with Smad4 that translocates into the nucleus and binds to DNA to modulate the transcription of multiple target genes. Smad3 has been shown to promote transcription of multiple genes involved in fibrosis, including PAI-1, proteoglycans, integrins, CTGF, TIMP-1, and collagen types 1, 5, and 6. SMAD 3 also plays a role in EMT.[448] In addition, TGF-β/Smad3 modulates the production of micro RNA molecules that promote fibrosis; levels of profibrotic miR-21, miR-192, and miR-433 are increased, whereas antifibrotic miR-29 and miR-200 are decreased.[449] Smad4 is a key modulator of the profibrotic actions of TGF-β and reduces the profibrotic actions of Smad3 by inhibiting its binding to the promoter regions of profibrotic genes. Smad7 is an important negative feedback regulator of TGF-β signaling, which acts by preventing the recruitment and phosphorylation of Smad2 and Smad3. Smad7 also increases the expression of inhibitor of kappa B (IκBα), an inhibitor of NF-κB, and may therefore be important in mediating the antiinflammatory actions of TGF-β. Finally, Smad signaling may also be activated by TGF-β–independent mechanisms that are relevant to CKD progression, including Ang II and advanced glycation end products (AGEs).[448]

In vitro TGF-β elicits overproduction of ECM constituents by mesangial cells, and its expression is increased in several experimental models of renal disease. These include diabetic nephropathy,[450] anti–Thy-1 glomerulonephritis,[451] adriamycin-induced nephropathy,[452] and chronic allograft nephropathy,[453] as well as human glomerulonephritis,[454,455] HIV nephropathy,[456] diabetic nephropathy,[457] and chronic allograft nephropathy.[458] The role of TGF-β in renal fibrosis has been further illustrated by experiments in which transfection of the gene for TGF-β into one renal artery produced ipsilateral renal fibrosis.[459] In 5/6 nephrectomized rats, a two- to threefold increase in remnant kidney mRNA levels for TGF-β was observed, and in situ hybridization revealed elevations in TGF-β mRNA throughout the glomeruli, tubules, and interstitium. Treatment with an ACE inhibitor or ARB resulted in substantial renal protection and prevented upregulation of TGF-β.[346,347] Furthermore, in rats treated with an ACE inhibitor or ARB, the extent of glomerulosclerosis correlated closely with remnant kidney TGF-β mRNA levels.[270]

Several interventions that inhibit the effects of TGF-β have been shown to afford renoprotection in animal models of renal disease. Transfection of the gene for decorin, a naturally occurring inhibitor of TGF-β, into skeletal muscle limited the progression of renal injury in anti–Thy-1 glomerulonephritis.[460] Administration of anti–TGF-β antibodies to salt-loaded, Dahl salt–sensitive rats ameliorated the hypertension, proteinuria, glomerulosclerosis, and interstitial fibrosis typical of this model.[461] Treatment with tranilast (N-[3′,4′-dimethoxycinnamoyl]-anthranilic acid; Pharm Chemical, Shanghai Lansheng, Shanghai, China), an inhibitor of TGF-β–induced ECM production, significantly reduced albuminuria, macrophage infiltration, glomerulosclerosis, and interstitial fibrosis in 5/6 nephrectomized rats.[462] Transfer of an inducible gene for Smad 7, which blocks TGF-β signaling by inhibiting Smad 2/3 activation, inhibited proteinuria, fibrosis, and myofibroblast accumulation after 5/6 nephrectomy.[463] Two weeks of treatment with a polyamide compound designed to suppress transcription of the TGF-β gene significantly reduced proteinuria and prevented upregulation of TGF-β, CTGF, collagen type I α1, and fibronectin mRNA in the renal cortex. This also suppressed urinary TGF-β excretion and staining for TGF-β by immunofluorescence in salt-loaded, Dhal salt–sensitive rats.[464] Another fibrogenic molecule, CTGF, has also been observed to be overexpressed in kidney biopsies from patients with a variety of renal diseases.[465] The specific induction of CTGF expression by exogenous TGF-β in mesangial cells[314,466] and fibroblasts,[467] together with the finding that blocking antibodies to TGF-β inhibits increased CTGF expression in mesangial cells exposed to high glucose concentrations,[466] suggests that CTGF may serve as a downstream mediator of the profibrotic effects of TGF-β.[468] Further experiments have shown that the fibrotic effects of TGF-β in the remnant kidney are mediated, at least in part, by the induction of the micro RNAs miR-21, miR-192, and miR-433 via Smad 3.[449] In vitro overexpression of miR-192 promotes, but inhibition of miR-192, attenuates, TGF-β–induced production of collagen type I in rat tubule cells.[469]

It is clear that interventions to inhibit the multiple mechanisms of renal fibrosis show promise as treatments for slowing the progression of CKD. However, despite the promising results of many preclinical studies inhibiting the expression or actions of TGF-β discussed previously, a phase 2 randomized trial of anti–TGF-β_1 monoclonal antibody therapy—added to RAAS inhibitor treatment in persons with diabetic kidney disease, GFR of 20 to 60 mL/min/1.73 m^2, and urine polymerase chain reaction (PCR) value higher than 800 mg/g, found no benefit with respect to change in serum creatinine levels after a median of 315 days of therapy.[470] Secondary analyses also found no differences in proteinuria or biomarkers of renal injury between groups receiving anti–TGF-β_1 antibody versus placebo.

This negative result may have been in part attributed to the relatively short trial duration, inadequate dosing of the antibody, or selection of participants with relatively severe and advanced nephropathy. Other possible explanations include redundancy in fibrotic mechanisms and the fact that TGF-β has potentially beneficial antiinflammatory actions. Nevertheless, it is likely that the development of interventions to inhibit renal fibrosis will remain an area of active research and result in multiple potential novel therapies.[471]

HEPATOCYTE GROWTH FACTOR

Investigations have shed light on the role of HGF as a potential antifibrotic factor in CKD. Initial studies focused on the property of HGF to ameliorate tubule cell injury in models of renal ischemia,[472,473] but studies in models of CKD have suggested that HGF may also ameliorate chronic renal injury through its mitogenic, motogenic, morphogenic, and antiapoptotic actions.[474] As discussed, HGF is upregulated in the remaining kidney after uninephrectomy and may play a role in compensatory renal hypertrophy.[168] Further studies have confirmed that HGF and its receptor, c-met, are also upregulated in the remnant kidney after 5/6 nephrectomy.[475] Furthermore, blockade of HGF action with anti-HGF antibodies resulted in a more rapid decline in the GFR and more severe renal fibrosis, which was associated with increased ECM accumulation and a greater number of myofibroblasts in the interstitium and tubules. Other studies have identified multiple mechanisms whereby HGF may contribute to renoprotection, including amelioration of podocyte injury and apoptosis, as well as proteinuria,[476] increased apoptosis of myofibroblasts,[477] decreased ECM accumulation in association with increased expression of MMP-9, decreased expression of endogenous inhibitors of MMPs, TIMP-1 and TIMP-2, in proximal tubule cell cultues,[475] disruption of NF-κB signaling in tubule cells,[478,479] and suppression of TGF-β–induced CTGF expression in tubule cells.[480]

Multiple experiments have confirmed the renoprotective effects of HGF. The renoprotective effects of ACE inhibitor and ARB treatment were associated with increased renal expression on HGF mRNA.[481]Treatment with anti-HGF antibodies resulted in increased TGF-β levels in a mouse model of chronic glomerulonephritis.[482] HGF treatment ameliorated the progression of chronic allograft nephropathy in a renal transplant model.[483] HGF blocked the TGF-β–induced transdifferentiation of tubule epithelial cells to myofibroblasts.[484] Exogenous HGF administration[484] or HGF overexpression[485] blocked myofibroblast activation and prevented interstitial fibrosis in the unilateral ureteric obstruction model. HGF gene transfer into skeletal muscle ameliorated glomerulosclerosis and interstitial fibrosis after 5/6 nephrectomy.[486] HGF treatment suppressed CTGF expression and attenuated renal fibrosis after 5/6 nephrectomy.[487] In contrast, other studies have reported adverse renal effects associated with excess HGF exposure. Transgenic mice that overexpressed HGF developed progressive renal disease characterized by tubular hypertrophy, glomerulosclerosis, and cyst formation,[488] and HGF administration resulted in more rapid deterioration of creatinine clearance, as well as increased albuminuria in obese db diabetic mice.[489] Available evidence thus has suggested that HGF may play a role in ameliorating chronic renal injury but inappropriate or excessive exposure to HGF may have adverse renal effects.

BONE MORPHOGENETIC PROTEIN-7

Bone morphogenetic protein (BMP)-7, also termed *osteogenic protein-1*, is a bone morphogen involved in embryonic development and tissue repair. Preliminary evidence has suggested that BMP-7 may also play a role in renal repair. BMP-7 is downregulated after acute renal ischemia,[490] early in the course of experimental diabetes,[491] and after 5/6 nephrectomy.[492] Furthermore, the administration of exogenous BMP-7 increased tubular regeneration after 5/6 nephrectomy,[492] attenuated interstitial inflammation and fibrosis after UUO,[493] and ameliorated glomerulosclerosis in rats with diabetic nephropathy.[494] In one model of diabetic nephropathy, BMP-7 was noted to be most effective at inhibiting tubular inflammation and TIF.[495] In vitro experiments have identified several potential renoprotective effects attributable to BMP-7, including inhibition of proinflammatory cytokine, as well as endothelin expression in tubule cells exposed to TNF-α,[496] reversal of renal tubule EMT,[497] antagonism of the fibrogenic effects of TGF-β in mesangial cells,[498] and protection from injury in podocytes exposed to high levels of glucose.[499] In a mouse model of UUO, administration of exogenous BMP-7 was associated with reduced accumulation of collagen type I, an effect attributed to reduced phosphorylation of Smad3 (canonical signaling of TGF-β) and Akt (noncanonical signaling TGF-β).[500] The renoprotective effect of BMP-7 was further illustrated by the observation that expression of a transgene for BMP-7 in podocytes and proximal renal tubule cells was associated with the prevention of podocyte dropout, as well as amelioration of albuminuria, glomerulosclerosis, and interstitial fibrosis after the induction of diabetes.[501] Further studies evaluating the effects of chronic treatment with BMP-7 or small- molecule BMP-7 agonists are still awaited.

MICRO RNAS

Micro RNAs (miRNAs) are small noncoding RNAs that have been shown to have important posttranscriptional gene regulatory function. Attention has been focused on the potential role of miRs to modulate renal fibrosis. miR-21 and miR-214 have been shown to be upregulated in several experimental models of CKD,[502,503] and miR-21 has been reported to be upregulated in human transplant kidneys with nephropathy.[502] Deletion of miR-21 or treatment with anti–miR-21 oligonucleotides was associated with amelioration of interstitial fibrosis in animal models.[502] Similarly, deletion of miR-214 and treatment with anti–miR-214 were each associated with attenuation of interstitial fibrosis in the unilateral ureteric obstruction model. Importantly, the effects of miR-214 appear to be independent of TGF-β signaling, and TGF-β blockade had an additive antifibrotic effect with miR-214 deletion.[504] As discussed previously, TGF-β/Smad3 signaling increases the production of profibrotic miRs and decreases the production of antifibrotic miRs. Multiple other miRs have been implicated in the pathogenesis of different forms of CKD, including miRs 23b, 29b, 29c, and 129 in diabetic nephropathy and miRs 30c,129-5p, 145, 196a, 196b, 200b, 200c, 215, and 433 in renal fibrosis.[505] Studies have also identified that miRs can be transferred between cells in extracellular vesicles to exert autocrine or paracrine effects.[505] The potential to administer beneficial miRs or antagonize the effects of pathogenic miRs with antagomirs represents a further potential therapeutic strategy for CKD.

OXIDATIVE STRESS

CKD is associated with increased oxidative stress that likely contributes to the progression of renal damage, as well as the pathogenesis of the associated cardiovascular disease.[506] Superoxide is the primary reactive oxygen species (ROS) that accounts for oxidative stress. Its major source is production by nicotinamide adenine dinucleotide phosphate (NADPH) oxidase, and it is removed (by conversion to hydrogen peroxide by superoxide dismutase (SOD). Following 5/6 nephrectomy, significant upregulation of NADPH oxidase and downregulation of SOD were observed in the liver and kidneys, indicating that the increase in superoxide is due to a combination of increased production and decreased removal.[507] Similarly, in a mouse model of proteinuric CKD induced by the administration of doxorubicin (ADR), NADPH oxidase was upregulated, and extracellular (EC-SOD) was downregulated. Furthermore, EC-SOD knockout mice showed greater albuminuria and more renal fibrosis after doxorubicin than wild-type mice, confirming that the antioxidant effects of EC-SOD protect against kidney damage.[508] Examination of human kidney biopsies confirmed that EC-SOD is similarly downregulated in human CKD.[508] BP was elevated, and nitrotyrosine levels were increased, whereas urine nitric oxide metabolites were decreased, observations consistent with increased NO inactivation by superoxide. The effects of increased levels of ROS are further compounded by the reduced abundance and activity of antioxidant enzymes (e.g., catalase, glutathione peroxidase, glutathione),[509] as well as reduced HDL, apolipoprotein A-1, and thiols.[510]

Adverse consequences of oxidative stress that may contribute to CKD progression include hypertension (due to inactivation of NO and oxidation of arachidonic acid to generate vasoconstrictive isoprostanes),[511] inflammation (due to activation of NF-κB),[509] fibrosis, and apoptosis,[510] as well as glomerular filtration barrier damage.[512] Inflammation may in turn increase oxidative stress due to the generation of ROS by activated leukocytes, thus establishing a vicious cycle of oxidative stress and inflammation.[506]

Other factors that may contribute to increased ROS production in CKD include Ang II,[513] reduced production of NO,[71] and hypertension.[514] The importance of ROS in the progression of CKD has been shown by experiments in which antioxidant therapies, including melatonin,[515] niacin,[516] and omega-3 fatty acids,[517] have reduced oxidative stress and ameliorated renal damage in the 5/6 nephrectomy model. On the other hand, treatment with tempol, an SOD mimetic, reduced plasma malondialdehyde levels and the number of superoxide-positive cells but did not reduce overall renal oxidative stress, inflammation, or renal damage.[518]

ACIDOSIS

As the GFR declines, the ability of kidneys to excrete hydrogen ions becomes impaired, and a new steady state is achieved that allows excretion of acid at the cost of a persistent metabolic acidosis (see earlier, "Acid-Base Regulation"). Acidosis is present in most patients when the GFR falls below 20% to 25% of normal.[519] Chronic metabolic acidosis has multiple adverse consequences, including increased protein catabolism, increased bone turnover, induction of inflammatory mediators, insulin resistance, and increased production of corticosteroids and PTH.[519,520] These observations make it reasonable to consider whether acidosis may also contribute to progressive renal damage in CKD. Early experiments found no persisting renal damage after dietary acid loading in rats with normal renal function and no renoprotection associated with sodium bicarbonate treatment in the 5/6 nephrectomy model, suggesting that acidosis does not initiate or exacerbate renal damage.[521]

Later experiments, however, have found that an acid-generating diet of casein protein induces tubulointerstitial injury in rats with normal renal function, whereas a non–acid-inducing diet of soy protein does not.[522] Furthermore, dietary acid supplementation with $(NH_4)_2SO_4$ in soy protein–fed rats was associated with tubulointerstitial injury. Similarly, in the 5/6 nephrectomy model, a casein-rich diet was associated with metabolic acidosis and a progressive decline in the GFR and increasing albuminuria, whereas a soy protein diet was not.[523]

Dietary acid supplementation with $(NH_4)_2SO_4$ in soy protein–fed rats provoked a decline in the GFR and albuminuria. Treatment with sodium bicarbonate or $Ca(HCO_3)_2$, but not sodium chloride, was renoprotective in the casein-fed rats but only if the resultant hypertension—in sodium bicarbonate but not $Ca(HCO_3)_2$–treated rats—was adequately treated. Furthermore, when rats were subject to 2/3 rather than 5/6 nephrectomy, a level of renal mass reduction not associated with metabolic acidosis, microdialysis identified tissue acid accumulation in muscle and kidney that correlated with the subsequent GFR decline.[524] Amelioration of tissue acid accumulation by a low acid-generating diet or alkali supplementation was associated with abrogation of the GFR decline, whereas dietary acid supplementation was associated with exacerbation of the GFR decline. Treatment with calcium citrate has also been shown to improve acidosis and reduce glomerular as well as interstitial injury in the 5/6 nephrectomy model.[525] Mechanisms whereby acidosis may contribute to renal damage after nephron loss include activation of the alternative complement pathway by increased ammoniagenesis,[526] increased levels of plasma and kidney Ang II,[527] and induction of ET, as well as aldosterone production.[528]

Observational clinical studies have identified acidosis as an independent risk factor for CKD progression,[529,530] but to date only relatively small studies have investigated the renoprotective potential of alkali supplementation in human subjects. In the first randomized study in adults with a creatinine clearance of 15 to 30 mL/min, randomization to treatment of acidosis (serum bicarbonate, 16–20 mmol/L) with sodium bicarbonate was associated with a lower decline in creatinine clearance (1.88 vs. 5.93 mL/min) and lower incidence of ESKD (6.5% vs. 33%).[531] The authors conceded, however, that the study was not blinded or placebo- controlled. In a second nonrandomized study, treatment of 30 subjects with sodium citrate was associated with reduced urinary excretion of ET-1 and *N*-acetyl-β-D-glucosaminidase (a marker of tubulointerstitial injury), as well as a lower rate of estimated

GFR decline than that observed in 29 untreated controls who were unwilling or unable to take sodium citrate.[532] One study has reported renoprotective effects following sodium bicarbonate treatment in early CKD. In a randomized placebo-controlled trial in subjects with a mean estimated GFR of 75 mL/min/1.73 m^2, treatment with sodium bicarbonate for 5 years was associated with a slower reduction in the estimated GFR (derived from plasma cystatin C measurements) than placebo or treatment with sodium chloride.[533]

Further studies have reported that correction of acidosis with a diet rich in fruit and vegetables is as effective in ameliorating kidney damage in early (CKD stage 1 or 2)[534] and more advanced (CKD stage 4) disease.[535] In a further trial, people with CKD stage 3 and a serum bicarbonate level of 22 to 24 mmol/L were randomized to oral bicarbonate supplementation, a diet rich in fruit and vegetables, or usual care. All participants received treatment with RAAS inhibition (RAASi), and SBP was controlled to less than 130 mm Hg. Both interventions achieved an increase in serum bicarbonate levels and were associated with a decrease in urinary angiotensinogen. After 3 years, both interventions were associated with less albuminuria and GFR decline than the usual care group.[536] Bicarbonate supplementation is already recommended for patients with levels below 22 mEq/L, but further studies are required to further investigate whether it is beneficial in the setting of less severe acidosis.[537]

HYPERTROPHY

The consistent observation of renal, and in particular glomerular, hypertrophy after renal mass reduction has prompted investigators to propose that processes involved in, or resulting from, hypertrophy may contribute to progressive renal injury in CKD.[538] The well- documented observation that renal and glomerular hypertrophy precede the development of diabetic nephropathy and the finding of a positive association between glomerular size and early sclerosis in rats subjected to renal mass ablation[539] further suggests that hypertrophy may play a direct role in the pathogenesis of glomerulosclerosis. Several clinical observations have also supported an association between glomerular hypertrophy and renal injury. Oligomeganephronia, a rare congenital condition with a nephron number 25% of normal or less, is characterized by marked hypertrophy of the remaining glomeruli and the development of proteinuria and renal failure in adolescence, with FSGS as the typical renal biopsy finding.[540] In children with minimal change disease, a glomerulopathy generally associated with spontaneous remission and lack of progression to renal failure, investigators have noted an association between glomerular size and the risk of developing FSGS and renal failure.[541]

Several interventions have been used in experiments to interrupt the development of glomerular hypertrophy after renal mass reduction and thereby assess its role in renal disease progression, but have produced contradictory results. Rats subjected to 5/6 nephrectomy were compared with rats in which 2/3 of the left kidney was infarcted and the right ureter drained into the peritoneal cavity, an intervention that apparently results in decreased renal clearance without compensatory renal hypertrophy.[538] Micropuncture studies confirmed similar degrees of elevation of P_{GC} and SNGFR in both models. At 4 weeks, however, the maximal planar area of the glomerulus was significantly less, and glomerular injury, as assessed by sclerosis index, was significantly reduced in ureteroperitoneostomized rats versus 5/6 nephrectomized controls. Accordingly, the authors have concluded that glomerular hypertrophy is more important than glomerular capillary hypertension in the progression of glomerular injury in this model.

Dietary sodium restriction has also been used to inhibit renal hypertrophy after 5/6 nephrectomy. Although sodium restriction had no effect on glomerular hemodynamics, glomerular volume was significantly reduced in 5/6 nephrectomized rats fed low versus normal sodium diets.[134] Moreover, urinary protein excretion was lower, and glomerulosclerosis was less severe, in rats on restricted sodium intake. These findings were extended by another study in which the effect of sodium restriction in preventing glomerular hypertrophy and ameliorating glomerular injury was confirmed, but which also found that these benefits were overcome by the administration of an androgen that stimulated glomerular hypertrophy, despite sodium restriction. Glomerular hemodynamics were similar among the groups.[542] On the other hand, treatment with seliciclib, a cyclin-dependent kinase inhibitor, reduced renal hypertrophy by 45% after 5/6 nephrectomy but had no effect on kidney damage.[543]

Glomerular hypertrophy may contribute to glomerulosclerosis through a number of different mechanisms. According to the law of Laplace, the increase in glomerular volume could result in an increase in capillary wall tension only if the capillary wall diameter were also increased (Fig. 51.8).

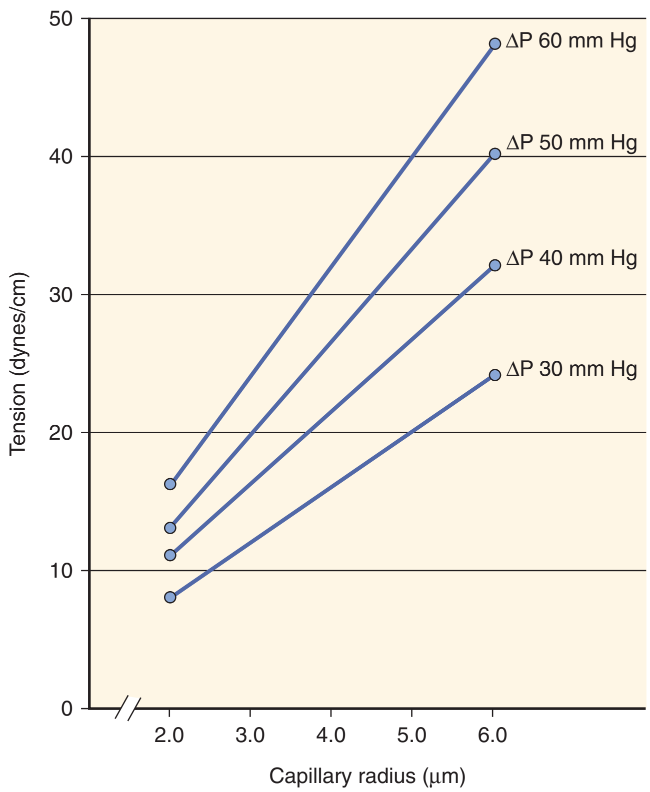

Fig. 51.8 Illustration of the synergistic effects of changes in transcapillary hydrostatic pressure difference (ΔP, mm Hg) and mean glomerular capillary radius (μm) on the calculated capillary wall tension (dynes/cm). (From Bidani AK, Mitchell KD, Schwartz MM, et al. Absence of progressive glomerular injury in a normotensive rat remnant kidney model. *Kidney Int.* 1990;38:28–38.)

Cyclic stretch would then exert stress capable of damaging epithelial, mesangial, and endothelial cells, as described previously. Alternatively, glomerulosclerosis may be viewed as a maladaptive growth response following loss of renal mass and resulting in excessive mesangial proliferation and extracellular matrix production.[538] The identification of TGF-β as a key mediator of renal hypertrophy, as well as an important promoter of renal fibrosis, provides an obvious link between renal hypertrophy and fibrosis.[175]

Detailed studies of podocyte hypertrophy have found that impaired podocyte hypertrophy in response to glomerular enlargement is a further mechanism that may contribute to the development of proteinuria and FSGS. Using a transgenic rat model (dominant negative AA-4E-BP1 transgene) that resulted in impaired podocyte hypertrophy, the investigators observed a remarkable linear relationship among gain in body weight, proteinuria, and glomerulosclerosis that was accelerated after uninephrectomy. Dietary caloric restriction that prevented weight gain and glomerular enlargement also prevented the proteinuria. Analysis of the kidneys from rats with proteinuria demonstrated a mismatch between glomerular tuft volume and total podocyte volume, and electron microscopy revealed a pulling apart of podocyte foot process, resulting in exposed areas of glomerular basement membrane and adhesions to Bowman's capsule.[544] These findings were extended in the same model by showing that the prevention of glomerular hypertrophy after uninephrectomy by dietary caloric restriction, mTORC1 kinase pathway inhibition (with rapamycin), or treatment with an ACE inhibitor each prevented proteinuria and FSGS. Proliferation of glomerular cells other than podocytes was identified and has been proposed to contribute to the development of FSGS. Moreover, transcriptomic analysis of gene expression in isolated glomeruli from rats that developed FSGS evidenced a similar pattern of gene expression to that identified in glomeruli from human biopsies of persons with progressive CKD, suggesting that similar factors play a role in human CKD progression.[545]

ANGIOTENSIN II

As noted, Ang II plays a central role in the glomerular hemodynamic adaptations observed after renal mass ablation. Ang subtype 1 receptors are, however, distributed on many cell types in the kidney, including mesangial, glomerular epithelial, endothelial, tubule epithelial, and vascular smooth muscle, cells suggesting multiple potential actions of Ang II in the kidney.[546] Experimental studies have revealed several nonhemodynamic effects of Ang II that may be important in CKD progression (Fig. 51.9). In isolated perfused kidneys, the infusion of Ang II results in loss of glomerular size permselectivity and proteinuria, an effect that has been attributed to both the hemodynamic effects of Ang II, resulting in elevations in P_{GC}, and a direct effect of Ang II on glomerular permselectivity.[372] Furthermore, overexpression of angiotensin subtype 1 receptors on podocytes resulted in albuminuria and FSGS in the absence of hypertension in transgenic rats.[547]

In vitro, Ang II has been shown to stimulate mesangial cell proliferation and induce expression of TGF-β, resulting in increased synthesis of ECM.[316] In vivo, transfection

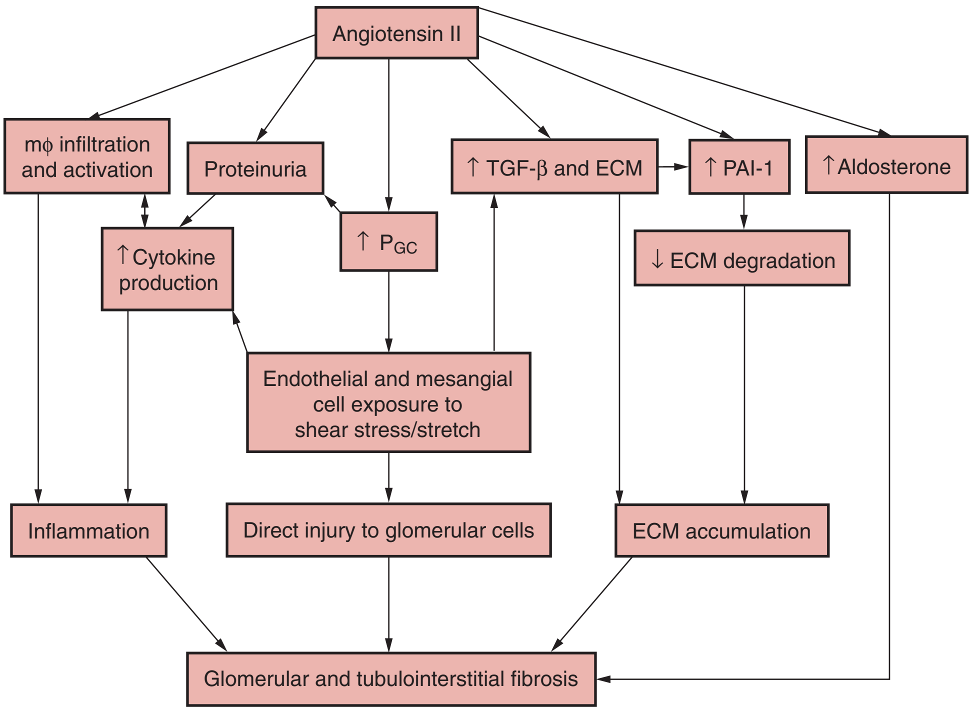

Fig. 51.9 Schematic depicting the central role of angiotensin II through hemodynamic and nonhemodynamic effects in the pathogenesis of progressive renal injury and fibrosis following nephron loss. *ECM,* Extracellular matrix; *m*ϕ, macrophage; *PAI-1,* plasminogen activator inhibitor-1; *P_{GC},* glomerular capillary hydraulic pressure; *TGF-β,* transforming growth factor-β). (From Taal MW, Brenner BM. Renoprotective benefits of RAS inhibition: from ACEI to angiotensin II antagonists. *Kidney Int.* 2000;57:1803–1817,)

of rat kidneys with human genes for renin and angiotensinogen, resulted in glomerular ECM expansion within 7 days.[548] Ang II also stimulates the production of PAI-1 by endothelial cells and vascular smooth muscle cells[549–551] and may therefore further increase the accumulation of ECM through inhibition of ECM breakdown by MMPs that require conversion to an active form by plasmin. Other reports have indicated that Ang II may directly induce the transcription of a variety of cell adhesion molecules and cytokines, as well as activating the transcription factor NF-κB[552–554] and directly stimulating monocyte activation.[555] Ang II infusion provoked upregulation of COX-2 expression in rats that was not dependent on BP elevation,[556] and 5/6 nephrectomized rats evidenced Ang II-dependent upregulation of interstitial COX-2 expression.[557]

In other experiments, Ang II infusion has been shown to induce interstitial macrophage infiltration and increased expression of MCP-1 and TGF-β, effects that were sustained for up to 6 days after cessation of the infusion.[558] In cell culture and animal experiments, Ang II has been shown to promote tubule cell EMT by suppressing expression of miR-429, which inhibits EMT.[559] Finally, Ang II may have fibrogenic effects via mineralocorticoids (see below). Interestingly Ang II may also have antifibrotic effects via the angiotensin subtype 2 receptor (AT_2). Ang II appears to upregulate AT_2 receptor expression via an AT_2 receptor–dependent mechanism after 5/6 nephrectomy, and treatment with an AT_2 receptor antagonist exacerbates renal damage[560] and increased renal PAI-1 expression.[561] Furthermore, overexpression of AT_2 receptors in transgenic mice was associated with reduced albuminuria, as well as decreased glomerular expression of PDGF-BB chain and TGF-β after 5/6 nephrectomy.[562]

ALDOSTERONE

Observations that aldosterone stimulates collagen synthesis in the myocardium, and that spironolactone treatment affords survival benefit in addition to that achieved with ACE inhibitor alone in heart failure patients,[563] have provided impetus to studies investigating the potential role of aldosterone in renal fibrosis. In the remnant kidney model, adrenal hypertrophy and markedly elevated plasma aldosterone levels have been reported. Furthermore, the administration of exogenous aldosterone during inhibition of the RAAS with combination ACE inhibitor–ARB therapy in the 5/6 nephrectomy model negates the renal protective effects of the latter.[564] Further evidence of the role of aldosterone has been provided by experiments in which rats subjected to adrenalectomy after 5/6 nephrectomy received replacement glucocorticoid but not mineralocorticoid therapy, resulting in less severe renal injury than rats with intact adrenal glands.[565]

Mechanisms whereby aldosterone may contribute to renal damage include hemodynamic effects (see previously), mesangial cell proliferation,[566] apoptosis,[567] hypertrophy and transdifferentiation,[568] podocyte injury and apoptosis associated with reduced expression of nephrin and podocin, resulting in proteinuria,[374,569,570] proximal tubule cell transdifferentiation and increased production of collagen types III and IV,[571] and increased renal production of ROS,[374,569] PAI-1,[572,573] TGF-β,[574] and CTGF.[575,576] Early experimental use of aldosterone receptor blockers in 5/6 nephrectomized rats has yielded only modest renoprotective effects,[564,577] but other studies have found significant amelioration of glomerulosclerosis in 5/6 nephrectomized rats treated with spironolactone, alone or in combination with triple antihypertensive therapy or an ARB.[572,578,579] In some rats, spironolactone was associated with the apparent regression of glomerulosclerosis. Furthermore, the observed renoprotection was associated with inhibition of PAI-1 mRNA expression in the renal cortex.[572]

In another experiment, renoprotection, achieved with combination ACE inhibitor and spironolactone treatment, was associated with abrogation of the increase in mesangial cells and decrease in podocytes observed in untreated rats. Combination treatment also attenuated the increase in expression of type IV collagen, TGF-β, and desmin.[578] Spironolactone has also been shown to ameliorate renal damage in other experimental models, including diabetic nephropathy,[575,580] Ren2 transgenic rats,[581] radiation nephritis,[582] and stroke-prone hypertension.[583] Several small clinical trials have reported an additional reduction of proteinuria by 15% to 54%, BP by approximately 40%, and the GFR by approximately 25% when aldosterone receptor blockers were added to ACE inhibitor or ARB treatment.[584]

More novel nonsteroidal aldosterone antagonists are also being evaluated. In one randomized trial that enrolled 336 subjects with hypertension, urine ACR of 30 to 599 mg/g, and an eGFR of 50 mL/min/1.73 m^2 or higher, treatment with eplerenone for 52 weeks, added to ACE inhibitor or ARB therapy, resulted in a 17.3% decrease in urine ACR, whereas a 10.3% increase was observed in those who received placebo (P = .02).[585] The serum potassium level was higher with eplerenone treatment, but no episodes of severe hyperkalemia (serum potassium >5.5 mmol/L) were observed. Nevertheless, large trials are required to assess the potential benefits of aldosterone antagonists in CKD fully; their use is currently limited by the associated risk of hyperkalemia.[586]

A UNIFIED HYPOTHESIS OF CHRONIC KIDNEY DISEASE PROGRESSION

In the past, there has tended to be a dichotomy of viewpoints regarding the relative importance of hemodynamic and nonhemodynamic factors in the pathogenesis of glomerulosclerosis and TIF after nephron loss.[538,587] Proponents of the so-called hypertrophy hypothesis have pointed out that in some experiments, a dissociation between glomerular hemodynamic changes and glomerulosclerosis has been observed, and that in one study, antihypertensive therapy was renoprotective without lowering P_{GC}.[538] On the other hand, those favoring the hemodynamic hypothesis have noted that treatment with an ACE inhibitor[20] or ARB[22] in 5/6 nephrectomized rats results in renoprotection without preventing renal or glomerular hypertrophy, and that many of the studies purporting to show a positive association between glomerular hypertrophy and sclerosis failed to report glomerular hemodynamic data. Furthermore, rats subjected to ureteroperitoneostomy developed significantly more glomerulosclerosis than sham-operated controls, despite a lack of increase in glomerular size.[587]

Several other observations have suggested that hemodynamic factors override the potential role of hypertrophy in progressive renal damage. The renoprotection achieved after

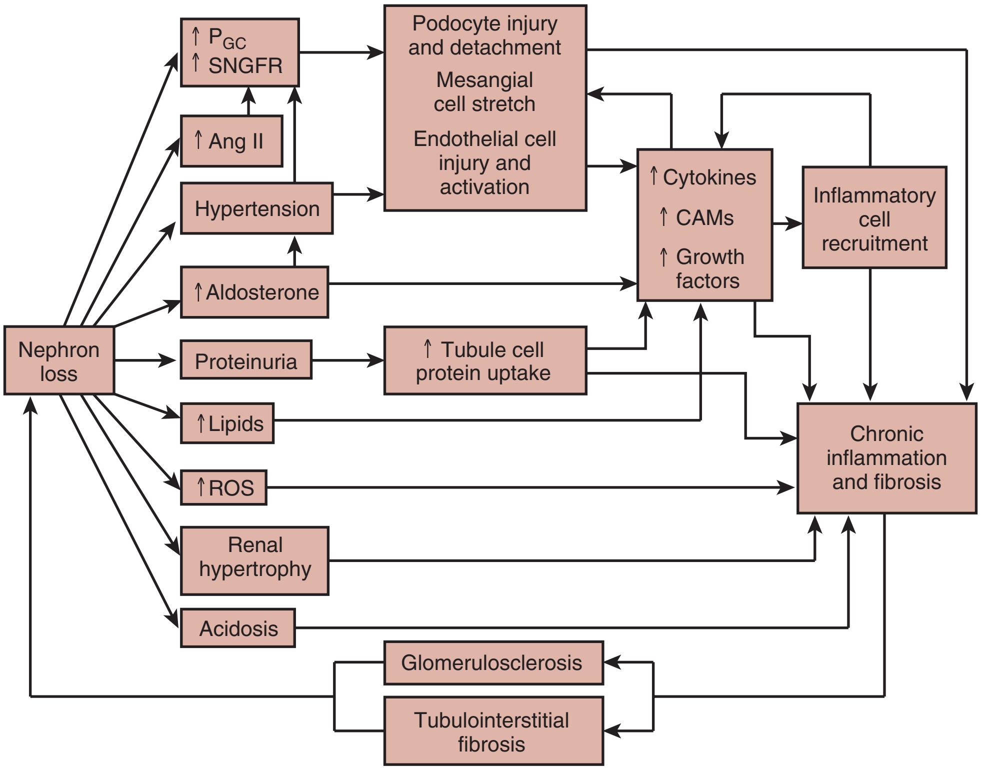

Fig. 51.10 Schematic illustrating the hypothesized interaction of multiple hemodynamic and nonhemodynamic factors in the pathogenesis of progressive nephron injury in chronic kidney disease. *CAM,* Cell adhesion molecules; *GFR* Glomerular filtration rate; P_{GC}, glomerular capillary hydraulic pressure; *ROS,* reactive oxygen species; *SNGFR*, single nephron GFR.

5/6 nephrectomy by a low-protein diet (associated with prevention of glomerular hypertrophy) can be reversed by treatment with calcium channel blockers that inhibit renal autoregulation but have no effect on glomerular size.[588] Comparison of rats subjected to 5/6 nephrectomy by excision versus infarction of two-thirds of the remaining kidney has shown similar increases in glomerular volume, but the infarction model is associated with more severe glomerular hypertension and glomerulosclerosis.[10]

Despite these apparently conflicting views, it is clear from the earlier discussion that hemodynamic and nonhemodynamic mechanisms often overlap. For example, Ang II, a key mediator of glomerular hemodynamic adaptations after nephron loss, also exerts multiple nonhemodynamic deleterious effects. Furthermore, the inflammatory and profibrotic mechanisms that eventuate in glomerulosclerosis and tubulointerstitial fibrosis may be provoked by both hemodynamic and nonhemodynamic stimuli. A growing appreciation of the complexity of the multiple adaptations that follow nephron loss has facilitated the development of a consensus view that continues to regard raised glomerular capillary pressure as a central factor in initiating glomerulosclerosis, but also acknowledges that other nonhemodynamic pathogenic mechanisms may act in concert with hemodynamic factors in a complex interplay that results in a vicious cycle of progressive renal damage (Fig. 51.10). An appreciation of the many mechanisms involved in CKD progression is essential to inform strategies for achieving optimal renoprotection.

INSIGHTS FROM MODIFIERS OF CHRONIC KIDNEY DISEASE PROGRESSION

> ***Clinical Relevance***
>
> **Pharmacological Inhibition of the RAAS**
>
> The treatment of hypertension with an optimal dose of a RAAS inhibitor is the mainstay of the approach to achieve renoprotection in proteinuric CKD. Blood pressure should be reduced to less than 130/80 mmHg and even lower targets should be considered in persons with persistent proteinuria, judged to be at low risk of adverse effects from lower blood pressue.

PHARMACOLOGIC INHIBITION OF THE RENIN-ANGIOTENSIN- ALDOSTERONE SYSTEM

Experimental evidence showing a central role for Ang II in mechanisms of CKD progression through hemodynamic and nonhemodynamic effects has been borne out in randomized clinical trials of ACE inhibitor and ARB treatment in patients with all forms of CKD. ACE inhibitor treatment has been shown to be renoprotective in patients with microalbuminuria and type 2 diabetes mellitus,[589] type 1 diabetes, and overt nephropathy,[281] as well as nondiabetic CKD.[282,285,590,591] Treatment with an ARB affords renoprotection in patients with type 2 diabetes and microalbuminuria[592] or overt nephropathy.[283,284]

More complete inhibition of the RAAS with combination ACE inhibitor–ARB treatment was associated with additional lowering of proteinuria in several small clinical studies.[593] However, the largest randomized study of combination ACE inhibitor–ARB treatment versus monotherapy in CKD was withdrawn because of serious concerns regarding the integrity of the data.[594]

A large trial of combination therapy in patients with hypertension and increased cardiovascular risk has reported no additional benefit with respect to cardiovascular outcomes, but combination therapy was associated with greater reductions in proteinuria than monotherapy.[595] However, the trial reported an increase in the combined endpoint of creatinine doubling, ESKD, or death with combination therapy, indicating that combination ACE inhibitor–ARB therapy may be associated with adverse outcomes in some patient groups. It should be noted, however that subjects were selected on the basis of cardiovascular risk profile and that the majority did not have a reduced GFR or proteinuria. A similar study in 1448 people with type 2 diabetes and urine ACR of 300 mg/g or higher also found no benefit with respect to the primary outcome of CKD progression, ESKD, or death in participants randomized to combination ACE inhibitor–ARB therapy versus monotherapy. However, those receiving combination therapy evidenced a significantly higher incidence of acute kidney injury (AKI) and hyperkalemia.[596]

Thus in two large randomized trials, combination ACE inhibitor–ARB therapy did not confer additional benefit and was associated with an increased risk of adverse effects. Combination therapy should therefore not be used in the patient groups included in these studies. Further studies are required to assess the risks versus benefits of combination therapy in those with proteinuric nondiabetic CKD. The development of direct renin inhibitors (DRIs) has made it possible to inhibit the RAAS at its rate-limiting step—the conversion of angiotensinogen to angiotensin I—and thereby achieve more complete blockade. DRIs were effective as antihypertensive agents and reduced proteinuria in animal models.[597,598] Two early randomized trials have reported additional lowering of albuminuria in subjects with diabetic nephropathy receiving combination DRI and ARB therapy versus ARB therapy alone.[599,600] However, a large randomized trial that included 8561 people with type 2 diabetes and albuminuria or cardiovascular disease was stopped prematurely after the second interim efficacy analysis. Despite greater lowering of BP and albuminuria with combination DRI-ARB therapy, no benefit was observed with respect to the composite primary endpoint of cardiovascular events, CKD progression, ESKD, or death versus ARB monotherapy, but combination therapy was associated with a higher incidence of hyperkalemia and hypotension.[601] Taken together, published data from large randomized trials have to date failed to show additional renoprotective benefits with combination RAAS inhibitor therapy, and all have reported increased adverse effects. This implies that near-complete blockade of the RAAS may be undesirable in many patient groups, but this does not exclude the possibility that it may be beneficial in selected patients who are at high risk of CKD progression, are resistant to monotherapy, and are at a low risk of the reported adverse effects. Further studies are required to investigate this possibility before combination RAAS inhibitor therapy can be recommended in clinical practice.

One metaanalysis has called into question the importance of RAAS inhibition.[602] It should be noted that this study was dominated by data from the Antihypertensive and Lipid Lowering Treatment to Prevent Heart Attack (ALLHAT) study, which found no difference in fatal coronary heart disease or nonfatal myocardial infarction among hypertensive patients with at least one cardiovascular risk factor randomized to treatment with a thiazide diuretic, a calcium channel blocker, or an ACE inhibitor.[603] In a posthoc analysis, there was also no difference in the secondary outcome of ESKD or more than a 50% decrement in the GFR, although patients with a serum creatinine level higher than 2 mg/dL were specifically excluded, resulting in only a minority of patients (5662 of 33,357) having renal disease (eGFR <60 mL/min/1.73 m^2). Furthermore, there was no assessment of proteinuria.[604] Thus inclusion of the ALLHAT data was inappropriate and significantly affected the results of the metaanalysis.[605]

Other metaanalyses that did not include ALLHAT data have shown significant renoprotective benefits in patients receiving ACE inhibitor treatment.[606,607] In summary, there is now evidence from multiple randomized trails showing significant renoprotection associated with pharmacologic inhibition of the RAAS in a wide variety of forms of CKD. This confirms that Ang II is a critical mediator of mechanisms of CKD progression in humans and provides support for the consensus that RAAS inhibition should be central to treatment strategies for slowing CKD progression.[608] The role of RAAS inhibitor treatment in achieving optimal renoprotection is discussed further in Chapter 59.

ARTERIAL HYPERTENSION

Malignant hypertension frequently leads to renal injury, but whether or not less severe forms of hypertension cause hypertensive nephrosclerosis remains a subject of debate.[609,610] An increased risk of developing progressive renal failure with higher levels of BP has been observed in several population-based studies[611–614] and is exemplified by findings from the Multiple Risk Factors Intervention Trial (MRFIT).[615] In a population of 332,544 men, there was a strong graded relationship between BP and the risk of developing or dying with ESKD over a 15- to 17-year follow-up period. Renal function was not assessed at screening or during follow-up, however, so it is not possible to establish with any certainty whether higher BP initiated renal disease or accelerated a nephropathy that was already present. In one study, the importance of hypertension as a risk factor for ESKD was further illustrated by the observation that lowering SBP by 20 mm Hg reduced the risk of ESKD by two-thirds.[612] Even small increases in BP, below the threshold usually used to define hypertension, are associated with an increased risk of ESKD.[611,613,616] Hypertension has also been identified as a risk factor for developing albuminuria or renal impairment among patients with type 2 diabetes mellitus.[617]

Whereas the role of hypertension in initiating renal disease requires further clarification, there is clear evidence that hypertension accelerates the rate of progression of preexisting renal disease, most likely through the transmission of raised hydraulic BP to the glomerulus, resulting in the exacerbation of glomerular capillary hypertension associated with nephron loss.[3] Among patients with diabetic nephropathy and nondiabetic CKD, the initiation of antihypertensive therapy results in

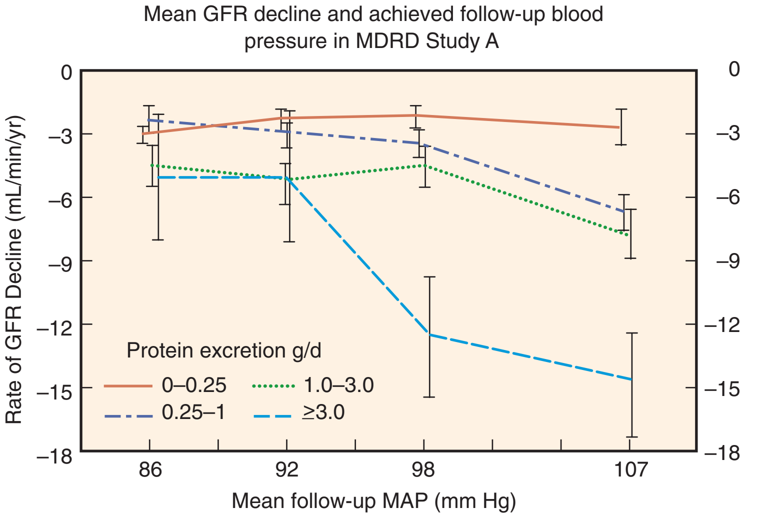

Fig. 51.11 Interaction of blood pressure reduction and proteinuria at baseline on the rate of decline in the glomerular filtration rate. *GFR,* Glomerular filtration rate; *MAP,* mean arterial pressure; *MDRD,* Modification of Diet in Renal Disease trial. (From Peterson JC, Adler S, Burkart JM, et al. Blood pressure control, proteinuria, and the progression of renal disease. The Modification of Diet in Renal Disease Study. *Ann Intern Med*. 1995;123:754–762.)

significant reductions in rates of GFR decline, implying that hypertension, an almost universal consequence of impaired renal function, also contributes to the progression of CKD.[618] The potential impact of hypertension on the kidney has been exemplified by case reports of patients with unilateral renal artery stenosis who manifested diabetic nephropathy or FSGS only in the nonstenotic kidney and not in the stenotic side that was shielded from the hypertension.[619,620] Analysis of data from the Chronic Renal Insufficiency Cohort (CRIC) study has emphasized the importance of BP control over time. Time-updated SBP higher than 130 mm Hg was more strongly associated with an increased risk of CKD progression than a single baseline measurement.[621]

Uncertainty remains, however, as to the level of BP lowering that is required to achieve optimal renoprotection. Several randomized trials have sought to resolve this issue. In the Modification of Diet in Renal Disease (MDRD) study, patients with predominantly nondiabetic CKD were randomized to a target mean arterial pressure (MAP) of less than 92 mm Hg (equivalent to <125/75 mm Hg) versus less than 107 mm Hg (equivalent to 140/90 mm Hg). Whereas there was no difference between the overall rate of change in the GFR during a mean of 2.2 years follow-up, patients randomized to the low BP target evidenced an early rapid decrease in the GFR, likely attributed to associated renal hemodynamic effects, which obscured a later significantly slower rate of GFR decline. Furthermore the effect of BP control was strongly modulated by the severity of the proteinuria. Among patients with greater than 3 g/day of proteinuria at baseline, randomization to the low BP target was associated with a significantly slower rate of GFR decline.[622]

Secondary analysis also revealed significant correlations between the rate of GFR decline and achieved BP, an effect that was more marked among those with greater baseline proteinuria.[623] In study group 1 (patients with GFR of 25–55 mL/min/1.73 m^2), rates of the GFR decline increased above a MAP of 98 mm Hg among patients with baseline proteinuria of 0.25 to 3.0g/day, and above 92 mm Hg in those with baseline proteinuria more than 3.0g/day. In study group 2 (patients with GFR of 13–24 ml/min/m^2), higher achieved BP was associated with greater rates of GFR decline at all levels among patients with baseline proteinuria more than 1 g/day (Fig. 51.11). That the benefits of lower BP may become evident only over a longer period is illustrated by the observation that further follow-up (mean 6.6 years) of patients from the MDRD study revealed a significant reduction in the risk of ESKD (adjusted hazard ratio [HR]. 0.68; 95% CI. 0.57–0.82) or a combined endpoint of ESKD or death (adjusted HR. 0.77; 95% CI, 0.65–0.91) among patients randomized to the low BP target, even though treatment and BP data were not available beyond the 2.2 years of the original trial.[624]

In contrast, in the African American Study of Kidney Disease and Hypertension (AASK), no significant difference in the rate of GFR decline was observed between patients randomized to MAP goals of 92 mm Hg or lower versus 102 to 107 mm Hg. Even after prolonged follow-up of the randomized groups, no difference in the combined endpoints of creatinine doubling, ESKD, or death was observed.[625] It should be noted, however, that patients in AASK generally had low levels of baseline proteinuria (mean urine protein, 0.38–0.63 g/day)[591] and after prolonged follow-up, some reduction in the risk of the primary endpoint was observed among those with a baseline urine protein to creatinine ratio more than 0.22 mg/mg, a threshold chosen posthoc by the investigators to be roughly equivalent to 300 mg/day of protein excretion, a value commonly used to define proteinuria.[625] Thus the MDRD and AASK study results support the notion that lower BP targets afford additional renoprotection in patients with more severe proteinuria. Because not all the patients in the MDRD study received ACE inhibitor treatment, it remained unclear to what extent the level of BP attained is important in CKD patients receiving ACE inhibitor or ARB treatment.

Experimental studies have found SBP to be a major determinant of glomerular injury in rats receiving ACE inhibitor or ARB treatment.[270,626] Moreover, among patients with type 1 diabetes and established nephropathy receiving ACE inhibitor treatment, randomization to a low (MAP <92 mm Hg) versus usual (MAP = 100–107 mm Hg) BP target was associated with significantly lower levels of proteinuria after 2 years, although there was no significant difference in GFR decline.[627] Furthermore, secondary analysis of data from the Irbesartan Diabetic Nephropathy Trial (IDNT) showed greater renoprotection among patients who achieved lower BP targets, so that achieved SBP more than 149 mm Hg was associated with a 2.2-fold increased risk of developing ESKD or a doubling of serum creatinine levels versus achieved a SBP less than 134 mm Hg.[628] Importantly, the relationship between improved outcomes and lower achieved SBP persisted among those patients treated with irbesartan. Similarly, in a metaanalysis of data from 1860 nondiabetic patients with CKD, the lowest risk of progression was observed in patients with a SBP of 110 to 129 mm Hg.[629] The ESCAPE Trial investigated the optimal level of BP control for renoprotection in the setting of ACE inhibitor treatment. The study found that among children with CKD receiving treatment with an ACE inhibitor, those randomized to a lower BP target had a significantly reduced risk of reaching ESKD or doubling of serum creatinine levels.[630]

A further trial has reported significant benefit associated with a lower BP target in young people (age <50 years) with autosomal dominant polycystic kidney disease and GFR more than 60 mL/min/1.73 m^2. Participants randomized to a low BP target (110/75–95/60 mm Hg) in addition to RAAS inhibitor treatment showed a slower rate of increase in kidney volume, as well as a greater decrease in albuminuria and left ventricular mass index, than those randomized to usual BP control (120/70–130/80 mm Hg).[631] However, these findings cannot be applied to patients who do not meet the inclusion criteria for this study. On the other hand, additional BP reduction with a calcium channel blocker in patients with nondiabetic CKD on ACE inhibitor treatment failed to produce additional renoprotection, although the degree of additional BP reduction was modest (4.1/2.8 mm Hg) and may have been insufficient to improve outcomes in patients already receiving optimal ACE inhibitor therapy.[632]

Several lines of evidence have drawn attention to the possibility that aggressive BP lowering may be associated with adverse effects in some patients. In the IDNT study, an achieved SBP less than 120 mm Hg was associated with increased all-cause mortality and no further improvement in renal outcomes[628]; in the metaanalysis described earlier, achieved SBP less than 110 mm Hg was associated with a higher risk of CKD progression.[629] Furthermore, secondary analysis of data from the Ongoing Telmisartan Alone and in combination with Ramipril Global EndpoinT (ONTARGET) trial has found that subjects who achieved a SBP of less than 120 mm Hg had a significantly higher cardiovascular mortality than those who achieved a SBP of 120 to 129 mm Hg.[633] Similarly, the ACCORD study reported no additional benefit with respect to cardiovascular endpoints among patients with diabetes randomized to achieve a SBP less than 120 mm Hg (vs. conventional control to <130/80 mm Hg), but the lower BP target was associated with more treatment-related adverse events and a greater decline in the GFR.[634]

On the other hand, the largest randomized trial to date to compare the effects of different BP targets on cardiovascular and renal outcomes (The Systolic Blood Pressure Intervention Trial [SPRINT]) was stopped early due to evidence of significant benefit in the primary outcome of cardiovascular events (CVE) and all-cause mortality in the group randomized to a SBP target of less than 120 mm Hg versus less than 140 mm Hg.[635] Analysis of the main trial showed no evidence of effect modification by CKD status on the primary outcome of CVE or all-cause mortality.

A prespecified subgroup analysis was conducted in 2646 participants with CKD at baseline. There was no difference between BP target groups in the renal outcome of 50% decline in the GFR or ESKD after a median of 3.3 years. The lower BP target was associated with some initial decline in the eGFR, likely due to increased use of RAASi, whereas a small initial increase in the GFR was observed with the higher BP. After excluding the first 6 months of observation, the low BP target was associated with a slightly higher rate of GFR decline (0.47 vs. 0.32 mL/min/1.73 m^2 per year; $P < .03$). The lower BP target was associated with a lower urine ACR at all time points to 48 months. There was no difference in serious adverse events between the groups, but the low BP target was associated with a higher incidence of hyperkalemia, hypokalemia, and AKI.[636] The results of the SPRINT study therefore indicate survival and cardiovascular benefits but no renoprotective benefit with lower BP targets, and question the notion that the risk of lower BP targets outweighs the benefits in all older adults.

Whereas the results of randomized trials comparing low and usual BP targets among persons with CKD have not yielded unequivocal results, the overall picture is one of lower BP targets being associated with more effective renoprotection among those with more severe proteinuria. These observations have led to a consensus that BP should be lowered to less than 130/80 mm Hg in all patients with diabetic or proteinuric CKD and less than 140/90 mm Hg in patients without these risk factors,[637] although lower targets should be considered to optimize cardiovascular risk reduction in patients not judged to be at high risk of adverse effects from intensive BP lowering.

DIETARY PROTEIN INTAKE

Increased dietary protein intake and intravenous protein loading in animals or humans with intact kidneys are associated with increases in renal mass, RBF, and GFR, as well as a decrease in renal vascular resistance. The magnitude of the increases in the GFR and RBF in response to a protein load is a function of renal reserve. In patients with a reduced GFR, some studies have shown that the percentage increase in the GFR in response to a protein meal is reduced in those with a lower baseline GFR,.[638] In contrast, a study comparing the renal response to an oral protein load in patients with moderate and advanced CKD found a similar percentage increase in the GFR over baseline in both groups, demonstrating that even with advanced renal disease, some renal reserve is still present, and that elevated intake of dietary protein may have undesirable effects on glomerular hemodynamics at all levels of renal function.[639]

To understand the mechanisms whereby protein loading acutely augments renal function, various components of protein diets have been examined individually. The administration of equivalent quantities of urea, sulfate, acid, and vegetable protein to dogs or humans all failed to reproduce a meat protein–induced rise in the GFR.[640–642] In contrast, feeding or infusion of mixed or individual amino acids (e.g., glycine, L-arginine) was shown to effect increases in the GFR of similar magnitude to those seen with meat ingestion.[643,644] Micropuncture experiments have demonstrated that amino acid infusion results in increases in glomerular plasma flow and transcapillary hydraulic pressure difference, thereby raising SNGFR without affecting the ultrafiltration coefficient.[643] Interestingly, however, perfusion of the isolated kidney with an amino acid mixture resulted in only a modest increase in the GFR.[645]

Taken together, these observations suggest that amino acids themselves do not have a major direct effect on renal hemodynamics, but their effects appear to be mediated by an intermediate compound generated only in the intact organism. Glucagon, the secretion of which is stimulated by protein feeding, has been proposed as such a mediator. The GFR and renal blood flow increase in response to glucagon infusion in dogs.[644] Furthermore, administration of the glucagon antagonist, somatostatin, consistently blocks amino acid–induced augmentation in renal function both in humans and rats.[643,646] Large protein meals are also rich in minerals, potassium, phosphate, and acids. Indeed, after feeding a protein meal to dogs, the excretions of sodium, potassium, phosphorus, and urea were found to increase in parallel to the increase in the GFR.[640] On the other hand, sodium chloride reabsorption in the proximal tubule and loop of Henle was found to be increased in rats maintained on a high-protein diet.[647] As result, less sodium and chloride would be delivered to the macula densa, thereby inhibiting tubuloglomerular feedback and adding a further stimulus to glomerular hyperperfusion. Because dietary protein does not affect systemic BP,[643] other factors have been suggested to contribute to the renal hemodynamic changes following a protein load. Administration of the nitric oxide inhibitor L-NMMA or nonsteroidal antiinflammatory drugs (NSAIDs) have been shown to blunt the renal hyperemic response to an oral protein load in both rats and humans, invoking a role for nitric oxide and PGs.[647,648] In addition, Ang II and ET have been proposed as mediators of protein-induced renal injury because low-protein diets have been shown to reduce renal ET-1, ET receptors A and B, and AT_1 receptor mRNA expression in PAN-injected and normal rats.[649,650]

It has been proposed that the augmented renal function induced by dietary protein may be an evolutionary adaptation of the kidney to the intermittent heavy protein intake of the hunter-gatherer.[4] Renal hyperfunction following a protein load would serve to facilitate excretion of the waste products of protein catabolism and other dietary components, thereby achieving homeostasis in the presence of an abrupt increase in consumption in times of nutritional plenty; the subsequent decline of the GFR to baseline during the intervals between meals would then favor mechanisms suited to conservation of fluid and electrolytes in times of scarcity. Persistent renal hyperfunction due to continuous excessive protein intake, however, leads to renal injury in experimental models. Laboratory animals with intact kidneys and ingesting food ad libitum become proteinuric and develop glomerulosclerosis with age.[4,196,651] This progression is significantly attenuated by feeding animals on alternate days only.[196]

Furthermore, aging rats fed a high-protein diet ad libitum showed accelerated and more severe renal injury compared with rats receiving a normal protein diet, whereas rats fed a low-protein diet were protected from renal injury.[651] Similarly, in diabetic rats, progression of nephropathy was markedly accelerated in the setting of a high-protein diet and substantially attenuated by a low-protein diet.[254] In this study, kidney weight in high-protein-diet-fed diabetic rats was significantly greater than in diabetic rats receiving normal protein diets, suggesting that protein-induced renal hypertrophy may itself contribute to acceleration of renal functional deterioration. As discussed earlier, the renoprotective effects of dietary protein restriction in experimental animals are associated with virtual normalization of P_{GC} and SNGFR.[9]

Despite unambiguous evidence from experimental studies, confirmation of a beneficial effect of dietary protein restriction in clinical trials has proved elusive. Several small studies generally have suggested a beneficial effect from protein restriction, but suffered from deficiencies in design or patient compliance. A large, multicenter randomized study, the MDRD study, was therefore conducted to resolve the issue.[622] In this study, 585 patients with moderate chronic renal failure (GFR = 25–55 mL/min/1.73 m^2) were randomized to a usual (1.3 g/kg/day) or low (0.58 g/kg/day) protein diet (LPD; study 1), and 255 patients with severe chronic renal failure (GFR = 13–24 mL/min/1.73 m^2) were randomized to a to low (0.58 g/kg/day) or very low (0.28 g/kg/day) protein diet (VLPD; study 2), supplemented with keto–amino acids to prevent malnutrition. All causes of CKD were included, but patients with diabetes mellitus requiring insulin therapy were excluded. Patients were also assigned to different levels of BP control. After a mean of a 2.2-year follow-up, the primary analysis revealed no difference in the mean rate of GFR decline in study 1 and only a trend toward a slower rate of decline in the VLPD group in study 2.

Secondary analyses of the MDRD data, however, revealed that dietary protein restriction probably did achieve beneficial effects. In study 1, a LPD was associated with an initial reduction in the GFR that likely resulted from the functional effects of decreased protein intake and not from loss of nephrons. This initial reduction in the GFR obscured a later reduction in the rate of the GFR decline that was evident after 4 months in the LPD group, which may have resulted in more robust evidence of renoprotection had follow-up been continued for a longer period.[652] Disappointingly, long-term follow-up of 255 participants in study 2 of the MDRD trial found no renoprotective benefit associated with randomization to a VLPD in the original study but did report a higher risk of death in this group (HR, 1.92; 95% CI, 1.15–3.20).[653] In a more recent trial, 207 well-nourished persons without diabetes and with eGFR less than 30 mL/min/1.73 m^2 and urine PCR less than 1 g/g were randomized to a vegetarian VLPD (<0.3 g/kg/day) with ketoanalogue supplementation (ketoanalogue diet, KD) or a standard (nonvegetarian) LPD (<0.6 g/kg/day). After 15 months, the primary outcome of renal replacement therapy (RRT) or 50% reduction in eGFR was reached in 13% of those on a KD versus 42% of those on a LPD (adjusted HR 0.1; 95% CI, 0.05–0.2). Those on the KD also showed a 3.2-mL/min/year slower rate of eGFR decline. Metabolic factors were also improved by the KD; serum bicarbonate and calcium levels were higher and phosphate levels lower. There was no change in nutritional status in either group, and no adverse reactions were reported.[654]

Despite inconclusive findings in several of the individual studies, three metaanalyses have each concluded that dietary protein restriction is associated with a reduced risk of ESKD (odds ratio [OR] of 0.62 and 0.67, respectively)[655,656] as well as a modest reduction in the rate of eGFR decline (0.53 mL/min/1.73 m^2 per year).[657] A further recent metaanalysis included 16 randomized controlled trials (RCTs) of persons with relatively advanced CKD (stages 4 and 5 in most cases). In studies comparing an LPD (<0.8 g/day) with higher protein intake, an LPD was associated with a small but significant absolute risk reduction for ESKD (4%) and higher serum bicarbonate level at 1 year (weighted mean difference 1.46 mEq/L) versus higher protein diets. Similarly, in studies comparing a VLPD to LPD, a VLPD was associated with an absolute risk reduction for ESKD (13%) and higher GFR at 1 year (weighted mean difference, 3.95 mL/min/1.73 m^2) versus LPD. No studies reported an increased risk of protein energy wasting or other safety concerns associated with dietary protein restriction.[658] An important caveat to these findings is that the proportion of the participants treated with RAAS inhibitors was variable. Although evidence suggests that LPD and RAAS inhibitors may have synergistic renoprotective effects, this has not yet been adequately evaluated in RCTs.

In summary, there is evidence from some small trials and several metaanalyses that dietary protein restriction may slow the progression of human CKD, particularly in more advanced and proteinuric disease. Whereas the renoprotective benefit appears modest, such dietary restriction is associated with other benefits, including improvement in acidosis, as well as a reduction in the phosphorus and potassium load. Thus comprehensive dietary intervention with a moderate restriction in dietary protein intake should be considered as part of the management of persons with CKD.[659] The potential benefit of VLPDs in advanced CKD requires further evaluation in RCTs. The role of dietary intervention in the management of CKD is discussed further in Chapter 60.

GENDER DIFFERENCES

Laboratory studies have indicated that male animals appear to be at greater risk of developing renal disease and disease progression than females. Age-associated glomerulosclerosis is much more pronounced in male than in female rats, and it is notable that the male propensity for age-related glomerulosclerosis can be prevented by castration.[660] This gender difference was found to be independent of P_{GC} or glomerular hypertrophy, suggesting a role for the sex hormones as modulators of renal injury. Ovariectomy had no effect on the development of glomerular injury in some animal models[660,661] but did accelerate progression in glomerulosclerosis-prone mice.[662]

In contrast, in the hypercholesterolemic Imai rat, the development of spontaneous glomerulosclerosis in males can be significantly reduced by castration or by the administration of exogenous estrogens.[663,664] These data suggest an important role for androgens in the development of renal injury and raise the possibility that estrogens may, to some extent, counteract the adverse effects of androgens. In an apparently conflicting observation, female Nagase analbuminemic rats developed more severe renal injury than males, a characteristic that is ameliorated by ovariectomy.[665] These rats may be unique, however, in that triglyceride levels, which are higher in females, may have an independent and overriding effect on renal disease propensity. Glomerulosclerosis also develops to a significantly greater extent in male versus female rats subjected to extensive renal ablation.[666] This difference was independent of BP and glomerular hypertrophy, but the degree of glomerulosclerosis and the extent of mesangial expansion were each found to correlate significantly with an increased expression of glomerular procollagen α_1(IV) mRNA in males. Similarly, in aging Munich-Wistar rats, glomerular metalloproteinase activity was found to decrease with age in males but not in females or castrated rats, suggesting that suppression of metalloproteinase activity by androgens could account for the gender difference in disease susceptibility.[667] Finally, estrogens, but not androgens, possess antioxidant activity and have been shown to inhibit mesangial cell LDL oxidation,[668] a property that may contribute to renoprotection.

Clinical studies have suggest that a gender difference with respect to CKD progression also exists in humans. Data from the US Renal Data System (USRDS) have shown a substantially higher incidence of ESKD among males (413/million population in 2003) versus females (280/million population),[669] and several studies have reported worse renal outcomes in males. In a Japanese community-based mass screening program, the risk of developing ESKD (if baseline serum creatinine level was >1.2 mg/dL for males or >1 mg/dL for females) was almost 50% higher in men than in women.[670] In a large population-based study in the United States, male gender was associated with a significantly increased risk of ESKD or death associated with CKD.[613] Similarly, in France, studies of factors influencing the development of ESKD in patients with moderate and severe renal disease found that disease progression was accelerated in males versus females, especially in those with chronic glomerulonephritis or autosomal dominant polycystic kidney disease (ADPKD). Furthermore, the effect of hypertension as a risk factor for CKD progression appeared to be greater in males.[671,672] Other studies of patients with CKD have reported a lower risk of ESKD among females with CKD stage 3[673] and a shorter time to RRT among males with CKD stages 4 and 5.[674]

One metaanalysis of 68 studies that included 11,345 persons with CKD has reported a higher rate of decline of renal function in men,[675] but another metaanalysis of individual participant data from 11 randomized trials evaluating the efficacy of ACE inhibitor treatment in CKD did not show an increased risk of doubling the serum creatinine level or ESKD or ESKD alone in men.[676] On the contrary, after adjustment for baseline variables, including BP and urinary protein excretion, women had a significantly higher risk of these endpoints than men.[676] One limitation of these studies is that the menopausal status of the women was often not documented. Interestingly in one study, bilateral oophorectomy in 1653 premenopausal women was associated with a higher risk of developing CKD (defined by eGFR<60 mL/min/1.73 m^2 on two occasions >90 days apart) over a median of 14 years of observation when compared with age-matched women who did not undergo oophorectomy (adjusted HR, 1.42; 95% CI, 1.14–1.77; absolute risk increase, 6.6%).[677]

In general, the prevalence of hypertension and uncontrolled hypertension is higher among men. Men tend to consume more protein than women; the prevalence of dyslipidemias is greater in men than premenopausal women.

All these factors may contribute to the increased severity of renal disease observed in men but do not explain all the differences.[678,679] A recent review has also highlighted that disparities in access to health care between men and women in many regions may contribute to the observed differences in outcomes.[680] The impact of gender differences on the epidemiology of kidney disease is reviewed in more detail in Chapter 19.

NEPHRON ENDOWMENT

Experimental and clinical studies have shown that the number of nephrons per kidney is variable and may be influenced by several factors during development in utero. Furthermore low nephron endowment predisposes individuals to hypertension and CKD. This has been confirmed in studies using a rat model of spontaneous renal agenesis. Rats born with a single kidney had 19% fewer nephrons per kidney than their two-kidney littermates, resulting in a 60.2% reduction in nephron endowment that was associated with subsequent renal and glomerular hypertrophy, proteinuria, glomerular sclerosis, and TIF.[341] It has been proposed that reduced nephron endowment results in an increase in the single-nephron GFR and therefore a reduction in renal reserve.[681] Whereas the glomerular hemodynamic changes associated with mild to moderate congenital nephron deficiencies may not in themselves be sufficient to provoke renal injury, they could be predicted to compound the effects of an acquired nephron loss and predispose an individual to progressive renal damage. Thus CKD should be viewed as a multihit process, in which the first hit may be reduced nephron endowment.[682] The impact of developmental programming on nephron endowment, BP, and kidney function is discussed in detail in Chapter 21.

RACE AND ETHNICITY

Data from the USRDS have shown a substantially higher incidence of ESKD among African Americans, Hispanics, and Native Americans versus Caucasians. In 2015, the adjusted ESKD incidence rate ratios for Native Hawaiians/Pacific Islanders, African Americans, Native Americans/Alaska Natives, and Asians as compared with Caucasians were 8.4, 3.0, 1.2, and 1.0, respectively. Interestingly, these represent reductions in the relative risk of ESKD for these minorities compared with Caucasians over the past 15 years (Fig. 51.12). The rate ratio for Hispanics versus non-Hispanics was 1.3. Similarly, in 2015, the prevalence of ESKD per million population (pmp) was 14,448 among Native Hawaiians/Pacific Islanders, 5705 among African Americans, 2315 among Native Americans/Alaska Natives, 1905 among Asians, and 1519 among Caucasians.[683] The reasons for these obvious discrepancies are complex and include both social and biologic factors.[684,685] Interestingly, data from the Reasons for Geographic and Racial Differences in Stroke (REGARDS) Cohort Study have shown a lower prevalence of eGFR 50 to 59 mL/min/1.73 m^2, among African American versus Caucasian subjects but a higher prevalence of eGFR, 10 to 19 mL/min/1.73 m^2, suggesting that African Americans have a lower risk of developing CKD but a higher risk of progression of CKD to ESKD.[686]

African Americans appear to be more susceptible to FSGS. One retrospective analysis of 340 routine kidney biopsies

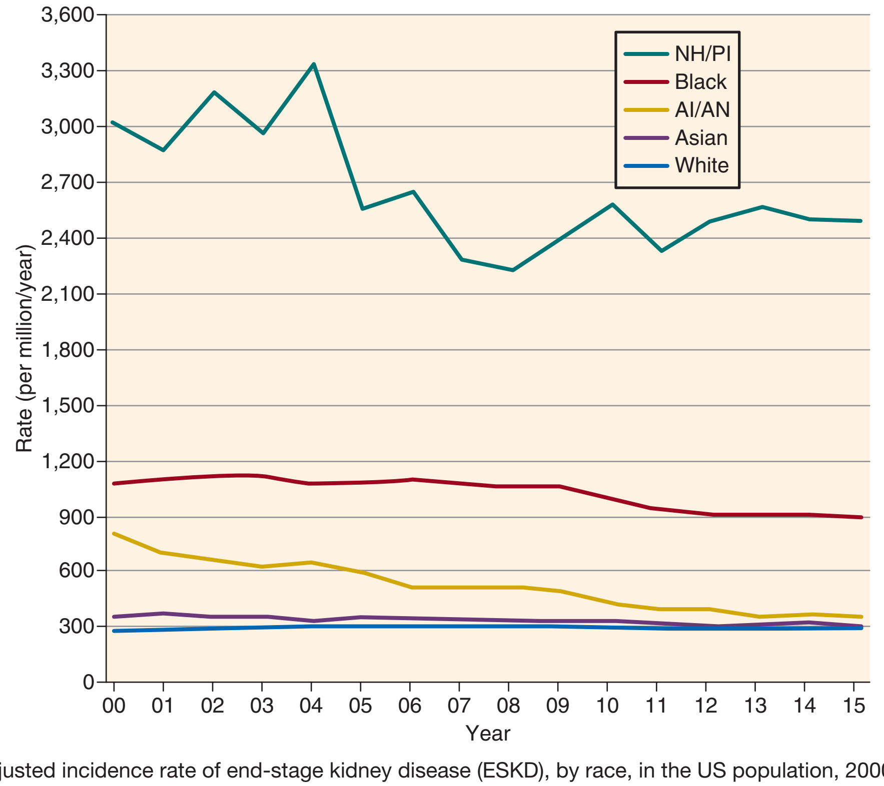

Fig. 51.12 Trends in adjusted incidence rate of end-stage kidney disease (ESKD), by race, in the US population, 2000–2015. *AI/AN,* American Indian/Alaska Native; *NH/PI,* Native Hawaiian/Pacific Islander. (From the US Renal Data System. Annual Data Report 2017, Chapter 1: Incidence, Prevalence, Patient Characteristics, and Treatment Modalities. https://www.usrds.org/2017.)

has detected a significantly higher prevalence of FSGS and a significantly lower prevalence of membranous glomerulonephritis, IgA, and immunotactoid nephropathies among African American versus Caucasian patients.[687] Similarly, among pediatric transplant recipients, a higher proportion of African American and Hispanic children had FSGS as a primary diagnosis versus Caucasians.[688] The same investigators found that despite similar treatment modalities and similar durations of nephrotic syndrome, African American children with FSGS reached ESKD almost twice as frequently as Caucasian children.[688]

More significant in terms of patient numbers and morbidity, however, are the racial discrepancies in the incidence of ESKD due to hypertensive and diabetic nephropathies. One longitudinal study that examined data from 1,306,825 Medicare beneficiaries has reported substantially increased risks of developing ESKD in African American versus Caucasian participants in all categories—among persons with diabetes, a 2.4- to 2.7-fold increase; among hypertensive persons, a 2.5- to 2.9-fold increase; and among persons with neither hypertension or diabetes, a 3.5-fold increase.[689] MDRD study data showed the prevalence of hypertension to be higher in African Americans versus Caucasians among persons with CKD, despite a higher mean GFR in the African American participants.[679] Hypertensive persons were found to have had more rapid progression of renal disease prior to entry into the study, suggesting that the higher prevalence of hypertension in African American patients is likely to be a significant contributor to the accelerated progression of CKD. On the other hand, both higher MAP and African American race/ethnicity were independent predictors of a faster decline in the GFR in the MDRD study.[690]

In a large community-based epidemiologic study, African Americans were found to have a 5.6 times higher unadjusted incidence of hypertensive ESKD.[691] This increased incidence was directly related to the prevalence of hypertension, severe hypertension, and diabetes in the study population and was inversely related to age at diagnosis of hypertension and socioeconomic status. After adjustment for these factors, the risk of hypertensive ESKD remained 4.5 times greater among African Americans compared with Caucasians, providing further evidence that African Americans have an increased susceptibility to renal disease beyond that attributable to their increased prevalence of hypertension and diabetes. Salt-sensitive hypertension, in particular, is more prevalent in the African American population than in the Caucasian population.[692] Comparing renal responses to a high sodium intake in salt-sensitive versus salt-resistant persons, RBF was found to decrease in the presence of an increased filtration fraction (implying an increased P_{GC}) in salt-sensitive patients, whereas the converse occurred in salt-resistant patients.[609]

These observations are consistent with the notion that salt loading injures the glomerulus through glomerular capillary hypertension and that salt-sensitive individuals, and African American subjects in particular, are at added risk of this form of injury. The incidence of ESKD due to diabetic nephropathy is fourfold higher in African Americans than among Caucasian Americans.[693] It is notable that after controlling for the higher prevalence of diabetes and hypertension, as well as age, socioeconomic status, and access to health care, the excess incidence of ESKD due to diabetes in African Americans versus Caucasians was confined to type 2 diabetics.[694] Among type 1 diabetics, African Americans were not found to be at higher risk than Caucasians. Indeed, most African Americans with diabetic ESKD (77%) had type 2 diabetes, whereas most Caucasians with diabetic ESKD (58%) had type 1 diabetes.[695] African American race/ethnicity was also found to be associated with a threefold higher risk of early renal function decline (increase in serum creatinine of ≥0.4 mg/dL) among adults with diabetes.[696]

Several potential factors contributing to the different prevalence and severity of renal disease among population groups have been analyzed. Adjustment for socioeconomic factors reduces, but does not eliminate, the increased risk of African Americans to develop ESKD.[685,693,696] African Americans generally have lower birth weights than their Caucasian counterparts and may therefore have programmed or genetically determined deficits in nephron number, rendering them more susceptible to hypertension and subsequent ESKD.[697,698] Finally, 40% of African American patients with hypertensive ESKD and 35% with type 2 diabetes-associated ESKD have a first-, second-, or third-degree relative with ESKD, implying a strong familial susceptibility to ESKD and therefore a genetic predisposition.[699]

Evidence of a genetic explanation for the high incidence of ESKD observed in African Americans was provided by research that identified a strong association between ESKD and two coding variants of the gene for apolipoprotein L1 (*APOL1*).[700] These gene variants confer resistance to infection with *Trypanosoma brucei rhodesiense*, which causes sleeping sickness, providing an explanation for how selection likely resulted in a high prevalence of these variants in the population.[701] Subsequent studies have identified associations between *APOL1* risk variants and several renal pathologies, including FSGS, HIV-associated nephropathy (HIVAN), sickle cell kidney disease, and severe lupus nephritis. *APOL1* risk variants have also been associated with HIVAN in black South Africans.[702] Moreover, cohort studies have reported associations between *APOL1* risk variants and the risk of progression to ESKD. The risk of progression was the lowest in European Americans (with no risk variants), intermediate in African Americans with no or one risk variant, and highest in African Americans with two risk variants.[703] It is estimated that *APOL1* variants account for 40% of disease burden due to CKD in African Americans.

The biologic role of *APOL1* in the progression of CKD remains to be fully elucidated. Expression of the *APOL1* risk variants in divergent species, *Drosophila* and *Saccharomyces* (a yeast), has identified impairment of conserved core intracellular endosomal trafficking processes as a key mechanism of cellular injury.[704] The APOL1 protein is expressed in the kidney but is also secreted and is bound to circulating HDL particles. Current evidence suggests that it is the locally expressed form of APOL1 that is involved in CKD pathogenesis.[705] Despite the strong association between the inheritance of two *APOL1* risk variants and ESKD, only a minority of people with this genotype actually develop kidney disease, suggesting that the action of a second factor is required to cause disease in genetically susceptible individuals. HIV is one example of such a second hit, but it has been proposed that other viruses and other gene variants may also be important.[701]

Other ethnic and racial minority groups, including Asians,[617,706] Hispanics,[707] Native Americans,[708] Mexican

Americans,[709] and Australian Aborigines,[710] have also been found to be at increased risk of developing CKD and ESKD. See Chapter 19 for a further discussion of ethnicity and the epidemiology of CKD and Chapters 43 and 44 for a detailed discussion of genetic factors in the pathogenesis of CKD.

OBESITY AND METABOLIC SYNDROME

Obesity may directly cause a glomerulopathy characterized by proteinuria and histologic features of focal and segmental glomerulosclerosis,[711] but it is likely that it also exacerbates the progression of other forms of CKD. Micropuncture studies have confirmed that obesity is another cause of glomerular hypertension and hyperfiltration that may contribute to the progression of CKD.[712,713] Griffin and colleagues have pointed out that whereas obesity is widespread, only a minority of obese individuals develop obesity-related glomerulopathy.[714] They have proposed that low nephron endowment (associated with low birth weight, which is also associated with an increased risk of later life obesity) or acquired nephron loss constitute a necessary additional factor that increases glomerular hypertrophy and preglomerular vasodilation and transmission of elevated systemic BP to the glomerulus, leading to glomerulosclerosis. A detailed investigation of adipocyte function has revealed that they are not merely storage cells but produce a variety of hormones and proinflammatory molecules that may contribute to progressive renal damage.[715,716] In addition, adiponectin, an adipokine produced by adipocytes, may exert a protective effect on podocyte function. Adiponectin levels are reduced in obesity and are inversely correlated with the magnitude of albuminuria.[717] Moreover, adiponectin knockout mice develop albuminuria associated with podocyte foot process effacement that is corrected by the administration of exogenous adiponectin.[718]

Obesity is also associated with increased production of aldosterone and expanded extracellular volume, both of which may contribute to progressive renal damage.[719]

In humans, severe obesity is associated with increased renal plasma flow, glomerular hyperfiltration, and albuminuria, abnormalities that are reversed by weight loss.[720] Several large population-based studies have identified obesity as an independent risk factor for developing CKD,[614,721] and one study has found a progressive increase in relative risk (RR) of developing ESKD associated with increasing body mass index (BMI; RR, 3.57; 95% CI, 3.05–4.18 for BMI 30.0–34.9 kg/m^2 vs. BMI 18.5–24.9 kg/m^2) among 320,252 subjects with no evidence of CKD at initial screening.[722]

On the other hand, analysis of data from the Framingham Heart Study has confirmed an increased risk of developing CKD stage 3 associated with obesity but found that this was no longer significant after adjustment for known cardiovascular risk factors. Nevertheless, the increased risk of incident proteinuria persisted in multivariable models.[723]

Change in body weight has also been identified as a risk factor for incident CKD. In one study of 8792 previously healthy men, an increase in body weight of 0.75 or more kg/year (and a decrease of <0.75 kg/year) was associated with an increased risk of developing CKD in previously obese and nonobese subjects.[724] The metabolic syndrome (insulin resistance) defined by the presence of abdominal obesity, dyslipidemia, hypertension, and fasting hyperglycemia is also associated with an increased risk of developing CKD. Analysis of the Third National Health and Nutrition Examination Survey (NHANES) data has revealed a significantly increased risk of CKD and microalbuminuria in subjects with the metabolic syndrome, as well as a progressive increase in risk associated with the number of components of the metabolic syndrome present.[725] Furthermore, a longitudinal study of 10,096 patients without diabetes or CKD at baseline has identified metabolic syndrome as an independent risk factor for the development of CKD over 9 years (adjusted OR, 1.43; 95% CI, 1.18–1.73). Again, there was a progressive increase in risk associated with the number of traits of the metabolic syndrome present (OR, 1.13; 95% CI, 0.89–1.45 for one trait versus OR, 2.45; 95% CI, 1.32–4.54 for five traits).[726] The metabolic syndrome has also been identified as a risk factor for incident CKD among Native Americans[727] and in Taiwan.[728] The waist-hip ratio (WHR), a marker of central fat distribution and insulin resistance, was independently associated with impaired renal function, even in lean individuals (BMI <25 kg/m^2) among a population-based cohort of 7676 subjects.[729] Furthermore, analysis of data from the Atherosclerosis Risk in Communities (ARIC) study has found that WHR but not BMI is associated with an increased risk of incident CKD and mortality,[730] suggesting that accumulation of visceral fat is more important as a risk factor than obesity per se.

The effect of obesity on progression in cohorts of patients with established CKD has been less well documented. In one study, increased BMI was an independent predictor of CKD progression among 162 patients with IgA nephropathy,[731] and BMI was independently associated with more rapid CKD progression in a cohort of CKD subjects (predominantly CKD stage 3).[732] On the other hand, obesity may be less relevant for progression in more advanced stages of CKD, as evidenced by the observation that BMI was unrelated to the risk of ESKD among a cohort of patients with CKD stages 4 and 5.[674] The association of obesity with CKD and increased risk of CKD progression suggests that weight loss may represent an important intervention for achieving renoprotection. Sustained weight loss is notoriously difficult to achieve, but a metaanalysis of several small, short-duration studies has found that weight loss achieved through diet or medication is associated with a reduction in proteinuria and BP. Surgical procedures to achieve weight loss in the morbidly obese were associated with normalization of glomerular hyperfiltration and reductions in microalbuminuria and BP.[733] A more recent systematic review has analyzed the effects of weight loss achieved by bariatric surgery, medication, or diet in 31 studies and found that in most studies, weight loss is associated with reductions in proteinuria. In people with glomerular hyperfiltration, the GFR tended to decrease with weight loss and, in those with a reduced GFR, it tended to increase.[734]

Clinical Relevance

Obesity and Metabolic Syndrome

Weight loss, because it is difficult to achieve, is an underutilized intervention to improve renoprotection. Nevertheless, available evidence indicates that in persons with obesity, weight loss affords renoprotection. In persons with severe obesity who are unable to achieve adequate weight loss with diet and exercise, bariatric surgery should be considered.

The increasing use of bariatric surgery to manage obesity has allowed investigation of the impact of surgically induced weight loss on CKD. In a long-term study of 2144 persons who underwent bariatric surgery, change in CKD risk category, as defined by the KDIGO classification, was assessed after 7 years. Among those with moderate risk at baseline, the risk category improved in 53% and deteriorated in 5% to 8%, in the high-risk group, improvement was observed in 56% and deterioration in 3% to 10%, and, in the very-high-risk group, 23% improved. The eGFR initially improved, with a peak at 2 years, and then gradually declined. Albuminuria showed large and sustained decreases in the moderate- (median urine ACR decreased from 48 to 14 mg/g) and high-risk groups (median urine ACR decreased from 326 to 26 mg/g).[735]

In another study comparing outcomes in persons with CKD stages 3 and 4 undergoing bariatric surgery with similarly obese propensity-matched persons who did not undergo surgery, the eGFR was significantly higher in the surgery group after 3 years by a mean of 9.84 mL/min/1.73 m^2.[736] In a metaanalysis that included 23 cohort studies and 3015 persons who underwent all forms of bariatric surgery, small but statistically significant improvements were observed in serum creatinine levels (mean decrease, 0.08 mg/dL) and proteinuria (mean decrease, 0.04 g/day).[737] Obesity therefore appears to be a modifiable risk factor for CKD progression.

SYMPATHETIC NERVOUS SYSTEM

Overactivity of the sympathetic nervous system has been observed in patients with CKD, and several lines of evidence have suggested that this may be a factor contributing to progressive renal injury.[738] The kidneys are richly supplied with afferent sensory, and efferent sympathetic innervation and may therefore act as both a source and target of sympathetic activation. That the former is true has been suggested by a study that compared postganglionic sympathetic nerve activity (SNA) measured via microelectrodes in the peroneal nerve in normal individuals and hemodialysis patients who retained their native kidneys with those who had undergone bilateral nephrectomy.[739] SNA was 2.5 times higher in nonnephrectomized dialysis patients compared with both normals and nephrectomized patients in whom SNA was similar. Furthermore, increased SNA was associated with increased vascular tone and mean arterial BP in nonnephrectomized patients. SNA did not vary as a function of age, BP, antihypertensive agents, or body fluid status. The authors speculated that intrarenal accumulation of uremic compounds stimulates renal afferent nerves via chemoreceptors, leading to reflex activation of efferent sympathetic nerves and increased SNA. Other studies, however, have observed increased SNA in the absence of uremia in patients with renovascular disease,[740] hypertensive ADPKD,[741] and nondiabetic CKD[742] or increased noradrenaline secretion in patients with nephrotic syndrome[743] and ADPKD.[741,744] Furthermore, correction of uremia by renal transplantation does not abrogate the increased SNA.[745] Interestingly, investigation of eight living kidney donors found no increase in SNA after donor nephrectomy, suggesting that the rise in SNA is related to renal damage rather than nephron loss.[742]

Together, these findings suggest that that a variety of forms of renal injury may provoke increased SNA and that uremia is not required for this response. Evidence from experimental studies has indicated that sympathetic overactivity resulting from renal disease may also accelerate renal injury. Ablation of afferent sensory signals from the kidneys by bilateral dorsal rhizotomy in 5/6 nephrectomized rats prevented the expected rise in systemic BP, attenuated the rise in serum creatinine levels, and reduced the severity of glomerulosclerosis in the remnant kidneys when compared with sham rhizotomized controls.[746] Moreover, the renoprotective effects of rhizotomy were found to be additive to that of ACE inhibitor treatment.[747]

To investigate further whether the benefits of rhizotomy were solely attributable to the prevention of hypertension, 5/6 nephrectomized rats were treated with nonhypotensive doses of the sympatholytic drug moxonidine.[748] Despite the lack of effect on BP, moxonidine treatment was associated with lower levels of proteinuria and less severe glomerulosclerosis in treated compared with untreated rats. In a similar study, 5/6 nephrectomized rats were treated with the alpha blocker phenoxybenzamine, the beta blocker metoprolol, or a combination.[749] As in the previous study, the doses used did not lower BP, but all three treatments significantly lowered albuminuria and almost normalized the reductions in capillary length density (an index of glomerular capillary obliteration) and podocyte number. Metoprolol and combination therapy significantly lowered the glomerulosclerosis index versus untreated controls. Taken together, these results indicate that increased SNA accelerates renal injury independently of its effect on BP and that the adverse effects are mediated by catecholamines.[750] Furthermore, sympathetic nerve overactivity has been proposed to contribute to the development of tubulointerstitial injury by reducing peritubular capillary perfusion to the extent that tubular and interstitial ischemia result.[751]

Preliminary evidence has suggested that sympathetic overactivity may also be important in the progression of human CKD. Among patients with type 1 diabetes mellitus and proteinuria, evidence of parasympathetic dysfunction—which permits unopposed sympathetic tone—was associated with an increase in serum creatinine levels over the next 12 months.[752] Analysis of data from the Atherosclerosis Risk in Communities (ARIC) study has found that a higher resting heart rate and reduced heart rate variability, markers of autonomic dysfunction, are each independently associated with an increased risk of developing ESKD or a CKD-related hospital admission during 16 years of follow-up.[753] Several drug treatments may improve sympathetic overactivity in CKD patients. Among 15 normotensive type 1 diabetics, 3 weeks of treatment with moxonidine significantly lowered albumin excretion rates without affecting BP.[754] In other studies, chronic treatment with an ACE inhibitor or ARB, of proven benefit in renoprotection, was associated with reduction but not normalization of sympathetic overactivity.[755–757] In contrast, treatment with amlodipine was associated with increased SNA. Because ACE inhibitors and ARBs do not readily enter the central nervous system (CNS), it is possible that RAAS inhibition modulates neurotransmitter release in the kidney and reduces afferent signaling.

Finally, renal denervation achieved by radiofrequency catheter ablation through the renal artery has been proposed as an intervention to treat resistant hypertension and possibly slow CKD progression. Following several small studies that reported reductions in BP following renal denervation, a large randomized trial was undertaken in which 535 participants were randomly assigned to renal denervation or a sham procedure. After 6 months, a significant reduction in office BP was observed in both groups, but there was no difference between groups. There was also no difference in 24-hour ambulatory BP between groups.[758]

Several questions remain to be answered regarding the role of increased SNA in CKD progression. Whereas the renoprotective effects of sympatholytic drugs appear to be independent of effects on systemic BP, it is as yet unknown what effect they have on glomerular hemodynamics. Further studies are also required to determine the extent to which chronic inhibition of sympathetic overactivity may be beneficial in a variety of forms of human CKD and whether this benefit is additive to that derived from inhibition of the RAAS.

DYSLIPIDEMIA

Moorhead and colleagues advanced the hypothesis that abnormalities in lipid metabolism may contribute to the progression of CKD.[759] Glomerular injury, accompanied by an alteration in basement membrane permeability, was envisaged as the initiator of a vicious cycle of hyperlipidemia and progressive glomerular injury. They proposed that urinary losses of albumin and lipoprotein lipase activators result in an increase in circulating LDLs, which in turn bind to the glomerular basement membrane, further impairing its permselectivity. Filtered lipoproteins accumulate in the mesangium, stimulating extracellular matrix synthesis and mesangial cell proliferation; filtered LDL is taken up and metabolized by the tubules, leading to cell injury and interstitial disease. Notably, this hypothesis did not propose hyperlipidemia as an initiating factor in renal injury, but rather as a participant in a self-sustaining mechanism of disease progression.

Several lines of experimental evidence have confirmed the association between dyslipidemia and renal injury. Both intact and uninephrectomized rats with dietary-induced hypercholesterolemia developed more extensive glomerulosclerosis than their normocholesterolemic controls, and the severity of glomerulosclerosis correlated with serum cholesterol levels[760]; aging female Nagase analbuminemic rats (NARs) have endogenous hypertriglyceridemia and hypercholesterolemia and develop proteinuria and glomerulosclerosis by 9 and 18 months of age, respectively, whereas male NARs have lower lipid levels and no glomerulosclerosis by 22 months of age.[665] Interestingly, ovariectomy in female NARs lowers triglyceride levels and reduces their renal injury. In seeming contradiction, however, young and aging male Sprague-Dawley rats developed more extensive glomerulosclerosis than age- and gender-matched NARs, despite higher cholesterol levels in the NARs.[761] Triglyceride levels, however, were lower in the NARs, again suggesting an independent role for triglycerides in lipid-mediated renal injury.

Whereas data regarding the role of lipids in initiating renal disease are conflicting, several studies have supported the notion that dyslipidemia may promote renal damage. Cholesterol feeding has been shown to exacerbate glomerulosclerosis in uninephrectomized rats, prediabetic rabbits, rats with puromycin aminonucleoside nephropathy, and in the unclipped kidney of rats with two-kidney, one-clip (2-K,1C) hypertension. When hypertension and dyslipidemia are superimposed, a synergistic effect that dramatically accelerates renal functional deterioration is observed.[762,763] In the 5/6 nephrectomy model, progressive renal damage is associated with renal tissue accumulation of lipids as well as upregulation of pathways involved in the tubular reabsorption of protein-bound lipids and downregulation of pathways involved in lipid catabolism.[764]

In humans, the role of lipids in the initiation and progression of renal disease remains unclear. At autopsy, a highly significant correlation was found between the presence of systemic atherosclerosis and the percentage of sclerotic glomeruli in normal individuals, fostering speculation that the development of glomerulosclerosis may be analogous to that of atherosclerosis.[765] A study designed to identify the clinical correlates of hypertensive ESKD has found a strong association between atherosclerosis and hypertensive ESKD among older Caucasian patients.[766] Furthermore, dyslipidemia has been identified in several large studies as a risk factor for the subsequent development of CKD in apparently healthy individuals.[614,767,768] The common forms of primary hypercholesterolemia are not associated with an increased incidence of renal disease in the general population, but renal injury has been described in association with rare inherited disorders of lipoprotein metabolism.[769,770]

Whereas primary lipid-mediated renal injury is rare among patients with CKD, the latter is frequently accompanied by elevations in serum lipid levels as a result of urinary loss of albumin and lipoprotein lipase activators, defective clearance of triglycerides, modification of LDLs by advanced glycation end products, reduced plasma oncotic pressure, adverse effects of medication, and underlying systemic diseases.[771,772] Among a cohort of adult patients with CKD, the most frequent lipid abnormalities noted were hypertriglyceridemia, low HDL levels, and increased apolipoprotein levels.[773] Furthermore, in a study of 631 routine renal biopsies, lipid deposits were detected in nonsclerotic glomeruli in 8.4% of kidneys, and staining for apolipoprotein B was positive in approximately 25% of biopsies, suggesting that lipid deposition is not infrequent in diverse renal diseases.[404] Several epidemiologic studies have found a strong association between CKD progression and dyslipidemia.

- In the MDRD study, low serum HDL cholesterol was found to be an independent predictor of more rapid rates of decline in the GFR.[774]
- Elevated total cholesterol, LDL cholesterol, and apolipoprotein B have been found to correlate strongly with the GFR decline in CKD patients.[775]
- Hypercholesterolemia was shown to be a predictor of loss of renal function in type 1 and 2 diabetes.[776,777]
- Among nondiabetic patients, CKD advanced more rapidly in patients with hypercholesterolemia and hypertriglyceridemia, independently of BP control.[778]
- Among patients with IgA nephropathy, hypertriglyceridemia was independently predictive of progression.[779]

However, not all studies have confirmed these findings:

- In the Multiple Risk Factor Intervention Trial (MRFIT), dyslipidemias were not associated with a decline in renal function.[780]

- After 10 years of follow-up in the MDRD study, measures of dyslipidemia were not predictive of cardiovascular events or ESKD.[781]
- In a retrospective analysis of patients with nephrotic syndrome, hypercholesterolemia at diagnosis was not found to be a predictor of renal disease progression.[782]
- In the CRIC study. no association was observed between total or LDL cholesterol and the risk of ESKD or a 50% reduction in the eGFR.[783]

Interpretation of these data is complicated by the fact that in patients with renal insufficiency, dyslipidemias do not occur in isolation and are associated with other factors that also affect renal disease progression, including hypertension, hyperglycemia, and proteinuria. Levels of serum cholesterol and triglycerides have been found to correlate with BP and circulating Ang II levels in type 1 and 2 diabetes with renal disease and to rise with increasing proteinuria in patients with nephrotic syndrome.[770]

The possible mechanisms whereby hyperlipidemia may contribute to renal injury have not been fully elucidated. Cholesterol feeding has been associated with an increase in mesangial lipid content,[760] glomerular macrophages, and TGF-β, as well as fibronectin mRNA levels.[784,785] Furthermore, reduction of glomerular macrophages by whole-body x-irradiation in the setting of nephrotic syndrome significantly reduced albuminuria without affecting serum lipid levels, indicating that macrophages play a central role in hyperlipidemic glomerular injury.[785] Mesangial cells express receptors for LDL, and uptake is stimulated by vasoconstrictor and mitogenic peptides such as ET-1 and PDGF.[407] Metabolism of LDL by mesangial cells leads to increased synthesis of fibronectin and MCP-1, which may contribute to mesangial matrix expansion and the recruitment of circulating macrophage/monocytes into the glomerulus.[408] Moreover, triglyceride-rich lipoproteins (very-low-density lipoprotein, [VLDL] and intermediate-density lipoprotein [IDL]) induce mesangial cell proliferation and elaboration of IL-6, PDGF, and TGF-β in vitro.[786] Mesangial cells, macrophages and renal tubule cells all have the capacity to oxidize LDL via the formation of ROS, a step that may be inhibited by antioxidants and HDL.[396,787,788] Oxidized LDLs may induce dose-dependent mesangial cell proliferation or mesangial cell death, as well as production of TNF-α, eicosanoids, monocyte chemotaxins, and glomerular vasoconstriction. These pathways, together with free radicals generated during LDL oxidation, may each contribute to renal inflammation and injury.[786,787]

Hyperlipidemia is also associated with elevated P_{GC}, raising the possibility of a further pathway to glomerulosclerosis via hemodynamic injury.[760] The elevated P_{GC} appears to be mediated, in part, by an increase in renal vascular resistance that occurs in the context of increased plasma viscosity. In diabetic patients, circulating Ang II levels have been found to correlate with serum cholesterol,[789] and both oxidized LDLs and lipoprotein(a) have been shown to stimulate renin production by juxtaglomerular cells in vitro.[788] Moreover, oxidized LDL has been found to reduce nitric oxide synthesis by endothelial cells,[788] raising the possibility that alterations in activity of the RAAS and nitric oxide metabolism could also contribute to the increase in P_{GC} observed with hyperlipidemia.

It would follow that if hyperlipidemia exacerbates renal injury, interventions designed to lower serum lipid levels should ameliorate disease progression. Treatment with a 3-hydroxyl-3-methylglutaryl coenzyme A (HMG-CoA) reductase inhibitor (statin) or clofibric acid in the obese Zucker rat, a strain with endogenous hyperlipidemia and spontaneous glomerulosclerosis, and 5/6 nephrectomized rats, which develop hyperlipidemia secondary to renal insufficiency, resulted in lowering of serum lipid levels, reduction in albuminuria, reduction in mesangial cell DNA synthesis, and attenuation of glomerulosclerosis, despite a lack of effect on either systemic BP or P_{GC}.[12,790] Indeed, statin treatment resulted in additional lowering of proteinuria, regression of glomerulosclerosis, normalization of podocyte number, and abrogation of tubulointerstitial injury when added to combination ACE inhibitor and ARB treatment.[791] In rats in the nephrotic phase of PAN, statin treatment resulted in reduction of albuminuria and serum cholesterol levels, reduction of MCP-1 mRNA expression, and a 77% reduction in glomerular macrophage accumulation.[792] The statins may therefore exert beneficial effects on renal disease progression, not only by reducing serum lipid levels, but also by inhibiting mesangial cell proliferation and mechanisms for the recruitment of macrophages due to the decreased expression of chemotactic factors and cell adhesion molecules.[793] Cholesterol-fed rats with puromycin aminonucleoside nephropathy, treated with the antioxidants probucol or vitamin E, showed significant reductions in proteinuria and glomerulosclerosis compared with untreated controls.[794] Furthermore, plasma VLDL and LDL from the treated animals were less susceptible to in vitro oxidation and less renal lipid peroxidation was evident, implying that lipid peroxidation plays an important role in renal injury associated with hyperlipidemia.

Niacin treatment after 5/6 nephrectomy resulted in lower BP, less proteinuria, less renal tissue accumulation of lipids, and attenuation of tubulointerstitial injury. This indicates that lipid lowering through strategies other than statins may also be renoprotective.[795]

In some clinical studies, dietary or pharmacologic lowering of serum lipid levels has also been associated with a reduction in proteinuria and lower rates of decline in renal function, but other studies have failed to demonstrate similar benefits, despite adequate therapeutic reductions in serum lipid levels. A metaanalysis of 13 small studies that included both diabetic and nondiabetic renal disease has found that lipid-lowering therapy significantly reduces the rate of decline in the GFR (mean reduction of 1.9 mL/min/1.73 m^2 per year).[796] Several secondary analyses of data from clinical trials have suggested that lipid-lowering therapy may slow progression in human CKD, but these data should be interpreted with caution.[797,798] In a placebo-controlled open-label study, atorvastatin treatment in patients with CKD, proteinuria, and hypercholesterolemia was associated with preservation of creatinine clearance, whereas those receiving placebo evidenced a significant decline.[799] In a metaanalysis of studies in which predialysis patients with CKD were randomized to therapy with a statin, and analysis of data from a relatively small subgroup in which renal endpoints were available, statin therapy was associated with a reduction in proteinuria but no improvement in creatinine clearance.[800]

Whereas these renoprotective effects were associated with cholesterol lowering, it is possible that they may also be due to the direct pleiotropic effects of HMG-CoA reductase inhibitors. This notion is further supported by the observation that lipid lowering with fibrates was not associated with preservation of renal function,[801,802] although one study did show reduced progression to microalbuminuria among type 2 diabetics receiving fenofibrate.[803] The Study of Heart and Renal Protection (SHARP) investigated the cardiovascular and renoprotective effects of lipid lowering with simvastatin and ezetimibe in 9438 subjects with CKD and ESKD.[804] Whereas the treatment arm showed a mean reduction of 43 mg/dL in LDL cholesterol and a 17% reduction in major atherosclerotic events, no significant effect was observed on the incidence of the renal endpoints in ESKD (risk ratio, 0.97; 95% CI, 0.89–1.05) and ESKD or creatinine doubling (risk ratio, 0.93; 95% CI, 0.86–1.01). It should be noted, however, that the subjects with CKD had relatively advanced disease (mean eGFR, 27 ± 13 mL/min/1.73 m^2), and these observations therefore do not exclude the possibility that lipid lowering may have renoprotective effects in less advanced CKD. However, a metaanalysis of 38 studies that included 37,274 participants with CKD has found that statin therapy is associated with a reduction in mortality and cardiovascular events but no clear effect on CKD progression.[805]

Inhibitors of the enzyme, proprotein convertase subtilisin/kexin type 9 (PCSK9), a key regulator of hepatic LDL receptor recycling, have been developed as novel potent treatments to lower cholesterol levels in persons with familial hyperlipidemias or those resistant to treatment with statins. Randomized trials to assess the renoprotective effects of PCSK9 inhibitors have not yet been published, but a pooled analysis of participants with CKD (eGFR, 30–59 mL/min/1.73 m^2; mean eGFR, 51 mL/min.1.73 m^2) in eight trials evaluating a monoclonal antibody to PCSK9, alirocumab, reported that treatment was associated with substantial reductions in LDL cholesterol and triglyceride levels, as well as other lipids versus placebo or ezetimibe. Alirocumab treatment was not associated with increased adverse events versus control groups, and no difference in eGFR was observed between groups at 24 or 104 weeks.[806] It should be noted, however, that the GFR was monitored for safety and not to assess renoprotection. Ongoing long-term trials will investigate the potential benefits of PCSK9 inhibitors on cardiovascular outcomes and will likely include participants with CKD.

CALCIUM AND PHOSPHATE METABOLISM

As is the case with many of the adaptations that follow nephron loss, evidence has been accumulating that alterations in calcium and phosphate metabolism may also contribute to progressive renal damage. A retrospective analysis of 15 patients with nonprogressing CKD (GFR, 27–70 mL/min/1.73 m^2, followed for up to 17 years) revealed that the single feature common to all these patients was an enhanced capacity to excrete phosphate when compared with patients with a similar GFR but progressive renal disease.[807] In all the nonprogressors, serum phosphate and calcium levels remained within normal limits without the use of phosphate binders, calcium supplementation, or vitamin D. It is not yet clear which factors are most important, but evidence suggests that hyperphosphatemia, renal calcium deposition, hyperparathyroidism, and activated vitamin D deficiency may each play a role. FGF23 has recently been identified as a key mediator of bone mineral and vitamin D metabolism in CKD and may emerge as the dominant factor.

HYPERPHOSPHATEMIA

Uninephrectomized rats receiving a high-phosphate diet (1%) developed renal calcium and phosphate deposition and tubulointerstitial injury within 5 weeks of nephrectomy.[239] Similar changes were observed in a proportion of intact rats fed a 2% phosphate diet. In clinical studies, serum phosphate has been identified as an independent risk factor for CKD progression[808] that may also attenuate the efficacy of ACE inhibitor treatment in the context of proteinuric nephropathies.[809] Phosphate excess therefore does appear to have some intrinsic nephrotoxicity that is enhanced in the setting of reduced nephron number. A high-phosphate diet has also been associated with the development of parathyroid hyperplasia and hyperparathyroidism in remnant kidney rats.[810] Conversely, in both animals and humans with renal insufficiency, dietary phosphate restriction or treatment with oral phosphate binders has been associated with reductions in proteinuria and glomerulosclerosis and attenuation of disease progression, as well as prevention of hyperparathyroidism.[811–814] A further mechanism whereby hyperphosphatemia may contribute to kidney damage has been suggested by the observation that a high-phosphate diet increases, and phosphate binder therapy decreases, renal expression of ACE after 5/6 nephrectomy.[815] Dietary phosphate restriction, however, almost inevitably also imposes dietary protein restriction. It is therefore not clear whether the benefit was derived directly from the reduced phosphate intake or indirectly from protein restriction. One study in humans has reported additional renoprotection when phosphate restriction is superimposed on protein restriction.[816]

RENAL CALCIUM DEPOSITION

Calcium phosphate deposition is a frequent histologic finding in end-stage kidney biopsies, irrespective of the underlying cause of renal failure.[249,817] Calcium levels in end-stage kidneys have been found to be approximately nine times greater than levels in control kidneys.[817] Histologically, deposits were seen in cortical tubule cells, basement membranes, and the interstitium.[817,818] Furthermore, the severity of renal parenchymal calcification has been found to correlate with the degree of renal dysfunction, implicating calcium phosphate deposition in disease progression.[811,819] To determine whether the calcium deposits observed in end-stage kidneys precede or follow renal parenchymal fibrosis, rats with a reduced renal mass were maintained on a high-phosphate diet, thus ensuring a high calcium phosphate product. A subgroup was treated with 3-phosphocitrate, an inhibitor of calcium phosphate deposition.[819] Treatment with 3-phosphocitrate led to a significant reduction in renal injury compared with controls, indicating that calcium phosphate deposition in the kidney occurs during the evolution of renal injury and may exacerbate nephron loss. Calcium deposition in the renal parenchyma is associated with ultrastructural evidence of mitochondrial disorganization and calcium accumulation[818] and may therefore contribute to renal injury via the uncoupling of mitochondrial respiration and generation of ROS.[820] Mitochondrial calcium deposition was reduced by dietary protein restriction or calcium channel blocker therapy.[818,820] Other potential roles for calcium in renal disease progression include effects on vascular smooth muscle tone, mesangial cell contractility, cell growth and proliferation, extracellular matrix synthesis, and immune cell modulation.[821]

HYPERPARATHYROIDISM

Podocytes express a unique transcript of PTH receptor, and PTH has been shown to have several effects on the kidney, including decreasing SNGFR (without change in Q_A, P_{GC}, or ΔP), lowering K_f, and stimulating renin production.[814] Furthermore, increased PTH levels may exacerbate renal damage through effects on BP,[822] glucose intolerance, and lipid metabolism.[823,824] Two experimental studies have provided evidence that PTH may contribute to CKD progression. In the first study, parathyroidectomy was shown to improve survival, ameliorate the increase in renal mass, as well as renal calcium content, and attenuate the rise in serum creatinine levels observed in 5/6 nephrectomized rats fed a high-protein diet.[825] In the other, calcimimetic treatment and parathyroidectomy after 5/6 nephrectomy each abrogated TIF and glomerulosclerosis.[826] Interpretation of these data are, however, complicated by the observation in the latter study that both interventions also lowered BP.

ACTIVATED VITAMIN D DEFICIENCY

It is perhaps not surprising that vitamin D, normally 1-hydroxylated in the kidney and therefore reduced in CKD, has several potentially beneficial effects on the kidney. Several experiments have reported amelioration of renal damage in rats treated with $1,25(OH)_2D_3$ or vitamin D analogue after 5/6 nephrectomy.[827–829] Interestingly, a further study found that $1,25(OH)_2D_3$ treatment also preserved podocyte number, volume, and structure after 5/6 nephrectomy.[327] In other experimental models, vitamin D or vitamin D analogues have been shown to abrogate interstitial inflammation by promoting the sequestration of NF-κB signaling,[830] inhibit renal hypertrophy after uninephrectomy,[831] reduce renin[814,832] and TGF-β expression,[829] and restore glomerular filtration barrier structure as well as slit diaphragm protein expression.[832] Several small trials have reported reductions in proteinuria among patients with diabetic[833] and nondiabetic CKD[834,835] randomized to treatment with the vitamin D analogue paricalcitol, but larger long-term studies are required to evaluate the potential renoprotective effects of vitamin D replacement further.

FIBROBLAST GROWTH FACTOR 23

FGF23 has been identified as a key regulator of the bone mineral and vitamin D changes observed in CKD and may also contribute to CKD progression, as well as mediate some of the adverse cardiovascular consequences associated with CKD. It is produced by osteoblasts and osteocytes; levels rise early in the course of CKD. FGF23 is stimulated chiefly by $1,25(OH)_2D_3$ and dietary phosphate intake.[836] Its chief actions are to reduce phosphate reabsorption in the proximal tubule by downregulating sodium phosphate cotransporters and reduce $1,25(OH)_2D_3$ levels by inhibiting renal $25(OH)D_3$ 1α-hydroxylase, as well as stimulating the catabolic $25(OH)D_3$ 24-hydroxylase.[836] Thus decreased phosphate excretion early in the course of CKD stimulates FGF23 production, which increases phosphate excretion to prevent hyperphosphatemia until late in the course of CKD. This response is achieved at the expense of low $1,25(OH)_2D_3$ levels, which in turn facilitate the development of secondary hyperparathyroidism.[247,837] In addition to its role in bone mineral metabolism, longitudinal studies have identified FGF23 as an independent risk factor for mortality in hemodialysis patients[838,839] and in persons with earlier stage CKD,[840] although one group has suggested that the relationship may not be causal.[841]

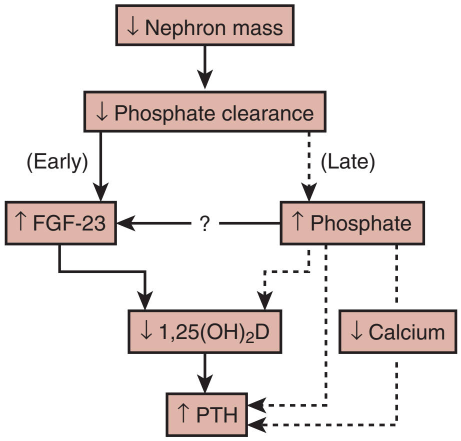

Fig. 51.13 Schematic explaining the possible interactions among phosphate, vitamin D ($1,25(OH)_2D$, fibroblast growth factor 23 (FGF-23), and parathyroid hormone *(PTH)* in regulating serum phosphate and calcium levels after nephron loss. (From Gutierrez OM: Fibroblast growth factor 23 and disordered vitamin D metabolism in chronic kidney disease: updating the "trade-off" hypothesis. *Clin J Am Soc Nephrol.* 2010;5: 1710–1716.)

Several studies have also identified FGF23 as an independent risk factor for CKD progression in diabetics[842] and nondiabetics,[843] including African Americans[844] and children with CKD.[845] Whether FGF23 contributes directly to CKD progression or is simply a risk marker remains to be elucidated. Indeed, overexpression of mutant FGF23 in the Thy1 model of glomerulonephritis resulted in lowering of serum phosphate levels and amelioration of glomerulosclerosis.[846] Possible mechanisms whereby FGF23 may contribute to CKD progression include exacerbation of $1,25(OH)_2D_3$ deficiency and secondary hyperparathyroidism (Fig. 51.13; see sections earlier).

ANEMIA

Anemia is a frequent consequence of CKD but may also influence its progression. Both acute and chronic anemia are associated with reversible increases in renal vascular resistance and a normal or reduced filtration fraction in animals and humans. Conversely, an increase in hematocrit is associated with an increase in filtration fraction. Thus hematocrit may influence renal hemodynamics and thereby affect the rate of progression of CKD. The effects of anemia on glomerular hemodynamics have been studied in rats subjected to 5/6 nephrectomy, DOCA-salt hypertension, and diabetes.[847–849] Irrespective of the model, anemia was associated with significant amelioration of glomerulosclerosis and a reduction in P_{GC}.

Reduced P_{GC} resulted predominantly from reductions in efferent arteriolar resistance in rats with renal ablation, lowered SBP in DOCA-salt rats, and increased afferent arteriolar resistance in diabetic rats. Similarly, in the MWF/Ztm rat, which develops spontaneous glomerulosclerosis with age, anemia induced by dietary iron deficiency was associated with lower BP, reduced urinary protein excretion, and less extensive glomerulosclerosis compared with controls fed diets of normal iron content.[850] In contrast, prevention of anemia by the administration of erythropoietin to remnant kidney rats to maintain a normal hematocrit resulted in increased systemic and glomerular BPs, as well as markedly increased glomerulosclerosis.[847] In another apparently contradictory study, treatment with epoetin delta after 5/6 nephrectomy was associated with slower rates of decline in renal function, decreased renal fibrosis, and less interstitial macrophage accumulation.[851] Interestingly, these effects were observed at subhemopoietic doses, indicating that they may have resulted from direct actions of the epoetin rather than anemia correction.

Despite the apparently favorable hemodynamic effects of anemia in experimental models of CKD, humans studies have suggested that anemia may accelerate CKD progression. In patients with inherited hemoglobinopathies, chronic anemia is associated with glomerular hyperfiltration that eventuates in proteinuria, hypertension, and ESKD.[852,853] Furthermore, reduced hemoglobin level was an independent predictor of increased risk of developing ESKD among patients with diabetic nephropathy in the RENAAL trial.[854] Several longitudinal studies of patients with other forms of CKD have identified lower hemoglobin levels as a risk factor for progression.[855,856]

Further confirmation that anemia has an adverse effect of CKD progression has been derived from two small randomized studies that reported renoprotective benefit when anemia is corrected with erythopoietin.[857,858] On the other hand, two other studies that had effects on left ventricular mass as their primary endpoint,[859,860] as well as the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT),[861] found no effect of high versus low hemoglobin target on rate of decline in the GFR. In the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, randomization to a higher hemoglobin target (13–15 mg/dL) was associated with a shorter time to initiation of dialysis than the lower target (10.5–11.5 mg/dL).[862]

The reasons for the apparent contradiction between the beneficial hemodynamic effects of anemia in experimental models, and the identification of anemia as a risk factor for CKD progression in clinical studies, are unknown. It is possible that the benefit of the hemodynamic effects is outweighed by other factors, such as increased renal hypoxia and ROS formation, which may contribute to progressive renal damage.[863] Nevertheless, several recent studies have indicated that normalization of hemoglobin in CKD may be associated with several serious adverse effects, including increased risk of stroke[861] and death.[864] Issues related to the treatment of anemia in CKD are discussed further in Chapter 55.

TOBACCO SMOKING

Smoking produces acute sympathetic nervous system activation resulting in tachycardia and an increase in SBP of up to 21 mm Hg.[865] Vasoconstriction occurs in several vascular beds, including the kidneys. Among healthy nonsmoking volunteers, acute exposure to cigarette smoke caused an 11% increase in renovascular resistance accompanied by a 15% reduction in the GFR and an 18% decrease in filtration fraction. These effects appear to be mediated, at least in part, by nicotine, because similar responses were observed after chewing nicotine gum.[866] The renal hemodynamic effects of smoking can be blocked by pretreatment with a beta blocker, indicating that β-adrenergic stimulation is also involved.[867]

The effects of chronic smoking on the normal kidney are less well defined. Renal plasma flow but not the GFR is reduced in chronic smokers, and plasma endothelin levels are elevated. In one population-based study, chronic smoking was associated with a small increase in creatinine clearance, implying that smoking may cause glomerular hyperfiltration.[868] That these functional abnormalities may result in structural changes to blood vessels is suggested by the observation of abnormal intrarenal vasculature in smokers.[869,870] Moreover, epidemiologic studies have found smoking to be an important predictor of albuminuria in the general population.[868,871] In one study, heavy smoking (>20 cigarettes/day) was associated with a relative risk for albuminuria of 1.92.[871]

Furthermore, in other epidemiologic studies, smoking has been identified as a significant risk factor for CKD[614,872,873] and the development of ESKD.[613] Smoking was associated with an increased risk of developing of glomerular hyperfiltration (eGFR ≥117 ml/min/1.73 m^2; OR, 1.32 vs. nonsmokers), as well as proteinuria (OR, 1.51 vs. nonsmokers) in one longitudinal study of 10,118 middle-aged Japanese workers.[874] Two other similar longitudinal studies from Japan have confirmed that smoking is associated with an increased risk of developing proteinuria but with a higher mean eGFR than in nonsmokers.[875,876] In one of the studies, smoking was associated with a reduced risk of developing CKD stage 3.[876]

Whereas more studies are required to elucidate the effects of smoking on healthy kidneys, several studies have suggested that smoking is a risk factor for disease progression in a variety of forms of CKD, including diabetic kidney disease,[877–884] autosomal dominant polycystic kidney disease, IgA nephropathy,[885] lupus nephritis,[886] primary glomerulonephritis,[887] and general population cohorts.[888–890] On the other hand, two large cohort studies of persons with CKD have reported no association between smoking status and CKD progression.[891,892]

Mechanisms whereby cigarette smoking may result in renal injury are the subject of ongoing research but are thought to include sympathetic nervous system activation, glomerular capillary hypertension, endothelial cell injury, and direct tubulotoxocity.[893] Furthemore, nicotine administration in Thy1 rats increased mesangial cell accumulation, and in vitro nicotine increased COX-2 expression and mesangial cell proliferation.[894] Similarly, in a mouse model of diabetic nephropathy, nicotine increased proteinuria, glomerular hypertrophy, and mesangial area. NOX4, nitrotyrosine, and Akt were also increased. In vitro nicotine and high glucose levels were found to have additive effects in stimulating the generation of ROS and Akt phosphorylation in mesangial cells.[895] Nicotine administration after 5/6 nephrectomy in rats was associated with a small increase in BP, as well as increased proteinuria (but not albuminuria). The glomerular injury score at 12 weeks was exacerbated by nicotine in association with increased expression of fibronectin, NADPH oxidase, and TGF-β.[896] Among patients with CKD, the hemodynamic effects of smoking were variable, but smoking was associated with a consistent increase in the urine ACR.[866] Analysis of urine from smokers and nonsmokers has revealed significantly higher excretions of thromboxane- and prostacyclin-derived products in smokers.[897] The authors have suggested that increased synthesis of thromboxanes and prostacyclins may have pathologic importance for vascular injury, given the biologic effects of these compounds on platelets and smooth muscle cells. An important role for sympathetic nervous system activation has been suggested by an experimental study in which sympathetic denervation abrogated renal injury induced by exposure to cigarette smoke condensate.[898] A growing body of evidence thus supports the notion that the kidney is yet another organ that is adversely affected by smoking, and that smoking cessation may contribute to slowing the rate of progression of CKD.[844,899]

ACUTE KIDNEY INJURY

A growing body of evidence has indicated that recovery from AKI is associated with a substantially increased risk of CKD, and AKI superimposed on CKD has been proposed as a previously underappreciated mechanism for CKD progression. Following publication of several individual cohort studies, a metaanalysis of 13 studies reported a significantly increased risk of developing CKD and ESKD in patients who had survived an episode of AKI versus participants without AKI (pooled adjusted HR for CKD, 8.8; 95% CI, 3.1–25.5; pooled HR for ESKD, 3.1; 95% CI, 1.9–5.0).[900]

Studies in animal models of AKI have identified failure of dedifferentiation of tubule cells as a key mechanism associated with progressive TIF after AKI. After acute tubular necrosis, regeneration of tubules is achieved by dedifferentiation of remaining tubule cells, followed by proliferation to replace lost cells and redifferentiation. If this process fails, tubules become arrested in the dedifferentiated state and continue to produce proinflammatory and profibrotic cytokines that drive progressive interstitial fibrosis. Activation of pericytes results in differentiation into myofibroblasts that contribute to fibrosis. The loss of pericytes contributes to loss of endothelial integrity and capillary rarefaction that exacerbates tissue hypoxia and fibrosis.[901] The specific factors that provoke progressive kidney damage after AKI remain to be elucidated. It has been proposed that a single episode of AKI normally heals without progressive kidney damage but that repeated episodes of AKI, a single very severe episode of AKI, or AKI superimposed on preexisting CKD may provoke the above mechanisms.[902] The interaction between AKI and CKD progression has been demonstrated by a study of 39,805 patients with eGFR less than 45 mL/min/1.73 m^2 before hospitalization.[903] Those who survived an episode of dialysis-requiring AKI had a very high risk of developing ESKD within 30 days of hospital discharge—that is, nonrecovery of AKI that was related to preadmission eGFR. For an eGFR of 30 to 44 mL/min/1.73 m^2, the incidence of ESKD was 42%, and for an eGFR of 15 to 29 mL/min/1.73 m^2, it was as high as 63%, whereas the incidence of ESKD was only 1.5% among those who did not have dialysis-requiring AKI. Among patients who survived more than 30 days after hospital discharge without ESKD, the incidence of ESKD and death at 6 months were 12.7% and 19.7%, respectively, versus 1.7% and 7.4% in the comparator group with CKD but no AKI. After adjustment for multiple risk factors AKI was associated with a 30% increase in long-term risk for death or ESKD (adjusted HR, 1.30; 95% CI 1.04–1.64).

The interaction between AKI and CKD has been explored in multiple animal models.[904] In rats subjected to renal ischemia 2 weeks after 3/4 nephrectomy, uninephrectomy, or sham operation, ischemia after 3/4 nephrectomy was associated with a sustained increase in serum creatinine levels and more tubules that failed to redifferentiate, associated with more severe capillary rarefaction and TIF. Furthermore, rats that were initially normotensive after 3/4 nephrectomy developed hypertension and proteinuria at 2 to 4 weeks after ischemia.[905] The investigators proposed that loss of autoregulation results in greater transmission of elevated systemic BP to the glomerulus that exacerbates glomerular damage and contributes to CKD progression.[902] In another animal model, investigators observed that moderate ischemia reperfusion injury resulted in AKI with transient activation of Wtn/β-catenin signaling and renal recovery, but severe ischemia reperfusion injury caused a sustained and exaggerated activation of Wtn/β-catenin signaling associated with progressive renal fibrosis.[906] Moreover, overexpression of Wtn1 accelerated the progression of AKI to CKD, and blockade of Wnt/β-catenin ameliorated the progression of AKI to CKD. Proposed interactions between mechanisms of kidney injury post AKI and mechanisms of CKD progression are illustrated in Fig. 51.14.

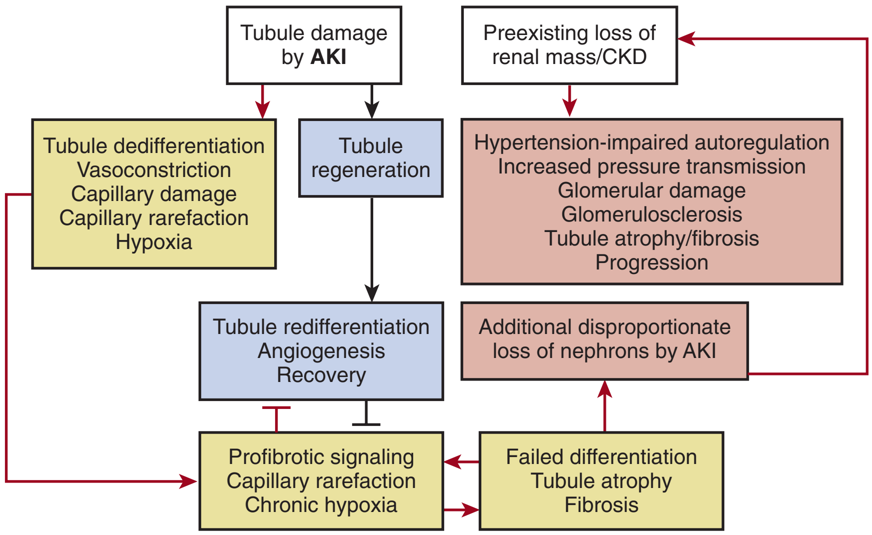

Fig. 51.14 Failed tubule differentiation and loss of renal mass after acute kidney injury *(AKI)* lead to hemodynamic abnormalities that cause chronic kidney disease *(CKD)* progression. This schematic diagram illustrates the effects of AKI that lead to tubulointerstitial fibrosis, the renal mass reduction that retards recovery of tubules regenerating after AKI, and the resulting disproportionate further reduction of renal mass that triggers hemodynamic mechanisms of renal disease progression. (From Venkatachalam MA, Weinberg JM, Kriz W, Bidani AK. Failed tubule recovery, AKI-CKD transition, and kidney disease progression. *J Am Soc Nephrol.* 2015;26(8):1765–1776.)

FUTURE DIRECTIONS

The development of pharmacologic inhibitors of the RAAS has provided powerful and incisive tools to explore renal hemodynamic and other associated adaptations in the setting of progressive renal injury. These insights have paved the way for clinical studies that provided clear evidence for the use of ACE inhibitor and ARB treatment as the mainstay of renoprotective strategies. Nevertheless, these studies have shown, at best, a 50% decrease in the rate of CKD progression. Ongoing research involving cell biology and molecular cloning, as well as genomics and proteomics, continues to yield novel insights into the mechanisms of progressive renal injury that promise to direct researchers to potential new molecular targets for renoprotective interventions. The development of the means to inhibit molecular targets specifically may provide new forms of therapy for those with CKD and enable physicians to realize the ultimate goal of achieving remission of progressive renal injury in most patients, and even regression of renal damage in some.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Micropuncture studies showed that after 5/6 nephrectomy in rats, glomerular capillary hydraulic pressure increased due to:
   a. An increase in efferent arteriolar resistance.
   b. An increase in afferent arteriolar resistance.
   c. A greater increase in afferent arteriolar resistance than efferent arteriolar resistance.
   d. A greater decrease in afferent arteriolar resistance than efferent arteriolar resistance.
   e. A greater decrease in efferent arteriolar resistance than afferent arteriolar resistance.

   Answer: d

   Rationale: After 5/6 nephrectomy, the glomerular hemodynamic changes observed are the result of the combined effects of multiple vasodilator and vasoconstrictor molecules. Overall, there is vasodilation of both afferent and efferent arterioles. However, because of the differential vasoconstrictor effects of angiotensin II, the efferent arteriole dilates relatively less than the afferent arteriole, resulting in an increase in glomerular capillary hydraulic pressure and an increase in the single-nephron glomerular filtration rate. (See Table 51.1)

2. Based on experimental evidence, which of the following statements about renal hypertrophy is false?
   a. Biochemical evidence of a hypertrophic response is evident within hours of uninephrectomy.
   b. Renal hypertrophy is accompanied by an increase in nephron number.
   c. The extent of hypertrophy increases with the amount of renal tissue removed.
   d. The extent of hypertrophy decreases with increasing age.
   e. Kidney weight is increased at 48 to 72 hours after uninephrectomy and achieves a 30% to 40% gain at 2 to 3 weeks.

   Answer: b

   Rationale: In humans and other mammals, no new nephrons are formed after birth (or shortly thereafter). Therefore renal hypertrophy after birth is achieved by growth of existing nephrons, not the development of new nephrons.

3. In rats subjected to 5/6 nephrectomy, treatment with an angiotensin-converting enzyme (ACE) inhibitor resulted in which of the following responses?
   a. Normalization of both glomerular capillary hydraulic pressure and single-nephron glomerular filtration rate (SNGFR).
   b. Normalization of glomerular capillary hydraulic pressure but no change in SNGFR.
   c. No change in glomerular capillary hydraulic pressure and normalization of SNGFR.
   d. No change in either glomerular capillary hydraulic pressure or SNGFR.
   e. Normalization of glomerular capillary hydraulic pressure and a further increase in SNGFR.

   Answer: b

   Rationale: Treatment with an ACE inhibitor reduces efferent arteriolar resistance to normalize glomerular capillary hydraulic pressure but the SNGFR remains elevated. Because treatment with an ACE inhibitor affords renoprotection, this implies that it is glomerular capillary hypertension rather than glomerular hyperfiltration that is the most important hemodynamic factor that drives chronic kidney disease progression.

4. As the glomerular filtration rate (GFR) decreases, decreased tubular reabsorption is the dominant mechanism that allows the nephron to maintain equilibrium for which of the following solutes?
   a. Sodium
   b. Potassium
   c. Phosphate
   d. Calcium
   e. Sodium and phosphate

   Answer: e

   Rationale: As the GFR decreases, sodium and phosphate balance is maintained largely through decreased reabsorption of filtered solute. Maintaining potassium balance requires increased secretion of potassium in the distal nephron of fewer nephrons. Fractional excretion of calcium remains unchanged until the GFR falls below 25 mL/min/1.73 m^2, when fractional excretion increases due to the obligatory solute diuresis.

5. Based on observations in experimental models of chronic kidney disease (CKD), which of the following statements is false?
   a. Glomerular capillary hypertension is a key factor driving a vicious cycle of progressive nephron loss.
   b. Hemodynamic and nonhemodynamic factors contribute to progressive kidney damage.
   c. Kidney function decreases due to a similar decline in the function of all nephrons.
   d. Angiotensin II is a key molecular mediator of mechanisms that contribute to CKD progression.
   e. Transforming growth factor-β is an important mediator of glomerulosclerosis and interstitial fibrosis.

   Answer: c

   Rationale: There is strong evidence from animal and clinical studies to support the Bricker hypothesis, which states that in CKD, kidney function declines due to a decreasing number of functioning (or hyperfunctioning) nephrons. This concept is central to our understanding of the mechanisms of CKD progression.